

A company hosts a website on Amazon EC2 instances behind an Application Load Balancer (ALB). The website serves static content. Website traffic is increasing. The company wants to minimize the website hosting costs.

Which solution will meet these requirements?

- A. Move the website to an Amazon S3 bucket. Configure an Amazon CloudFront distribution for the S3 bucket. **Most Voted**
- B. Move the website to an Amazon S3 bucket. Configure an Amazon ElastiCache cluster for the S3 bucket.
- C. Move the website to AWS Amplify. Configure an ALB to resolve to the Amplify website.
- D. Move the website to AWS Amplify. Configure EC2 instances to cache the website.

Correct Answer: A

Community vote distribution

A (80%)

B (20%)

Comments

Scheldon **Highly Voted** 2 years ago

Selected Answer: A

Answer A

Based on <https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/getting-started-secure-static-website-cloudformation-template.html>

Amazon CloudFront:

Uses the durable storage of Amazon Simple Storage Service (Amazon S3) – This solution creates an Amazon S3 bucket to host your static website's content. To update your website, just upload your new files to the S3 bucket.

upvoted 9 times

Dantecito **Most Recent** 1 year, 3 months ago

Selected Answer: A

I don't think anyone has mentioned it, but cloudfront helps reduce costs. If you want to use ElastiCache, it is possible but incredibly difficult, the question asks to reduce costs and cloudfront is cheaper than ElastiCache.

upvoted 1 times

joanna91 1 year, 4 months ago

Selected Answer: A

A vs B -> A. Because Elsticache is for DB, not for S3.

upvoted 1 times

Cpso 1 year, 6 months ago

Selected Answer: A

A. but questions should specify it is static or dynamic website

upvoted 1 times

j21kjf0o9ijhj 1 year, 7 months ago

Selected Answer: A

Why??? Guys, you'll fail the others. Don't troll or believe everything chatgpt says... 🤖

The question indicates "The website serves static content", why would you use elasticache for it? It may work, but it'll take a helluva effort.

S3 and CloudFront is straightforward and simple. Not only that, it also works great for static content. It's AWS's cdn. It can also serve dynamic content, but that's out of scope for this question.

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/getting-started-secure-static-website-cloudformation-template.html>

upvoted 2 times

mk168898 1 year, 7 months ago

Selected Answer: B

S3 bucket is correct, so it's between A or B.
B because cache to handle when traffic increase.
upvoted 1 times

mk168898 1 year, 7 months ago

i changed my answer to A because elasticache is only for dynamic content but question only requires static content
upvoted 2 times

Abdullah2004 1 year, 9 months ago

Selected Answer: B

website serves static content So we need to ElastiCache when traffic increases , no need for cloudfront
upvoted 1 times

muhammadahmer36 1 year, 11 months ago

Selected Answer: A

A. Move the website to an Amazon S3 bucket. Configure an Amazon CloudFront distribution for the S3 bucket.
upvoted 3 times

FrozenCarrot 1 year, 11 months ago

Selected Answer: B

S3 for static contents, and ElastiCache for traffic increasing. No need for cloudfront cuz there is no need for global deliver for the website
upvoted 3 times

FrozenCarrot 1 year, 11 months ago

" customers often complement S3 with an in-memory cache, such as Amazon ElastiCache for Redis, to reduce the S3 retrieval cost and to improve performance." --<https://aws.amazon.com/blogs/storage/turbocharge-amazon-s3-with-amazon-elasticache-for-redis/>
upvoted 1 times

trinh_le 2 years, 1 month ago

Selected Answer: A

static content -> S3
upvoted 3 times

A company is implementing a shared storage solution for a media application that the company hosts on AWS. The company needs the ability to use SMB clients to access stored data.

Which solution will meet these requirements with the LEAST administrative overhead?

- A. Create an AWS Storage Gateway Volume Gateway. Create a file share that uses the required client protocol. Connect the application server to the file share.
- B. Create an AWS Storage Gateway Tape Gateway. Configure tapes to use Amazon S3. Connect the application server to the Tape Gateway.
- C. Create an Amazon EC2 Windows instance. Install and configure a Windows file share role on the instance. Connect the application server to the file share.
- D. Create an Amazon FSx for Windows File Server file system. Connect the application server to the file system. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

LeonSauveterre 1 year, 5 months ago

Selected Answer: D

FSx for Windows File Server for sure. About other options:

A - Volume Gateway doesn't directly support SMB. SMB is more commonly supported in File Gateway or FSx for Windows.

B - Tape Gateway is for backup, not for file sharing via SMB.

C - This is actually a valid alternative if you're looking for more control, but it involves manual setup, maintenance, patches, backups and of course scaling, which increases administrative overhead in a big way.

upvoted 2 times

muhammadahmer36 1 year, 11 months ago

Selected Answer: D

D. Create an Amazon FSx for Windows File Server file system. Connect the application server to the file system.

upvoted 2 times

Scheldon 2 years ago

Selected Answer: D

Answer D

<https://aws.amazon.com/fsx/windows/>

upvoted 4 times

trinh_le 2 years, 1 month ago

Selected Answer: D

SMB protocol -> FSx windows

upvoted 3 times

A company is designing its production application's disaster recovery (DR) strategy. The application is backed by a MySQL database on an Amazon Aurora cluster in the us-east-1 Region. The company has chosen the us-west-1 Region as its DR Region.

The company's target recovery point objective (RPO) is 5 minutes and the target recovery time objective (RTO) is 20 minutes. The company wants to minimize configuration changes.

Which solution will meet these requirements with the MOST operational efficiency?

- A. Create an Aurora read replica in us-west-1 similar in size to the production application's Aurora MySQL cluster writer instance.
- B. Convert the Aurora cluster to an Aurora global database. Configure managed failover. **Most Voted**
- C. Create a new Aurora cluster in us-west-1 that has Cross-Region Replication.
- D. Create a new Aurora cluster in us-west-1. Use AWS Database Migration Service (AWS DMS) to sync both clusters.

Correct Answer: B

Community vote distribution

B (100%)

Comments

LeonSauveterre 1 year, 5 months ago

Selected Answer: B

A - Read replicas can't automatically provide failover in the event of a disaster.

B - Aurora Global Databases are designed for cross-region replication and are synchronous, which means they can meet a low RPO (within seconds). Then failover is automated, meaning that if the primary region fails, traffic can be redirected to the secondary region automatically.

C - Aurora does not automatically fail over with cross-region replication. You need to promote the replica manually, which could take longer than 20 minutes. The 5-minute RPO might not be met either.

D - Well, just too much work to do manually. Let's simply rule it out.

upvoted 3 times

muhammadahmer36 1 year, 11 months ago

Selected Answer: B

B. Convert the Aurora cluster to an Aurora global database. Configure managed failover.

upvoted 1 times

EdricHoang 1 year, 11 months ago

Selected Answer: B

I go for B. However, C is also a good option except manual failover intervention

upvoted 2 times

Scheldon 2 years ago

Selected Answer: B

Answer:B

<https://aws.amazon.com/rds/aurora/global-database/>

Cross-Region disaster recovery

If your primary Region suffers a performance degradation or outage, you can promote one of the secondary Regions to take read/write responsibilities. An Aurora cluster can recover in less than 1 minute, even in the event of a complete Regional outage. This provides your application with an effective recovery point objective (RPO) of 1 second and a recovery time objective (RTO) of less than 1 minute, providing a strong foundation for a global business continuity plan.

upvoted 3 times

sandordini 2 years, 1 month ago

Selected Answer: B

Aurora Global Database: allowing a single Amazon Aurora database to span multiple AWS Regions. It replicates your data with no impact on database performance, enables fast local reads with low latency in each Region, and provides disaster recovery from Region-wide outages.

upvoted 3 times

A company runs a critical data analysis job each week before the first day of the work week. The job requires at least 1 hour to complete the analysis. The job is stateful and cannot tolerate interruptions. The company needs a solution to run the job on AWS.

Which solution will meet these requirements?

A. Create a container for the job. Schedule the job to run as an AWS Fargate task on an Amazon Elastic Container Service (Amazon ECS) cluster by using Amazon EventBridge Scheduler. **Most Voted**

B. Configure the job to run in an AWS Lambda function. Create a scheduled rule in Amazon EventBridge to invoke the Lambda function.

C. Configure an Auto Scaling group of Amazon EC2 Spot Instances that run Amazon Linux. Configure a crontab entry on the instances to run the analysis.

D. Configure an AWS DataSync task to run the job. Configure a cron expression to run the task on a schedule.

Correct Answer: A

Community vote distribution

A (100%)

Comments

Hkayne **Highly Voted** 2 years, 1 month ago

Selected Answer: A

B is wrong because the job takes 1 hour and the lambda maximum execution time is 15 minutes.

C is wrong can't use spot instances because the job can not tolerate interruptions.

D is wrong too because DataSync is not designed to launch jobs.

Correct answer is A

upvoted 6 times

mk168898 **Most Recent** 1 year, 7 months ago

B is not because lambda only max 15mins but job requires 1 hour

C is not because cannot tolerate interruptions, spot instance might be gone anytime.

D datasync doesn't seem correct

A is most correct

upvoted 1 times

744fdad 1 year, 10 months ago

I know datasync and lambda is event based, there are interruptions. C, doesn't address the scheduling requirement. Has to be A

upvoted 1 times

muhammadahmer36 1 year, 11 months ago

Selected Answer: A

A. Create a container for the job. Schedule the job to run as an AWS Fargate task on an Amazon Elastic Container Service (Amazon ECS) cluster by using Amazon EventBridge Scheduler.

upvoted 1 times

Scheldon 2 years ago

Selected Answer: A

Answer: A

Fargate is compatible with ECS and is allowing for long running tasks

upvoted 4 times

A company runs workloads in the AWS Cloud. The company wants to centrally collect security data to assess security across the entire company and to improve workload protection.

Which solution will meet these requirements with the LEAST development effort?

- A. Configure a data lake in AWS Lake Formation. Use AWS Glue crawlers to ingest the security data into the data lake.
- B. Configure an AWS Lambda function to collect the security data in .csv format. Upload the data to an Amazon S3 bucket.
- C. Configure a data lake in Amazon Security Lake to collect the security data. Upload the data to an Amazon S3 bucket.
Most Voted
- D. Configure an AWS Database Migration Service (AWS DMS) replication instance to load the security data into an Amazon RDS cluster.

Correct Answer: C

Community vote distribution

C (100%)

Comments

sandordini **Highly Voted** 2 years, 1 month ago

Selected Answer: C

A, B, D are senseless +

Amazon Security Lake automatically centralizes security data from AWS environments, you can get a more complete understanding of your security data across your entire organization. You can also improve the protection.

upvoted 7 times

LeonSauveterre **Most Recent** 1 year, 5 months ago

Selected Answer: C

Let's just focus on Security Lake.

1. Purpose-built: Amazon Security Lake is specifically designed to collect and centralize security data with minimal configuration and development overhead.
2. It integrates easily with AWS security services, providing a streamlined solution with less manual configuration compared to using AWS Glue or Lambda.
3. This is also designed to scale automatically, making it the most efficient and effective solution for the needs mentioned in the question.

upvoted 1 times

muhammadahmer36 1 year, 11 months ago

Selected Answer: C

C. Configure a data lake in Amazon Security Lake to collect the security data. Upload the data to an Amazon S3 bucket.

upvoted 1 times

Scheldon 2 years ago

Selected Answer: C

Answer C

<https://aws.amazon.com/security-lake/>

Amazon Security Lake automatically centralizes security data from AWS environments, SaaS providers, on premises, and cloud sources into a purpose-built data lake stored in your account.

upvoted 3 times

A company is migrating five on-premises applications to VPCs in the AWS Cloud. Each application is currently deployed in isolated virtual networks on premises and should be deployed similarly in the AWS Cloud. The applications need to reach a shared services VPC. All the applications must be able to communicate with each other.

If the migration is successful, the company will repeat the migration process for more than 100 applications.

Which solution will meet these requirements with the LEAST administrative overhead?

- A. Deploy software VPN tunnels between the application VPCs and the shared services VPC. Add routes between the application VPCs in their subnets to the shared services VPC.
- B. Deploy VPC peering connections between the application VPCs and the shared services VPC. Add routes between the application VPCs in their subnets to the shared services VPC through the peering connection.
- C. Deploy an AWS Direct Connect connection between the application VPCs and the shared services VPC. Add routes from the application VPCs in their subnets to the shared services VPC and the applications VPCs. Add routes from the shared services VPC subnets to the applications VPCs.
- D. Deploy a transit gateway with associations between the transit gateway and the application VPCs and the shared services VPC. Add routes between the application VPCs in their subnets and the application VPCs to the shared services VPC through the transit gateway. **Most Voted**

Correct Answer: D

Community vote distribution

D (93%)

7%

Comments

Scheldon **Highly Voted** 2 years ago

Selected Answer: D

Answer: D

<https://aws.amazon.com/transit-gateway/>

Looks like the best solution would be transit gateway. It will allow for inter-VPC communication for all 5 applications/VPC, reach shared resource/VPC and in the future it will be easy to allow for inter-communication between even 100 VPCs (applications)

upvoted 5 times

LeonSauveterre **Most Recent** 1 year, 5 months ago

Selected Answer: D

A - VPN Tunnels: High administrative overhead. Requires individual tunnel setup and routing configuration for each VPC, making scaling to 100+ VPCs inefficient.

B - VPC Peering: Moderate overhead. Requires manual peering for each pair of VPCs and routing configurations, which becomes harder to manage as more VPCs are added.

C - Direct Connect: High cost and complexity. Direct Connect is typically used for hybrid cloud setups or data center connectivity but is costly and harder to manage at large scale. Not suitable for connecting numerous VPCs in the AWS Cloud.

D - Transit Gateway: Lowest overhead. Acts as a centralized hub for managing inter-VPC communication and simplifies the process of routing between VPCs. It is highly scalable and reduces the complexity of adding new VPCs, and optimal for managing large-scale environments.

upvoted 1 times

LeonSauveterre 1 year, 5 months ago

I copied this from somewhere else for your references.

upvoted 1 times

mk168898 1 year, 7 months ago

each application needs to be in their own VPC and can communicate with each other => transit gateway

upvoted 2 times

muhammadahmer36 1 year, 11 months ago

Selected Answer: D

D. Deploy a transit gateway with associations between the transit gateway and the application VPCs and the shared services VPC. Add routes between the application VPCs in their subnets and the application VPCs to the shared services VPC through the transit gateway.

upvoted 2 times

emakid 1 year, 11 months ago

Selected Answer: D

AWS Transit Gateway:

Centralized Connectivity: AWS Transit Gateway provides a hub-and-spoke model for connecting multiple VPCs, simplifying network management by providing a single point of connectivity for all VPCs.

Scalability: It is designed to handle many VPCs, making it suitable for scaling beyond the initial five applications to more than 100 applications.

Reduced Administrative Overhead: Managing VPC peering connections or VPN tunnels for each pair of VPCs would become complex and difficult to manage at scale. Transit Gateway simplifies this by providing centralized routing and connectivity.

upvoted 3 times

DanielWuTRT 1 year, 11 months ago

Selected Answer: D

the LEAST administrative overhead = transit gateway

upvoted 2 times

0bdf3af 2 years ago

D. <https://docs.aws.amazon.com/whitepapers/latest/building-scalable-secure-multi-vpc-network-infrastructure/transit-gateway.html>

upvoted 3 times

Hkayne 2 years, 1 month ago

Selected Answer: B

Correct answer is B

upvoted 1 times

dragosky 1 year, 5 months ago

not possible, vpc peerings can have up to 125 connections and the request is for 100 apps, meaning $100(100-1)/2$ connections...

upvoted 1 times

A company wants to use Amazon Elastic Container Service (Amazon ECS) to run its on-premises application in a hybrid environment. The application currently runs on containers on premises.

The company needs a single container solution that can scale in an on-premises, hybrid, or cloud environment. The company must run new application containers in the AWS Cloud and must use a load balancer for HTTP traffic.

Which combination of actions will meet these requirements? (Choose two.)

- A. Set up an ECS cluster that uses the AWS Fargate launch type for the cloud application containers. Use an Amazon ECS Anywhere external launch type for the on-premises application containers. **Most Voted**
- B. Set up an Application Load Balancer for cloud ECS services. **Most Voted**
- C. Set up a Network Load Balancer for cloud ECS services.
- D. Set up an ECS cluster that uses the AWS Fargate launch type. Use Fargate for the cloud application containers and the on-premises application containers.
- E. Set up an ECS cluster that uses the Amazon EC2 launch type for the cloud application containers. Use Amazon ECS Anywhere with an AWS Fargate launch type for the on-premises application containers.

Correct Answer: AB

Community vote distribution

AB (100%)

Comments

Scheldon **Highly Voted** 2 years ago

Answer: AB

We need to load-balance HTTP traffic hence Application Load Balancer is needed. Because Customer want to use container solution we need to use ECS with Fargate which will launch cloud applications. To run on-premises applications in containers we need to use ECS Anywhere.

https://docs.aws.amazon.com/AmazonECS/latest/developerguide/AWS_Fargate.html

<https://docs.aws.amazon.com/AmazonECS/latest/developerguide/Welcome.html>

upvoted 5 times

Scheldon 2 years ago

Amazon ECS services on AWS Fargate support the Application Load Balancer and Network Load Balancer load balancer types. Application Load Balancers are used to route HTTP/HTTPS (or layer 7) traffic.

Serverless (AWS Fargate (Fargate)) in the AWS cloud

Fargate is a serverless, pay-as-you-go compute engine. With Fargate you don't need to manage servers, handle capacity planning, or isolate container workloads for security.

On-premises virtual machines (VM) or servers

Amazon ECS Anywhere provides support for registering an external instance such as an on-premises server or virtual machine (VM), to your Amazon ECS cluster.

upvoted 2 times

LeonSauveterre **Most Recent** 1 year, 5 months ago

Selected Answer: AB

E is incorrect because the combination of EC2 for cloud containers and Fargate for on-premises containers is unnecessary and would complicate the solution unnecessarily. Using ECS Anywhere with Fargate for both environments is a simpler and more efficient approach (option A).

Also, HTTP => ALB; TCP/UDP => NLB.

upvoted 1 times

muhammadahmer36 1 year, 11 months ago

Selected Answer: AB

AB.the combination of setting up an ECS cluster with Fargate for the cloud and ECS Anywhere for on-premises, along with using an Application Load Balancer, provides a scalable, hybrid, and cloud-native solution with minimal operational overhead.

upvoted 2 times

emakid 1 year, 11 months ago

Selected Answer: AB

the combination of setting up an ECS cluster with Fargate for the cloud and ECS Anywhere for on-premises, along with using an Application Load Balancer, provides a scalable, hybrid, and cloud-native solution with minimal operational overhead.

upvoted 3 times

KennethNg923 1 year, 12 months ago

Selected Answer: AB

Have to use fargate, and since it has on-premises application so it need to use Amazon ECS Anywhere

upvoted 3 times

Obdf3af 2 years ago

BD.

<https://aws.amazon.com/blogs/aws/getting-started-with-amazon-ecs-anywhere-now-generally-available/>

upvoted 1 times

Hkayne 2 years, 1 month ago

Selected Answer: AB

AB is the correct answer,must lunch the cluster as external lunch

upvoted 2 times

A company is migrating its workloads to AWS. The company has sensitive and critical data in on-premises relational databases that run on SQL Server instances.

The company wants to use the AWS Cloud to increase security and reduce operational overhead for the databases.

Which solution will meet these requirements?

- A. Migrate the databases to Amazon EC2 instances. Use an AWS Key Management Service (AWS KMS) AWS managed key for encryption.
- B. Migrate the databases to a Multi-AZ Amazon RDS for SQL Server DB instance. Use an AWS Key Management Service (AWS KMS) AWS managed key for encryption. **Most Voted**
- C. Migrate the data to an Amazon S3 bucket. Use Amazon Macie to ensure data security.
- D. Migrate the databases to an Amazon DynamoDB table. Use Amazon CloudWatch Logs to ensure data security.

Correct Answer: B

Community vote distribution

B (100%)

Comments

trinh_le **Highly Voted** 2 years, 1 month ago

Selected Answer: B

Migrate the databases to a Multi-AZ Amazon RDS for SQL Server DB instance. Use an AWS Key Management Service (AWS KMS) AWS managed key for encryption.

upvoted 6 times

mk168898 **Most Recent** 1 year, 7 months ago

looking to migrate SQL server instance.

D is wrong because dynamoDB is noSQL

upvoted 2 times

muhammadahmer36 1 year, 11 months ago

Selected Answer: B

B. Migrate the databases to a Multi-AZ Amazon RDS for SQL Server DB instance. Use an AWS Key Management Service (AWS KMS) AWS managed key for encryption.

upvoted 2 times

Seb888 1 year, 11 months ago

Selected Answer: B

B. Migrate the databases to a Multi-AZ Amazon RDS for SQL Server DB instance. Use an AWS Key Management Service (AWS KMS) AWS managed key for encryption.

This option provides the best balance between increased security, reduced operational overhead, and maintaining the relational database functionalities that the company needs.

upvoted 4 times

A company wants to migrate an application to AWS. The company wants to increase the application's current availability. The company wants to use AWS WAF in the application's architecture.

Which solution will meet these requirements?

A. Create an Auto Scaling group that contains multiple Amazon EC2 instances that host the application across two Availability Zones. Configure an Application Load Balancer (ALB) and set the Auto Scaling group as the target. Connect a WAF to the ALB.

Most Voted

B. Create a cluster placement group that contains multiple Amazon EC2 instances that hosts the application. Configure an Application Load Balancer and set the EC2 instances as the targets. Connect a WAF to the placement group.

C. Create two Amazon EC2 instances that host the application across two Availability Zones. Configure the EC2 instances as the targets of an Application Load Balancer (ALB). Connect a WAF to the ALB.

D. Create an Auto Scaling group that contains multiple Amazon EC2 instances that host the application across two Availability Zones. Configure an Application Load Balancer (ALB) and set the Auto Scaling group as the target. Connect a WAF to the Auto Scaling group.

Correct Answer: A

Community vote distribution

A (100%)

Comments

trinh_le **Highly Voted** 2 years, 1 month ago

Selected Answer: A

Not D because AWS WAF cannot be directly connected to an Auto Scaling Group, it should be associated with the ALB which managing the incoming web traffic

upvoted 6 times

Scheldon **Highly Voted** 1 year, 12 months ago

Selected Answer: A

AnswerA

WAF > ALB > AutoScalingGroup(MultiAZ EC2 Instances)

Looks good

upvoted 5 times

mk168898 **Most Recent** 1 year, 7 months ago

WAF manage incoming traffic should be at ALB, because auto scaling group is behind ALB then it defeats purpose of WAF

upvoted 2 times

muhammadahmer36 1 year, 11 months ago

Selected Answer: A

A. Create an Auto Scaling group that contains multiple Amazon EC2 instances that host the application across two Availability Zones. Configure an Application Load Balancer (ALB) and set the Auto Scaling group as the target. Connect a WAF to the ALB.

upvoted 2 times

sandordini 2 years, 1 month ago

A: EC2 - MultiAZ > ALB > WAF

upvoted 3 times

A company manages a data lake in an Amazon S3 bucket that numerous applications access. The S3 bucket contains a unique prefix for each application. The company wants to restrict each application to its specific prefix and to have granular control of the objects under each prefix.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Create dedicated S3 access points and access point policies for each application. **Most Voted**
- B. Create an S3 Batch Operations job to set the ACL permissions for each object in the S3 bucket.
- C. Replicate the objects in the S3 bucket to new S3 buckets for each application. Create replication rules by prefix.
- D. Replicate the objects in the S3 bucket to new S3 buckets for each application. Create dedicated S3 access points for each application.

Correct Answer: A

Community vote distribution

A (85%)

B (15%)

Comments

anirudhsharma **Highly Voted** 2 years ago

Answer A

By creating separate access points for each application, you can enforce access controls specific to their respective prefixes while minimizing administrative complexity. This approach provides a clean separation of permissions and reduces the risk of misconfigurations.

Options B, C, and D are not as efficient or straightforward:

Option B (S3 Batch Operations) involves setting ACL permissions for each object individually, which can be cumbersome and time-consuming. Option C (replicating objects to new S3 buckets) introduces additional buckets and replication rules, increasing management overhead. Option D (replicating objects and creating dedicated S3 access points) adds unnecessary complexity by combining replication and access point creation.

upvoted 6 times

AMEJack **Most Recent** 1 year, 6 months ago

Selected Answer: A

"unique prefix for each application" hence access points. Also, ACL is not recommended by AWS.

upvoted 2 times

muhammadahmer36 1 year, 11 months ago

Selected Answer: A

A. Create dedicated S3 access points and access point policies for each application.

upvoted 2 times

emakid 1 year, 11 months ago

Selected Answer: A

Explanation:

S3 Access Points: These provide a way to manage access to shared data sets in Amazon S3. Each access point has a unique hostname and a policy that is specific to the use case, allowing for granular control over access to data.

Access Point Policies: These policies can be tailored to restrict access to specific prefixes within an S3 bucket, ensuring that each application only has access to its designated prefix.

upvoted 3 times

Scheldon 2 years ago

Answer B

Taking into consideration that we have "numerous applications" (10,100,1000?) and we need meet requirements with the LEAST operational overhead I would go into automatization of operations hence Batch Operations seems to be good choice.

<https://aws.amazon.com/blogs/storage/updating-amazon-s3-object-acls-at-scale-with-s3-batch-operations/>
upvoted 1 times

f07ed8f 2 years ago

However, answer B need to set ACL for each OBJECT inside each S3... and the Batch operation has to be executed where there another new object added to the S3. I am not sure the answer too :)

upvoted 2 times

Tomrr 2 years ago

Selected Answer: A

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/access-points-policies.html>
upvoted 4 times

trinh_le 2 years, 1 month ago

Selected Answer: B

Create an S3 Batch Operations job to set the ACL permissions for each object in the S3 bucket

upvoted 2 times

aditianand 2 years ago

I have 2 questions: Is a batch process there to set ACL permissions. Secondly, they are asking with least operational overhead. Isnt A CORRECT?

upvoted 4 times

A company has an application that customers use to upload images to an Amazon S3 bucket. Each night, the company launches an Amazon EC2 Spot Fleet that processes all the images that the company received that day. The processing for each image takes 2 minutes and requires 512 MB of memory.

A solutions architect needs to change the application to process the images when the images are uploaded.

Which change will meet these requirements MOST cost-effectively?

- A. Use S3 Event Notifications to write a message with image details to an Amazon Simple Queue Service (Amazon SQS) queue. Configure an AWS Lambda function to read the messages from the queue and to process the images. **Most Voted**
- B. Use S3 Event Notifications to write a message with image details to an Amazon Simple Queue Service (Amazon SQS) queue. Configure an EC2 Reserved Instance to read the messages from the queue and to process the images.
- C. Use S3 Event Notifications to publish a message with image details to an Amazon Simple Notification Service (Amazon SNS) topic. Configure a container instance in Amazon Elastic Container Service (Amazon ECS) to subscribe to the topic and to process the images.
- D. Use S3 Event Notifications to publish a message with image details to an Amazon Simple Notification Service (Amazon SNS) topic. Configure an AWS Elastic Beanstalk application to subscribe to the topic and to process the images.

Correct Answer: A

Community vote distribution

A (100%)

Comments

aesopx39078 **Highly Voted** 1 year, 11 months ago
Congratulations! Once again, You made it!
upvoted 14 times

Scheldon **Highly Voted** 2 years ago
Selected Answer: A
Answer A

I would go with Lambda and SQS.
when using SQS we will be sure that all images will be processed and hence to process we need 2 min and 512 MB of memory (Lambda is allowing upto 15 min and upto 10K MB) Lambda should be perfect scalable solution which will allow for almost in real time image processing.
upvoted 8 times

Scheldon 2 years ago
and it is cost effective ;)
upvoted 2 times

kevindu **Most Recent** 1 year, 10 months ago
Is there anyone who has recently passed the exam who can tell me approximately how many of the original questions are in the actual exam?
upvoted 4 times

muhammadahmer36 1 year, 11 months ago
Selected Answer: A

A. Use S3 Event Notifications to write a message with image details to an Amazon Simple Queue Service (Amazon SQS) queue. Configure an AWS Lambda function to read the messages from the queue and to process the images.
upvoted 2 times

DanielWuTRT 1 year, 11 months ago

Selected Answer: A

We made it to the last one! Good luck!

upvoted 5 times

NSA_Poker 1 year, 11 months ago

Selected Answer: A

(B) is eliminated. Reserved instances are more expensive than Spot Fleet.

(C) is eliminated. Container instance more expensive than Lambda. SQS needed NOT SNS.

(D) is eliminated. Elastic Beanstalk is more expensive than Spot Fleet. SQS needed NOT SNS.

(A) is correct. It's the most cost effective service & the scope of its capabilities are within the requirements.

upvoted 5 times

trinh_le 2 years, 1 month ago

Selected Answer: A

less than 5 minutes -> use lambda

upvoted 6 times

A company wants to improve the availability and performance of its hybrid application. The application consists of a stateful TCP-based workload hosted on Amazon EC2 instances in different AWS Regions and a stateless UDP-based workload hosted on premises.

Which combination of actions should a solutions architect take to improve availability and performance? (Choose two.)

- A. Create an accelerator using AWS Global Accelerator. Add the load balancers as endpoints. **Most Voted**
- B. Create an Amazon CloudFront distribution with an origin that uses Amazon Route 53 latency-based routing to route requests to the load balancers.
- C. Configure two Application Load Balancers in each Region. The first will route to the EC2 endpoints, and the second will route to the on-premises endpoints.
- D. Configure a Network Load Balancer in each Region to address the EC2 endpoints. Configure a Network Load Balancer in each Region that routes to the on-premises endpoints. **Most Voted**
- E. Configure a Network Load Balancer in each Region to address the EC2 endpoints. Configure an Application Load Balancer in each Region that routes to the on-premises endpoints.

Correct Answer: AD

Community vote distribution

AD (100%)

Comments

Ricksaurus **Highly Voted** 1 year, 10 months ago

Selected Answer: AD

TCP >> NLB
non-http >> accelerator.
upvoted 7 times

GOTJ **Most Recent** 1 year, 4 months ago

Selected Answer: AD

But, regarding to option "D", what's the reason behind the multi-regional deploy of the NLB for on-premises UDP endpoints? Shouldn't it be deployed on the nearest region only?
upvoted 2 times

muhammadahmer36 1 year, 10 months ago

Selected Answer: AD

TCP >> NLB
non-http >> accelerator
upvoted 3 times

swati1508 1 year, 10 months ago

AD
-TCP use NLB
FOR non-http use accelerator.
upvoted 3 times

A company runs a self-managed Microsoft SQL Server on Amazon EC2 instances and Amazon Elastic Block Store (Amazon EBS). Daily snapshots are taken of the EBS volumes.

Recently, all the company's EBS snapshots were accidentally deleted while running a snapshot cleaning script that deletes all expired EBS snapshots. A solutions architect needs to update the architecture to prevent data loss without retaining EBS snapshots indefinitely.

Which solution will meet these requirements with the LEAST development effort?

- A. Change the IAM policy of the user to deny EBS snapshot deletion.
- B. Copy the EBS snapshots to another AWS Region after completing the snapshots daily.
- C. Create a 7-day EBS snapshot retention rule in Recycle Bin and apply the rule for all snapshots. **Most Voted**
- D. Copy EBS snapshots to Amazon S3 Standard-Infrequent Access (S3 Standard-IA).

Correct Answer: C

Community vote distribution

C (100%)

Comments

FlyingHawk 1 year, 4 months ago

Selected Answer: C

AWS Recycle Bin allows you to set retention rules for EBS snapshots, preventing them from being permanently deleted for a specified period (e.g., 7 days).

If a snapshot is accidentally deleted, it is moved to the Recycle Bin and can be restored within the retention period.

<https://docs.aws.amazon.com/ebs/latest/userguide/recycle-bin.html>

upvoted 1 times

dhewa 1 year, 9 months ago

Selected Answer: C

Provides a safety net against accidental deletions

upvoted 1 times

progounick 1 year, 9 months ago

Selected Answer: C

C. Snapshot can be recovered if accidentally deleted

upvoted 1 times

pujithacg8 1 year, 10 months ago

C, AWS Recycle Bin allows you to recover resources like EBS snapshots that were accidentally deleted.

upvoted 1 times

AnasAWS 1 year, 10 months ago

Selected Answer: C

7 Days retention period will protect snapshots from being accidentally permanently deleted.

upvoted 1 times

Ricksaurus 1 year, 10 months ago

Selected Answer: C

Snapshot can be recovered if accidentally deleted

upvoted 1 times

example_ 1 year, 10 months ago

Selected Answer: C

<https://aws.amazon.com/blogs/aws/new-recycle-bin-for-ebs-snapshots/>
upvoted 1 times

A company wants to use an AWS CloudFormation stack for its application in a test environment. The company stores the CloudFormation template in an Amazon S3 bucket that blocks public access. The company wants to grant CloudFormation access to the template in the S3 bucket based on specific user requests to create the test environment. The solution must follow security best practices.

Which solution will meet these requirements?

- A. Create a gateway VPC endpoint for Amazon S3. Configure the CloudFormation stack to use the S3 object URL.
- B. Create an Amazon API Gateway REST API that has the S3 bucket as the target. Configure the CloudFormation stack to use the API Gateway URL.
- C. Create a presigned URL for the template object. Configure the CloudFormation stack to use the presigned URL. **Most Voted**
- D. Allow public access to the template object in the S3 bucket. Block the public access after the test environment is created.

Correct Answer: C

Community vote distribution

C (71%)

A (29%)

Comments

AnasAWS **Highly Voted** 1 year, 10 months ago

Selected Answer: C

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/ShareObjectPreSignedURL.html>

upvoted 6 times

blehbleh 1 year, 9 months ago

Thank you for posting this.

upvoted 2 times

Dantecito **Most Recent** 1 year, 3 months ago

Selected Answer: C

C. Not the most secure but the only that works.

A. Cloudformation is a regional service so is outside the vpc and gateway VPC endpoint only works inside a VPC.

B. Doesn't do anything, we need a lambda between api gateway and cloudformation.

D. public access is a no.

upvoted 2 times

FlyingHawk 1 year, 4 months ago

Selected Answer: A

Option A the gateway endpoint for S3 provides private access to the S3 template, aligns with security best practices, and incurs no additional costs. Option C (presigned URLs) potentially exposing the template to the public internet, even if temporarily. You only use C for the following reasons If you need a quick, temporary solution for granting access to the template.

If you are not using a VPC or do not want to set up a VPC endpoint.

If the template is not highly sensitive and temporary public exposure is acceptable.

upvoted 2 times

FlyingHawk 1 year, 4 months ago

Given the requirement to follow security best practices, Option A (VPC Endpoint) is the better choice. While it requires slightly more effort to set up initially, it provides a secure, private, and persistent solution that aligns with security best practices.

If development effort is a critical factor and the company is comfortable with the temporary public exposure of the template, Option C (Presigned URL) can be considered as a quick and easy alternative.

upvoted 1 times

robotgeek 1 year, 6 months ago

Selected Answer: A

The question states "specific user requests" meaning a shared URL does not meet the requirement. BUT with gateway VPC endpoint you can specify custom policies as in the following link

<https://docs.aws.amazon.com/vpc/latest/privatelink/vpc-endpoints-access.html>

upvoted 2 times

pujithacg8 1 year, 10 months ago

C, A presigned URL grants temporary access to an S3 object without making it publicly accessible.

upvoted 1 times

flaviobrf 1 year, 10 months ago

Selected Answer: C

For me C is the right answer

upvoted 2 times

A company has applications that run in an organization in AWS Organizations. The company outsources operational support of the applications. The company needs to provide access for the external support engineers without compromising security.

The external support engineers need access to the AWS Management Console. The external support engineers also need operating system access to the company's fleet of Amazon EC2 instances that run Amazon Linux in private subnets.

Which solution will meet these requirements MOST securely?

A. Confirm that AWS Systems Manager Agent (SSM Agent) is installed on all instances. Assign an instance profile with the necessary policy to connect to Systems Manager. Use AWS IAM Identity Center to provide the external support engineers console access. Use Systems Manager Session Manager to assign the required permissions. **Most Voted**

B. Confirm that AWS Systems Manager Agent (SSM Agent) is installed on all instances. Assign an instance profile with the necessary policy to connect to Systems Manager. Use Systems Manager Session Manager to provide local IAM user credentials in each AWS account to the external support engineers for console access.

C. Confirm that all instances have a security group that allows SSH access only from the external support engineers' source IP address ranges. Provide local IAM user credentials in each AWS account to the external support engineers for console access. Provide each external support engineer an SSH key pair to log in to the application instances.

D. Create a bastion host in a public subnet. Set up the bastion host security group to allow access from only the external engineers' IP address ranges. Ensure that all instances have a security group that allows SSH access from the bastion host. Provide each external support engineer an SSH key pair to log in to the application instances. Provide local account IAM user credentials to the engineers for console access.

Correct Answer: A

Community vote distribution

A (100%)

Comments

LeonSauveterre 1 year, 5 months ago

Selected Answer: A

A - It leverages AWS-managed solutions that simplify operations, enhance security, and meet the requirement for both console and OS-level access, while avoiding the pitfalls of SSH-based access or local user management, so this is what we're looking for.

B - Providing local IAM user credentials increases operational overhead and security risks. Managing these credentials across multiple AWS accounts can lead to inconsistency and potential vulnerabilities.

C - Allowing SSH access via security groups and distributing SSH key pairs introduces significant security risks. Keys could be lost or misused, and managing IP address ranges is not a piece of cake.

D - Bastion hosts require ongoing maintenance and monitoring to ensure security, which is unnecessary with Session Manager.

upvoted 2 times

EllenLiu 1 year, 5 months ago

Selected Answer: A

B: I believe the Session Manager may not have the capability to provide IAM user credentials. If it were to do so, it could potentially lead to security concerns.

upvoted 1 times

Cpso 1 year, 6 months ago

Selected Answer: A

A or B. Exam not tell if they have IDP and no. of outsource.

upvoted 2 times

Changwha 1 year, 6 months ago

A: This solution minimizes the attack surface by eliminating the need for SSH access, avoids bastion hosts, and provides secure, auditable, and federated access to AWS resources.

upvoted 2 times

swati1508 1 year, 10 months ago

A use MSM

upvoted 2 times

officedepotadmin 1 year, 10 months ago

Selected Answer: A

Systems Manager Session Manager allows secure, auditable, and controlled access to your EC2 instances without needing to open SSH ports or manage SSH keys, reducing the attack surface.

Local IAM user credentials are less secure and harder to manage at scale compared to using IAM Identity Center.

upvoted 3 times

A company uses Amazon RDS for PostgreSQL to run its applications in the us-east-1 Region. The company also uses machine learning (ML) models to forecast annual revenue based on near real-time reports. The reports are generated by using the same RDS for PostgreSQL database. The database performance slows during business hours. The company needs to improve database performance.

Which solution will meet these requirements MOST cost-effectively?

- A. Create a cross-Region read replica. Configure the reports to be generated from the read replica.
- B. Activate Multi-AZ DB instance deployment for RDS for PostgreSQL. Configure the reports to be generated from the standby database.
- C. Use AWS Data Migration Service (AWS DMS) to logically replicate data to a new database. Configure the reports to be generated from the new database.
- D. Create a read replica in us-east-1. Configure the reports to be generated from the read replica. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

hpirnaj 1 year, 5 months ago

Selected Answer: D

MOST cost-effectively

upvoted 1 times

muhammadahmer36 1 year, 10 months ago

Selected Answer: D

Read replicas are typically less expensive than setting up a cross-Region replica or activating Multi-AZ deployments. You only pay for the additional read replica, without the overhead costs associated with cross-Region data transfer or maintaining a synchronous standby in Multi-AZ setups.

upvoted 1 times

officedepotadmin 1 year, 10 months ago

Selected Answer: D

Read replicas are typically less expensive than setting up a cross-Region replica or activating Multi-AZ deployments. You only pay for the additional read replica, without the overhead costs associated with cross-Region data transfer or maintaining a synchronous standby in Multi-AZ setups.

upvoted 1 times

A company hosts its multi-tier, public web application in the AWS Cloud. The web application runs on Amazon EC2 instances, and its database runs on Amazon RDS. The company is anticipating a large increase in sales during an upcoming holiday weekend. A solutions architect needs to build a solution to analyze the performance of the web application with a granularity of no more than 2 minutes.

What should the solutions architect do to meet this requirement?

- A. Send Amazon CloudWatch logs to Amazon Redshift. Use Amazon QuickSight to perform further analysis.
- B. Enable detailed monitoring on all EC2 instances. Use Amazon CloudWatch metrics to perform further analysis. **Most Voted**
- C. Create an AWS Lambda function to fetch EC2 logs from Amazon CloudWatch Logs. Use Amazon CloudWatch metrics to perform further analysis.
- D. Send EC2 logs to Amazon S3. Use Amazon Redshift to fetch logs from the S3 bucket to process raw data for further analysis with Amazon QuickSight.

Correct Answer: B

Community vote distribution

B (100%)

Comments

muhammadahmer36 1 year, 10 months ago

Selected Answer: B

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/using-cloudwatch-new.html>

upvoted 2 times

officedepotadmin 1 year, 10 months ago

Selected Answer: B

Enabling detailed monitoring on EC2 instances provides metrics at a 1-minute granularity, which is well within the required 2-minute granularity for performance analysis.

upvoted 2 times

komorebi 1 year, 10 months ago

Selected Answer: B

Answer is B

upvoted 2 times

A company runs an application that stores and shares photos. Users upload the photos to an Amazon S3 bucket. Every day, users upload approximately 150 photos. The company wants to design a solution that creates a thumbnail of each new photo and stores the thumbnail in a second S3 bucket.

Which solution will meet these requirements MOST cost-effectively?

- A. Configure an Amazon EventBridge scheduled rule to invoke a script every minute on a long-running Amazon EMR cluster. Configure the script to generate thumbnails for the photos that do not have thumbnails. Configure the script to upload the thumbnails to the second S3 bucket.
- B. Configure an Amazon EventBridge scheduled rule to invoke a script every minute on a memory-optimized Amazon EC2 instance that is always on. Configure the script to generate thumbnails for the photos that do not have thumbnails. Configure the script to upload the thumbnails to the second S3 bucket.
- C. Configure an S3 event notification to invoke an AWS Lambda function each time a user uploads a new photo to the application. Configure the Lambda function to generate a thumbnail and to upload the thumbnail to the second S3 bucket.
Most Voted
- D. Configure S3 Storage Lens to invoke an AWS Lambda function each time a user uploads a new photo to the application. Configure the Lambda function to generate a thumbnail and to upload the thumbnail to a second S3 bucket.

Correct Answer: C

Community vote distribution

C (100%)

Comments

ieffiong 1 year, 3 months ago

Selected Answer: C

Take a sip of water and smile, you're almost there.

upvoted 4 times

swati1508 1 year, 10 months ago

C is correct

upvoted 2 times

komorebi 1 year, 10 months ago

Selected Answer: C

Answer is C

upvoted 2 times

A company has stored millions of objects across multiple prefixes in an Amazon S3 bucket by using the Amazon S3 Glacier Deep Archive storage class. The company needs to delete all data older than 3 years except for a subset of data that must be retained. The company has identified the data that must be retained and wants to implement a serverless solution.

Which solution will meet these requirements?

- A. Use S3 Inventory to list all objects. Use the AWS CLI to create a script that runs on an Amazon EC2 instance that deletes objects from the inventory list.
- B. Use AWS Batch to delete objects older than 3 years except for the data that must be retained.
- C. Provision an AWS Glue crawler to query objects older than 3 years. Save the manifest file of old objects. Create a script to delete objects in the manifest.
- D. Enable S3 Inventory. Create an AWS Lambda function to filter and delete objects. Invoke the Lambda function with S3 Batch Operations to delete objects by using the inventory reports. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

Salilgen 1 year, 5 months ago

Selected Answer: D

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/batch-ops.html>

upvoted 2 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: D

Takeaways:

1. S3 Inventory provides a flat file report of the objects in an S3 bucket, including metadata such as storage class, size, and last modified date. Now you can easily find files "older than 3 years".
2. The Lambda function can be used to filter objects based on the company's requirements (excluding the subset of data to retain) and to delete the objects programmatically.
3. S3 Batch Operations allow the execution of bulk operations (e.g., delete operations) on a large number of S3 objects.

A - Running a script on an EC2 instance is not serverless.

B - AWS Batch is designed for batch computing workloads, not operations like deleting S3 objects.

C - AWS Glue crawlers are designed to extract metadata and schema from data stored in S3 for querying and ETL processes, not for managing or deleting objects.

upvoted 3 times

agbor_tambe 1 year, 8 months ago

yes D is the right answer

upvoted 2 times

muhammadahmer36 1 year, 10 months ago

Selected Answer: D

Enable S3 Inventory. Create an AWS Lambda function to filter and delete objects. Invoke the Lambda function with S3 Batch Operations to delete objects by using the inventory reports.

upvoted 3 times

swati1508 1 year, 10 months ago

D is correct

upvoted 2 times

A company is building an application on AWS. The application uses multiple AWS Lambda functions to retrieve sensitive data from a single Amazon S3 bucket for processing. The company must ensure that only authorized Lambda functions can access the data. The solution must comply with the principle of least privilege.

Which solution will meet these requirements?

- A. Grant full S3 bucket access to all Lambda functions through a shared IAM role.
- B. Configure the Lambda functions to run within a VPC. Configure a bucket policy to grant access based on the Lambda functions' VPC endpoint IP addresses.
- C. Create individual IAM roles for each Lambda function. Grant the IAM roles access to the S3 bucket. Assign each IAM role as the Lambda execution role for its corresponding Lambda function. **Most Voted**
- D. Configure a bucket policy granting access to the Lambda functions based on their function ARNs.

Correct Answer: C

Community vote distribution

C (100%)

Comments

LeonSauveterre 1 year, 5 months ago

Selected Answer: C

A - Full access? No!

B - Theoretically possible but IP-based policies are harder to manage and not as secure as IAM roles.

C - By creating separate IAM roles for each Lambda function, permissions can be narrowly scoped to each function's specific needs.

D - This would actually work. It's just that bucket policies based on function ARNs are less flexible and scalable compared to IAM roles, especially when you need to manage permissions for multiple Lambda functions or add new functions.

upvoted 3 times

LeonSauveterre 1 year, 5 months ago

Out of my instinct, least privilege leads to individually managing each function.

upvoted 1 times

56ce46c 1 year, 8 months ago

i think D is also right

S3 Bucket Policy: Use an S3 bucket policy that grants access to the specific Lambda functions based on their function ARNs. This ensures that only the authorized Lambda functions can retrieve data from the S3 bucket.

upvoted 3 times

swati1508 1 year, 10 months ago

A, B and D wrong only C is right

upvoted 2 times

A company has developed a non-production application that is composed of multiple microservices for each of the company's business units. A single development team maintains all the microservices.

The current architecture uses a static web frontend and a Java-based backend that contains the application logic. The architecture also uses a MySQL database that the company hosts on an Amazon EC2 instance.

The company needs to ensure that the application is secure and available globally.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon CloudFront and AWS Amplify to host the static web frontend. Refactor the microservices to use AWS Lambda functions that the microservices access by using Amazon API Gateway. Migrate the MySQL database to an Amazon EC2 Reserved Instance.

B. Use Amazon CloudFront and Amazon S3 to host the static web frontend. Refactor the microservices to use AWS Lambda functions that the microservices access by using Amazon API Gateway. Migrate the MySQL database to Amazon RDS for MySQL. **Most Voted**

C. Use Amazon CloudFront and Amazon S3 to host the static web frontend. Refactor the microservices to use AWS Lambda functions that are in a target group behind a Network Load Balancer. Migrate the MySQL database to Amazon RDS for MySQL.

D. Use Amazon S3 to host the static web frontend. Refactor the microservices to use AWS Lambda functions that are in a target group behind an Application Load Balancer. Migrate the MySQL database to an Amazon EC2 Reserved Instance.

Correct Answer: B

Community vote distribution

B (100%)

Comments

kaleido 1 year ago

Selected Answer: B

NLB does not support lambda function as target, thus C is incorrect.

upvoted 2 times

muhammadahmer36 1 year, 10 months ago

Selected Answer: B

Answer is B

upvoted 2 times

komorebi 1 year, 10 months ago

Selected Answer: B

Answer is B

upvoted 3 times

swati1508 1 year, 10 months ago

Answer is D

upvoted 1 times

swati1508 1 year, 10 months ago

B is correct sorry

upvoted 4 times

A video game company is deploying a new gaming application to its global users. The company requires a solution that will provide near real-time reviews and rankings of the players.

A solutions architect must design a solution to provide fast access to the data. The solution must also ensure the data persists on disks in the event that the company restarts the application.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Configure an Amazon CloudFront distribution with an Amazon S3 bucket as the origin. Store the player data in the S3 bucket.
- B. Create Amazon EC2 instances in multiple AWS Regions. Store the player data on the EC2 instances. Configure Amazon Route 53 with geolocation records to direct users to the closest EC2 instance.
- C. Deploy an Amazon ElastiCache for Redis duster. Store the player data in the ElastiCache cluster. **Most Voted**
- D. Deploy an Amazon ElastiCache for Memcached duster. Store the player data in the ElastiCache cluster.

Correct Answer: C

Community vote distribution

C (100%)

Comments

officedepotadmin **Highly Voted** 1 year, 10 months ago

Selected Answer: C

Amazon ElastiCache for Redis provides in-memory caching which ensures low latency and high throughput, perfect for near real-time access to player reviews and rankings.

Redis supports data persistence by snapshotting data to disk (RDB snapshots) and appending changes to a log (AOF), ensuring that the data is not lost even if the application restarts.

upvoted 5 times

Saliigen **Most Recent** 1 year, 5 months ago

Selected Answer: C

<https://aws.amazon.com/blogs/database/building-a-real-time-gaming-leaderboard-with-amazon-elasticache-for-redis/>

upvoted 1 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: C

To be more exact, you should use Amazon ElastiCache for Redis with AOF (Append-Only File) persistence. BTW, all the "duster"s in the options should be "cluster"s.

upvoted 1 times

Bwhizzy 1 year, 8 months ago

Selected Answer: C

REDIS

- Multi AZ with Auto-Failover
- Read Replicas to scale reads and have high availability
- Data Durability using AOF persistence
- Backup and restore features
- Supports Sets and Sorted Sets

upvoted 2 times

A company is designing an application on AWS that processes sensitive data. The application stores and processes financial data for multiple customers.

To meet compliance requirements, the data for each customer must be encrypted separately at rest by using a secure, centralized key management solution. The company wants to use AWS Key Management Service (AWS KMS) to implement encryption.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Generate a unique encryption key for each customer. Store the keys in an Amazon S3 bucket. Enable server-side encryption.
- B. Deploy a hardware security appliance in the AWS environment that securely stores customer-provided encryption keys. Integrate the security appliance with AWS KMS to encrypt the sensitive data in the application.
- C. Create a single AWS KMS key to encrypt all sensitive data across the application.
- D. Create separate AWS KMS keys for each customer's data that have granular access control and logging enabled. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

officedepotadmin **Highly Voted** 1 year, 10 months ago

Selected Answer: D

While enabling server-side encryption in S3 can manage encryption, it does not offer the same level of control and auditing as AWS KMS. Managing individual keys manually in S3 would also increase operational overhead.

upvoted 7 times

Jeyaluxshan **Most Recent** 1 year, 9 months ago

D is with less management overhead

upvoted 2 times

dhewa 1 year, 9 months ago

Selected Answer: D

D is more secure

upvoted 2 times

progounick 1 year, 9 months ago

Selected Answer: D

it is obvious that D is correct

upvoted 1 times

muhammadahmer36 1 year, 10 months ago

Selected Answer: D

hile enabling server-side encryption in S3 can manage encryption, it does not offer the same level of control and auditing as AWS KMS. Managing individual keys manually in S3 would also increase operational overhead.

upvoted 2 times

nebajp 1 year, 10 months ago

Selected Answer: D

D is the correct Answer

upvoted 2 times

A company needs to design a resilient web application to process customer orders. The web application must automatically handle increases in web traffic and application usage without affecting the customer experience or losing customer orders.

Which solution will meet these requirements?

- A. Use a NAT gateway to manage web traffic. Use Amazon EC2 Auto Scaling groups to receive, process, and store processed customer orders. Use an AWS Lambda function to capture and store unprocessed orders.
- B. Use a Network Load Balancer (NLB) to manage web traffic. Use an Application Load Balancer to receive customer orders from the NLB. Use Amazon Redshift with a Multi-AZ deployment to store unprocessed and processed customer orders.
- C. Use a Gateway Load Balancer (GWLB) to manage web traffic. Use Amazon Elastic Container Service (Amazon ECS) to receive and process customer orders. Use the GWLB to capture and store unprocessed orders. Use Amazon DynamoDB to store processed customer orders.
- D. Use an Application Load Balancer to manage web traffic. Use Amazon EC2 Auto Scaling groups to receive and process customer orders. Use Amazon Simple Queue Service (Amazon SQS) to store unprocessed orders. Use Amazon RDS with a Multi-AZ deployment to store processed customer orders. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

Cpso 1 year, 6 months ago

Selected Answer: D

D is right. But explaining is unclear. should have 2 group of ec2. one for accept order to queue. another to process and store to DB.
upvoted 1 times

pujithacg8 1 year, 10 months ago

D is perfect
upvoted 2 times

komorebi 1 year, 10 months ago

Selected Answer: D

Answer is D
upvoted 3 times

swati1508 1 year, 10 months ago

Answer is D
upvoted 2 times

A company is using AWS DataSync to migrate millions of files from an on-premises system to AWS. The files are 10 KB in size on average.

The company wants to use Amazon S3 for file storage. For the first year after the migration, the files will be accessed once or twice and must be immediately available. After 1 year, the files must be archived for at least 7 years.

Which solution will meet these requirements MOST cost-effectively?

A. Use an archive tool to group the files into large objects. Use DataSync to migrate the objects. Store the objects in S3 Glacier Instant Retrieval for the first year. Use a lifecycle configuration to transition the files to S3 Glacier Deep Archive after 1 year with a retention period of 7 years. **Most Voted**

B. Use an archive tool to group the files into large objects. Use DataSync to copy the objects to S3 Standard-Infrequent Access (S3 Standard-IA). Use a lifecycle configuration to transition the files to S3 Glacier Instant Retrieval after 1 year with a retention period of 7 years.

C. Configure the destination storage class for the files as S3 Glacier Instant Retrieval. Use a lifecycle policy to transition the files to S3 Glacier Flexible Retrieval after 1 year with a retention period of 7 years.

D. Configure a DataSync task to transfer the files to S3 Standard-Infrequent Access (S3 Standard-IA). Use a lifecycle configuration to transition the files to S3 Deep Archive after 1 year with a retention period of 7 years.

Correct Answer: A

Community vote distribution



Comments

dhewa **Highly Voted** 1 year, 9 months ago

Selected Answer: D

D simplifies the process by directly using S3 Standard-IA and then transitioning to S3 Glacier Deep Archive, which aligns well with access patterns and cost requirements. Option A sounds right but using an archive tool to group files into large objects adds complexity and operational overhead. This step isn't necessary if you can directly manage the files with S3 lifecycle policies.

upvoted 13 times

GOTJ 1 year, 4 months ago

Correct during the first year (online period). Incorrect during years 2-8 (archive period). Go check the numbers in price calculator yourself with the following scenarios for the archive period (S3 Glacier Deep Archive):

* For small files: 100,000,000 files of 10KB (1000GB aprox). Considering both, the transition cost of 100,000,000 files to the Glacier Deep Archive class and the maintenance cost of 7 years for those 100,000,000 files: \$6,940,34

* For big files: 1000 files of 1000MB (1000GB aprox). Considering both, the transition cost of 1,000 files to the Glacier Deep Archive class and the maintenance cost of 7 years for those 1,000 files: \$83,23

upvoted 2 times

blehbleh **Highly Voted** 1 year, 9 months ago

Selected Answer: A

Its A, took some research but A is correct Per Amazon S# glacier page: "Amazon S3 Glacier Instant Retrieval is an archive storage class that delivers the lowest-cost storage for long-lived data that is rarely accessed and requires retrieval in milliseconds. With S3 Glacier Instant Retrieval, you can save up to 68% on storage costs compared to using the S3 Standard-Infrequent Access (S3 Standard-IA) storage class, when your data is accessed once per quarter." After the one year move it to Deep archive.

upvoted 6 times

je33 **Most Recent** 8 months, 2 weeks ago

Selected Answer: A

Group millions of small 10KB files into archives to optimize costs, and minimize long-term storage costs by transitioning from Glacier Instant Retrieval to Deep Archive.

Option D's Standard-IA costs more than Glacier Instant Retrieval during the first year.

upvoted 1 times

Odd48f2 11 months ago

Selected Answer: A

With Amazon S3 Glacier Instant Retrieval, you can save up to 68% on storage costs compared to using the S3 Standard-Infrequent Access (S3 Standard-IA) storage class, when your data is accessed once per quarter." After the one year move it to S3 Glacier Deep archive.

upvoted 1 times

TEC65 1 year, 1 month ago

Selected Answer: D

def d is the answer

upvoted 2 times

ad2dj28 1 year, 2 months ago

Selected Answer: D

10/10 TIMES STANDARD INFREQUENT ACCESS IS CHEAPER THAN USING GLACIER INSTANT RETRIEVAL WHEN COMPARING COSTS WITH PRICING CALCULATOR

upvoted 3 times

tch 1 year, 3 months ago

Selected Answer: A

B & C are out

upvoted 1 times

tch 1 year, 3 months ago

Amazon S3 Glacier Instant Retrieval is an archive storage class that delivers the lowest-cost storage for long-lived data that is rarely accessed and requires retrieval in milliseconds.

With Amazon S3 Glacier Instant Retrieval, you can save up to 68% on storage costs compared to using the S3 Standard-Infrequent Access (S3 Standard-IA) storage class, when your data is accessed once per quarter." After the one year move it to S3 Glacier Deep archive.

upvoted 1 times

Mistwalker 1 year, 4 months ago

Selected Answer: A

10KB/file x millions of files = insane unnecessary cost when each is charged at 128KB/object

No one is talking about operational overhead, they just want cost-effectiveness

upvoted 2 times

GOTJ 1 year, 4 months ago

Correct, but they mention that the files must be "immediately available" during the first year. So any option involving archive tools should be discarded for this period, due to the extra steps needed to accessing the files: figuring out which archive contains the file you are looking for, download the archive file and extract the desired file. I think this simple requirement explains the almost 50%-50% consensus for this particular question.

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Selected Answer: A

- Each file is 10 KB in size.
- S3 charges a minimum of 128 KB per object.
- Total number of files: millions of files.
- Files are accessed twice in the first year.
- After 1 year, files are archived for 7 years
- S3 Glacier: Minimum charge for 40 KB per object.

S3 Standard-IA and S3 One Zone-IA storage have a minimum billable object size of 128 KB. Smaller objects may be stored but will be charged for 128 KB of storage at the appropriate storage class rate Option D does not group those files into big object, so 1 million files will be charged as 1 million objects(128KB).

<https://aws.amazon.com/s3/pricing/>

upvoted 3 times

FlyingHawk 1 year, 4 months ago

Note

The S3 Standard-IA and S3 One Zone-IA storage classes are suitable for objects larger than 128 KB that you plan to store for at least 30 days. If an object is less than 128 KB, Amazon S3 charges you for 128 KB. If you delete an object before the end of the 30-day minimum storage duration period, you are charged for 30 days. Objects that are deleted, overwritten, or transitioned to a different storage class before 30 days will incur the normal storage usage charge plus a pro-rated charge for the remainder of the 30-day minimum. For pricing information, see Amazon S3 pricing. <https://docs.aws.amazon.com/AmazonS3/latest/userguide/storage-class-intro.html>

upvoted 2 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: D

About option A: For data stored in the S3 Glacier Instant Retrieval storage class, there is indeed a minimum object size of 128 KB. If the objects are smaller than 128 KB, AWS still charges for 128 KB per object, which can significantly impact cost efficiency when dealing with a large number of small files, such as the 10 KB files in the given scenario.

One more thing: once or twice a year is perfect for Infrequent Access. It also provides instant availability for the objects stored in it.

upvoted 2 times

LeonSauveterre 1 year, 5 months ago

I believe Standard-IA remains the most straightforward and cost-effective choice for small files (10 KB average) that are accessed 1~2 times in the first year without aggregation.

upvoted 1 times

EllenLiu 1 year, 5 months ago

Selected Answer: A

S3 Glacier Instant Retrieval is more expensive than S3 Standard-IA for the first year when immediate availability is required.

upvoted 2 times

EllenLiu 1 year, 5 months ago

sorry, choose D

upvoted 1 times

dragosky 1 year, 5 months ago

Selected Answer: A

For sure A, has amazing retrieval - Store the objects in S3 Glacier Instant Retrieval

upvoted 2 times

jfedotov 1 year, 6 months ago

Selected Answer: D

D is correct.

A is not correct, because Amazon S3 Glacier Instant Retrieval min object size is 128KB, in question 10KB

upvoted 2 times

AMEJack 1 year, 6 months ago

Selected Answer: D

There is no need to use archive tool to migrate small files, DataSync is enough.

upvoted 2 times

Ez1tap 1 year, 8 months ago

Selected Answer: A

correct answer is A you need 2 separate lifecycle policy

upvoted 4 times

Abhiinav 1 year, 9 months ago

Selected Answer: D

answer is D. Storing the objects in S3 Glacier Instant Retrieval for the first year is more expensive than S3 Standard-IA for data that is accessed infrequently.

upvoted 3 times

blehbleh 1 year, 8 months ago

Wrong, it says 1-2 times in the first year and Amazon states that s3 glacier isn't at retrieval can save up to 68% compared to s3 infrequent access.

upvoted 2 times

Abdullah2004 1 year, 9 months ago

Selected Answer: D

D is most cost effective

upvoted 3 times

A company recently performed a lift and shift migration of its on-premises Oracle database workload to run on an Amazon EC2 memory optimized Linux instance. The EC2 Linux instance uses a 1 TB Provisioned IOPS SSD (io1) EBS volume with 64,000 IOPS.

The database storage performance after the migration is slower than the performance of the on-premises database.

Which solution will improve storage performance?

A. Add more Provisioned IOPS SSD (io1) EBS volumes. Use OS commands to create a Logical Volume Management (LVM) stripe. **Most Voted**

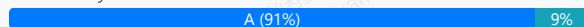
B. Increase the Provisioned IOPS SSD (io1) EBS volume to more than 64,000 IOPS.

C. Increase the size of the Provisioned IOPS SSD (io1) EBS volume to 2 TB.

D. Change the EC2 Linux instance to a storage optimized instance type. Do not change the Provisioned IOPS SSD (io1) EBS volume.

Correct Answer: A

Community vote distribution



Comments

FlyingHawk 1 year, 4 months ago

Selected Answer: A

<https://docs.aws.amazon.com/prescriptive-guidance/latest/sql-server-ec2-best-practices/stripping.html>

upvoted 2 times

FlyingHawk 1 year, 4 months ago

When the performance requirement cannot be met by a single EBS volume, consider striping using Logical Volume Management (LVM). For example, if a single volume has 250 MiB/s throughput capacity, having a stripe set across four volumes can deliver 1000 MiB/s throughput.

Volumes should be of the same size and performance characteristics.

<https://docs.aws.amazon.com/wellarchitected/latest/sap-lens/best-practice-14-2.html>

upvoted 3 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: A

A - For example, given two io1 volumes, each provisioned with 64,000 IOPS, would provide a combined 128,000 IOPS.

B - The maximum IOPS for an io1 volume is capped at 64,000 IOPS when attached to an EC2 instance.

C - While increasing the size of the volume increases baseline throughput (3 IOPS per GB), it won't exceed the 64,000 IOPS limit.

D - Storage optimized instances (like i3 or i4i) use instance store volumes, which are ephemeral (non-persistent) and cannot be used as a replacement for durable EBS volumes in this use case.

What is the deal with LVM striping?

LVM striping allows multiple volumes to be treated as a single logical volume. Reads and writes are distributed across the striped volumes, effectively increasing the available IOPS and throughput beyond the single-volume limits.

upvoted 4 times

JA2018 1 year, 5 months ago

Selected Answer: B

why I would choose Option B:

- Issue:

The problem is that the current IOPS (Input/Output Operations Per Second) provisioned on the EBS volume are not sufficient to meet the database workload demands, leading to slower performance compared to the on-premises setup.

- Solution:

Since the bottleneck is clearly identified as inadequate IOPS, the most effective solution is to simply increase the provisioned IOPS on the existing EBS volume, allowing for faster data read/write operations.

upvoted 1 times

jingen11 1 year, 7 months ago

Selected Answer: A

A

upvoted 2 times

elmyth 1 year, 9 months ago

Selected Answer: A

Increase the Provisioned IOPS SSD (io1) EBS volume to more than 64,000 IOPS.

upvoted 2 times

pujithacg8 1 year, 10 months ago

A is correct, The maximum provisioned IOPS for io1 is 64000 and hence you can achieve higher aggregate performance by adding more io1 volumes

upvoted 4 times

A company is migrating from a monolithic architecture for a web application that is hosted on Amazon EC2 to a serverless microservices architecture. The company wants to use AWS services that support an event-driven, loosely coupled architecture. The company wants to use the publish/subscribe (pub/sub) pattern.

Which solution will meet these requirements MOST cost-effectively?

- A. Configure an Amazon API Gateway REST API to invoke an AWS Lambda function that publishes events to an Amazon Simple Queue Service (Amazon SQS) queue. Configure one or more subscribers to read events from the SQS queue.
- B. Configure an Amazon API Gateway REST API to invoke an AWS Lambda function that publishes events to an Amazon Simple Notification Service (Amazon SNS) topic. Configure one or more subscribers to receive events from the SNS topic.
- C. Configure an Amazon API Gateway WebSocket API to write to a data stream in Amazon Kinesis Data Streams with enhanced fan-out. Configure one or more subscribers to receive events from the data stream.
- D. Configure an Amazon API Gateway HTTP API to invoke an AWS Lambda function that publishes events to an Amazon Simple Notification Service (Amazon SNS) topic. Configure one or more subscribers to receive events from the topic.

Most Voted

Correct Answer: D

Community vote distribution

D (74%)

B (26%)

Comments

srcntpc 1 year ago

Selected Answer: D

D for sure

upvoted 1 times

bora4motion 1 year ago

Selected Answer: D

I hate so much the "most cost effectively" questions. While B is also valid from a technical perspective apparently D is cheaper.

upvoted 2 times

tch 1 year, 3 months ago

REST APIs and HTTP APIs are both RESTful API products. REST APIs support more features than HTTP APIs, while HTTP APIs are designed with minimal features so that they can be offered at a lower price. Choose REST APIs if you need features such as API keys, per-client throttling, request validation, AWS WAF integration, or private API endpoints. Choose HTTP APIs if you don't need the features included with REST APIs.

upvoted 2 times

tomasmolinari 1 year, 4 months ago

Selected Answer: D

"HTTP APIs enable you to create RESTful APIs with lower latency and lower cost than REST APIs."

HTTP APIs are more cost-effective than REST APIs

upvoted 3 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: B

Amazon SNS is perfectly designed to support the pub/sub pattern efficiently. So B or D.

About option D: Using an HTTP API instead of a REST API is less common when using SNS for event publishing. Typically, REST APIs in API Gateway are used for interacting with Lambda functions, making option B a more conventional choice.

That said, scenarios vary based on your actual, detailed needs, like predicted frequencies and costs. For the SAA exam, I'll definitely go with B.

upvoted 1 times

jfedotov 1 year, 5 months ago

Selected Answer: D

Http api

upvoted 1 times

Denise123 1 year, 5 months ago

Selected Answer: B

While the Amazon API Gateway HTTP API (Option D) can be more cost-effective for simple APIs with low data transfer and minimal feature requirements, the Amazon API Gateway REST API (Option B) may still be more cost-effective in many real-world scenarios, especially for more complex web applications with additional requirements.

The choice between Options B and D should be based on a careful evaluation of your specific requirements, existing infrastructure, and potential cost savings or trade-offs involved.

upvoted 1 times

78b9037 1 year, 5 months ago

Selected Answer: D

HTTP API is cheaper. The difference in pricing is steep. REST APIs will run you USD \$3.50 per one million requests plus charges for data transferred out. By contrast, HTTP APIs cost a mere \$1.00 per request for the first million requests and then \$0.90 per million requests after that.

upvoted 3 times

jingen11 1 year, 7 months ago

Selected Answer: D

Question ask for cheapest.

Http api offers everything the requirements asked for. i.e Lambda function for serverless, event driven.

upvoted 2 times

fis_examtopic 1 year, 9 months ago

HTTP APIs with Lambda. I'm go with D

upvoted 1 times

progounick 1 year, 9 months ago

Selected Answer: B

rest is cheaper than http

upvoted 1 times

Noveo 1 year, 9 months ago

Quite the opposite: "REST APIs support more features than HTTP APIs, while HTTP APIs are designed with minimal features so that they can be offered at a lower price."

upvoted 3 times

mk168898 1 year, 7 months ago

based on this answer should be D

upvoted 1 times

Denise123 1 year, 5 months ago

The Amazon API Gateway HTTP API is a more recent addition to API Gateway and is designed to be a more cost-effective option for simple APIs. However, it has fewer features and capabilities compared to the REST API.

Here's a comparison of the costs between the two API Gateway options:

-REST API: \$3.50 per million API calls received, plus data transfer costs.

-HTTP API: \$0.90 per million API calls received, plus data transfer costs.

As you can see, the HTTP API is cheaper in terms of the API call pricing. However, the REST API may still be more cost-effective in certain scenarios I am not sure what is needed in this question though.

upvoted 2 times

dhewa 1 year, 9 months ago

Selected Answer: B

An HTTP API instead might not be necessary for this use case.

upvoted 2 times

muhammadahmer36 1 year, 10 months ago

Selected Answer: D

D is the right answer

upvoted 2 times

pujithacg8 1 year, 10 months ago

B or D, Will go with B

upvoted 2 times

muhammadahmer36 1 year, 10 months ago

Why B, not D?

upvoted 1 times

A company recently migrated a monolithic application to an Amazon EC2 instance and Amazon RDS. The application has tightly coupled modules. The existing design of the application gives the application the ability to run on only a single EC2 instance.

The company has noticed high CPU utilization on the EC2 instance during peak usage times. The high CPU utilization corresponds to degraded performance on Amazon RDS for read requests. The company wants to reduce the high CPU utilization and improve read request performance.

Which solution will meet these requirements?

- A. Resize the EC2 instance to an EC2 instance type that has more CPU capacity. Configure an Auto Scaling group with a minimum and maximum size of 1. Configure an RDS read replica for read requests. **Most Voted**
- B. Resize the EC2 instance to an EC2 instance type that has more CPU capacity. Configure an Auto Scaling group with a minimum and maximum size of 1. Add an RDS read replica and redirect all read/write traffic to the replica.
- C. Configure an Auto Scaling group with a minimum size of 1 and maximum size of 2. Resize the RDS DB instance to an instance type that has more CPU capacity.
- D. Resize the EC2 instance to an EC2 instance type that has more CPU capacity. Configure an Auto Scaling group with a minimum and maximum size of 1. Resize the RDS DB instance to an instance type that has more CPU capacity.

Correct Answer: A

Community vote distribution

A (100%)

Comments

GOTJ 1 year, 4 months ago

Selected Answer: A

Option "A", but... How does an auto scaling group with min = max = 1 instances contribute to the solution?

upvoted 2 times

dhewa 1 year, 9 months ago

Selected Answer: A

This approach addresses both the high CPU utilization on the EC2 instance and the degraded read performance on the RDS instance effectively.

upvoted 2 times

Abbas_Abi_AWS 1 year, 10 months ago

Selected Answer: A

A is correct

upvoted 2 times

Itetti 1 year, 10 months ago

Selected Answer: A

Option B incorrectly suggests redirecting all read/write traffic to the replica. RDS read replicas are designed to handle read operations only, not write operations. Writes must still be handled by the primary DB instance

upvoted 4 times

A company needs to grant a team of developers access to the company's AWS resources. The company must maintain a high level of security for the resources.

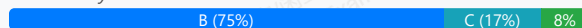
The company requires an access control solution that will prevent unauthorized access to the sensitive data.

Which solution will meet these requirements?

- A. Share the IAM user credentials for each development team member with the rest of the team to simplify access management and to streamline development workflows.
- B. Define IAM roles that have fine-grained permissions based on the principle of least privilege. Assign an IAM role to each developer. **Most Voted**
- C. Create IAM access keys to grant programmatic access to AWS resources. Allow only developers to interact with AWS resources through API calls by using the access keys.
- D. Create an AWS Cognito user pool. Grant developers access to AWS resources by using the user pool.

Correct Answer: B

Community vote distribution



Comments

In2718 10 months ago

Selected Answer: C

Even though at first answer B looks correct, answer C is better... these are developers and not anything else. It's a tricky question.
upvoted 1 times

Steve122 8 months, 3 weeks ago

AWS says:

"IAM users with access keys are an account security risk. Manage your access keys securely."

upvoted 1 times

Steve122 8 months, 3 weeks ago

I would go with B.

upvoted 1 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: B

A - Sharing IAM user credentials is a security risk and violates AWS best practices.

B - IAM roles allow for temporary credentials that can be automatically rotated, and that improves security.

C - Would work but managing static keys for multiple developers introduces significant operational overhead.

D - AWS Cognito is primarily designed for managing end-user authentication (for web or mobile apps or such), not for managing access to AWS resources for developers.

upvoted 1 times

hpirnaj 1 year, 5 months ago

it says " team of developers " . they are not in the company's AWS account. how do you want to apply IAM roles without defining users ? I think C is the answer

upvoted 1 times

hpirnaj 1 year, 5 months ago

also, roles can not assign to user/group . you need to make a policy from IAM roles first then assign policies to user/group

upvoted 1 times

KennethYY 1 year, 5 months ago

Selected Answer: D

A is wrong, common sense not security

B is wrong, role cannot assign to user/group

D is wrong, is designed for authentication and access control for web or mobile app users, not for internal developers accessing AWS resources. so remain C.

upvoted 1 times

EllenLiu 1 year, 5 months ago

Selected Answer: B

AWS Cognito is designed for authentication and access control for web or mobile app users, not for internal developers accessing AWS resources.

upvoted 1 times

EllenLiu 1 year, 5 months ago

Selected Answer: C

AWS Cognito is designed for authentication and access control for web or mobile app users, not for internal developers accessing AWS resources.

upvoted 1 times

EllenLiu 1 year, 5 months ago

Sorry, go with B.

upvoted 1 times

Cpso 1 year, 6 months ago

Selected Answer: B

good practice should B. but map role to Identity center federated to corporate IDP.

upvoted 2 times

Bwhizzy 1 year, 8 months ago

Selected Answer: B

B is the right answer. IAM Role

upvoted 2 times

aragon_saa 1 year, 10 months ago

Selected Answer: B

Answer is B

upvoted 3 times

muhammadahmer36 1 year, 10 months ago

Create an AWS Cognito user pool. Grant developers access to AWS resources by using the user pool.

upvoted 1 times

mk168898 1 year, 7 months ago

cognito for app user auth, qns asking for access to AWS resource. your answer is wrong

upvoted 2 times

A company hosts a monolithic web application on an Amazon EC2 instance. Application users have recently reported poor performance at specific times. Analysis of Amazon CloudWatch metrics shows that CPU utilization is 100% during the periods of poor performance.

The company wants to resolve this performance issue and improve application availability.

Which combination of steps will meet these requirements MOST cost-effectively? (Choose two.)

- A. Use AWS Compute Optimizer to obtain a recommendation for an instance type to scale vertically. **Most Voted**
- B. Create an Amazon Machine Image (AMI) from the web server. Reference the AMI in a new launch template.
- C. Create an Auto Scaling group and an Application Load Balancer to scale vertically.
- D. Use AWS Compute Optimizer to obtain a recommendation for an instance type to scale horizontally.
- E. Create an Auto Scaling group and an Application Load Balancer to scale horizontally. **Most Voted**

Correct Answer: AE

Community vote distribution

AE (49%) BE (46%) 6%

Comments

rpmaws **Highly Voted** 1 year, 8 months ago

Selected Answer: AE

AWS Compute Optimizer analyzes your current EC2 instance usage and recommends the most cost-effective instance type. In this case, the current instance may not have enough CPU capacity, so scaling vertically (upgrading to a larger instance type) could provide immediate relief from the 100% CPU utilization.

upvoted 6 times

FlyingHawk **Highly Voted** 1 year, 4 months ago

Selected Answer: BE

The question asks for a combination of steps, meaning the selected answers should work together to form a complete solution rather than being two independent actions.

B - prepared the instance for the scaling

E - Implement Auto Scaling for performance improvement and high availability

upvoted 5 times

In2718 **Most Recent** 10 months ago

Selected Answer: DE

Read JA2018's answer below

upvoted 1 times

bora4motion 1 year ago

Selected Answer: BE

You can't just scale vertically every single time you have a problem. I don't understand people who voted for A and E. You can't combine A and E. You can't stop the instance to improve memory/cpu of an instance and then add horizontal scaling to it. (spend even more). B and E

upvoted 2 times

7dcef09 1 year, 1 month ago

Selected Answer: AE

A: Using compute optimizer helps improve performance on the monolithic web application scale up

E: Enhances availability and scalability

upvoted 1 times

dfgdsfgdgreg 1 year, 4 months ago

Selected Answer: BE

Since options C and D are irrelevant, let's disregard them. That leaves us with ABE. B is a prerequisite for E, so you cannot have E with B. AB will be two disjointed choices. So you have to choose B/E.

upvoted 1 times

Salilgen 1 year, 5 months ago

Selected Answer: BE

The question talk about a "combination of steps" then IMO answer is BE.

BE are steps of a solution that resolve performance issue and improve application availability.

D is for improve the solution : <https://docs.aws.amazon.com/compute-optimizer/latest/ug/view-asg-recommendations.html>

A is a solution (not a step) but it is not the most cost effective

C is wrong: autoscaling and ALB are horizontal scaling

upvoted 2 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: AE

What you should know:

1. Horizontal Scaling (Scaling Out/In): Adding or removing instances to distribute the load across multiple servers, improving availability and fault tolerance. Example: Using an Auto Scaling group with a load balancer to add more EC2 instances dynamically as demand increases.

2. Vertical Scaling (Scaling Up/Down): Increasing or decreasing the size (compute power, memory, etc) of a single instance, which is easier to implement for monolithic applications but has limitations (limited instance types). Example: Upgrading an instance from "t3.medium" to "m5.large".

If you can understand above, then A & E are perfect choices.

B - The performance issue is still there!

C - Auto Scaling and Load Balancers are used for horizontal scaling, not vertical scaling.

D - Horizontal scaling involves adding more instances, not changing the instance type.

upvoted 2 times

EllenLiu 1 year, 5 months ago

Selected Answer: AE

it's truly difficult to answer as per the description.

there are two requirement:

1. resolve performance issue => A: use compute optimizer to scale vertically, it can provides an immediate fix for high CPU utilization

2. improve availability => B + D, B is the prerequisite for D, I prefer autoscaling if must choose only one

upvoted 1 times

Anyio 1 year, 5 months ago

Selected Answer: AE

The correct answers are A,E.

Option A: Correct. Using AWS Compute Optimizer to obtain a recommendation for an instance type to scale vertically is an effective first step. This service analyzes your specific application workload patterns and recommends an appropriate EC2 instance type that provides better performance for the given CPU load. This approach allows you to choose a more powerful instance that can handle peak loads more effectively without immediately shifting to a horizontal scaling architecture.

Option B: Incorrect by itself. Creating an AMI and a launch template is useful in setting up an Auto Scaling group or new instances but does not address the issue of CPU utilization or availability on its own. This process would be part of preparing for scaling but doesn't directly answer the requirement of resolving performance issues by itself.

upvoted 2 times

jfedotov 1 year, 6 months ago

Selected Answer: BE

horizontal scaling is a cost-effective solution

upvoted 1 times

JA2018 1 year, 6 months ago

Selected Answer: DE

What about options D & E?

1 Scaling horizontally is the most cost-effective solution for handling fluctuating traffic, especially when the bottleneck is CPU utilization reaching 100% during peak times. This allows adding more instances to handle the increased load instead of upgrading to a larger instance (vertical scaling) which might be unnecessary during non-peak periods.

2 Auto Scaling group and Application Load Balancer: These are essential components for horizontal scaling, where the Auto Scaling group automatically adjusts the number of instances based on demand, and the Application Load Balancer distributes traffic evenly across available instances.

#3 AWS Compute Optimizer: This service can analyze your application usage and recommend the most cost-effective instance type for your specific needs, including suggesting options for horizontal scaling.

upvoted 1 times

Abdullah2004 1 year, 7 months ago

Selected Answer: AE

Due to monolithic application we can't scale horizontal, the scale will be vertical

upvoted 2 times

jfedotov 1 year, 6 months ago

you can, you'll get for ex. 2 instances with the monolithic app, I see no issues here.

upvoted 1 times

Abdullah2004 1 year, 7 months ago

I mean the instance

upvoted 1 times

XXXXXINN 1 year, 8 months ago

BE makes more sense to me. Vertical scaling increases cost continuously even when the instance is in low demand. for a long run, the cost would be higher than just scale horizontally. The questions says the 'poor performance at specific times', so we just need to scale up at specific times.

Additionally, in order to scale up, we do need AMI and a launch template. Thus, a combination of B & E should be the correct answer

upvoted 2 times

Bwhizzy 1 year, 8 months ago

Selected Answer: AE

The answer is A and E. A for adding more CPU (Vertical scaling) and E for adding more Servers (Horizontal scaling)

upvoted 3 times

komorebi 1 year, 10 months ago

Selected Answer: BE

Answer is BE

upvoted 2 times

flaviobrf 1 year, 10 months ago

Selected Answer: BE

BE is the right choice

upvoted 3 times

A company runs all its business applications in the AWS Cloud. The company uses AWS Organizations to manage multiple AWS accounts.

A solutions architect needs to review all permissions that are granted to IAM users to determine which IAM users have more permissions than required.

Which solution will meet these requirements with the LEAST administrative overhead?

- A. Use Network Access Analyzer to review all access permissions in the company's AWS accounts.
- B. Create an AWS CloudWatch alarm that activates when an IAM user creates or modifies resources in an AWS account.
- C. Use AWS Identity and Access Management (IAM) Access Analyzer to review all the company's resources and accounts.
Most Voted
- D. Use Amazon Inspector to find vulnerabilities in existing IAM policies.

Correct Answer: C

Community vote distribution

C (100%)

Comments

muhammadahmer36 1 year, 10 months ago

Selected Answer: C

<https://aws.amazon.com/iam/access-analyzer/>

upvoted 2 times

swati1508 1 year, 10 months ago

C is correct

upvoted 2 times

komorebi 1 year, 10 months ago

Selected Answer: C

Answer is C

upvoted 3 times

A company needs to implement a new data retention policy for regulatory compliance. As part of this policy, sensitive documents that are stored in an Amazon S3 bucket must be protected from deletion or modification for a fixed period of time.

Which solution will meet these requirements?

- A. Activate S3 Object Lock on the required objects and enable governance mode.
- B. Activate S3 Object Lock on the required objects and enable compliance mode. **Most Voted**
- C. Enable versioning on the S3 bucket. Set a lifecycle policy to delete the objects after a specified period.
- D. Configure an S3 Lifecycle policy to transition objects to S3 Glacier Flexible Retrieval for the retention duration.

Correct Answer: B

Community vote distribution

B (88%) 13%

Comments

raghav2704 12 months ago

Selected Answer: B

we need the data to be protected from deletion, hence option B (it protects from both versioning and deletion) and ofcourse we dont want governance mode (option A)and option D is a sidetrack.

upvoted 1 times

Sarayounisaldossary 1 year, 2 months ago

Selected Answer: B

This is tricky between A and C but-- the question focused on compliance and FIXED Period of time... governance feature will let certain users to override B

upvoted 1 times

JA2018 1 year, 6 months ago

Selected Answer: B

S3 Object Lock:

- This feature allows you to lock objects in an S3 bucket, preventing them from being deleted or modified for a specified retention period.

Compliance mode:

- When enabled, S3 Object Lock in compliance mode provides the most stringent protection, ensuring that once an object is locked, even the account root user cannot delete or modify it, making it ideal for regulatory compliance scenarios where data must be preserved for a fixed period

upvoted 3 times

SmartGil2024 1 year, 6 months ago

Selected Answer: C

The question says : "for a fixed period of time.", why not C ? Because the items can be deleted after this fix period of time./ with compliance mode - no one can delete, it is true, but we need to also pay attention to the fixed period of time.

upvoted 1 times

JA2018 1 year, 6 months ago

While versioning helps maintain historical data, a lifecycle policy alone cannot guarantee that sensitive documents will not be deleted prematurely, as it can be modified or deleted.

upvoted 1 times

komorebi 1 year, 10 months ago

Selected Answer: B

Answer is B

upvoted 2 times

swati1508 1 year, 10 months ago

B compliance mode - no one can delete

upvoted 3 times

ccceb01 1 year, 9 months ago

Except god. lol

upvoted 2 times

A company runs its customer-facing web application on containers. The workload uses Amazon Elastic Container Service (Amazon ECS) on AWS Fargate. The web application is resource intensive.

The web application needs to be available 24 hours a day, 7 days a week for customers. The company expects the application to experience short bursts of high traffic. The workload must be highly available.

Which solution will meet these requirements MOST cost-effectively?

- A. Configure an ECS capacity provider with Fargate. Conduct load testing by using a third-party tool. Rightsize the Fargate tasks in Amazon CloudWatch.
- B. Configure an ECS capacity provider with Fargate for steady state and Fargate Spot for burst traffic. **Most Voted**
- C. Configure an ECS capacity provider with Fargate Spot for steady state and Fargate for burst traffic.
- D. Configure an ECS capacity provider with Fargate. Use AWS Compute Optimizer to rightsize the Fargate task.

Correct Answer: B

Community vote distribution

D (50%)

B (50%)

Comments

rosanna **Highly Voted** 1 year, 6 months ago

Selected Answer: D

spots might be interrupted and the questions says it has to be available 24/7, rightsizing addresses the cost optimization point but the spot is so risk so I'll go for D

upvoted 5 times

FlyingHawk 1 year, 4 months ago

If it is a stateless and fault tolerate web app, it should be okay to be interrupted.

<https://repost.aws/knowledge-center/fargate-spot-termination-notice>

upvoted 2 times

robst84 **Most Recent** 5 months ago

Selected Answer: B

B is the most cost-effective solution

upvoted 1 times

In2718 10 months ago

Selected Answer: D

AWS Compute Optimizer now works with Fargate. All your Spot instances could be deleted and your SOL.

upvoted 1 times

Clouddon 11 months ago

Selected Answer: B

✗ Why other options are suboptimal:

A. Fargate only + rightsizing via CloudWatch

Using Fargate alone for all workloads is more expensive.

While rightsizing helps, it doesn't address bursty traffic cost-effectively.

C. Fargate Spot for steady state

Fargate Spot is interruptible. It is not suitable for critical steady state workloads that need 24/7 availability.

D. Fargate only + AWS Compute Optimizer

Compute Optimizer does not yet support Fargate task-level recommendations.

Doesn't solve the cost-efficiency for burst traffic.

upvoted 1 times

7dcef09 1 year, 1 month ago

Selected Answer: B

B is a better answer

upvoted 2 times

Sarayounisaldossary 1 year, 2 months ago

Selected Answer: B

Use Fargate Spot for burst traffic — it's significantly cheaper (up to 70% savings)

ECS Capacity Providers let you define a mix strategy for both types of compute

This solution provides: High availability via regular Fargate- Cost savings (via Spot)- Scalability for traffic bursts without manual management

why other options are incorrect?

A.

Only uses standard Fargate with performance tuning tools.

Does not address cost optimization or scaling for traffic bursts.

No use of Fargate Spot, which misses out on potential savings.

C.

Using Fargate Spot for steady-state workloads is not reliable.

Spot tasks can be interrupted at any time, which is not acceptable for a 24/7 high-availability app.

D.

AWS Compute Optimizer helps with right-sizing but doesn't handle scaling or burst traffic.

upvoted 2 times

dfgdsfgfdgreg 1 year, 4 months ago

Selected Answer: B

Compute Optimizer can not optimise fargate tasks.

upvoted 2 times

FlyingHawk 1 year, 4 months ago

Selected Answer: B

<https://repost.aws/knowledge-center/fargate-spot-termination-notice>

B is correct as you can use ECS task draining to gracefully handle Spot interruptions.

upvoted 1 times

BugsyWarribwoy 1 year, 5 months ago

Selected Answer: B

B is the valid answer. Use Fargate for steady-state tasks to ensure high availability and reliability, and Fargate Spot for burst tasks to handle high traffic at a reduced cost. This combination meets the requirements most cost-effectively.

D is wrong because rightsizing alone does not optimize for burst traffic scenarios or reduce costs as effectively as incorporating Fargate Spot.

upvoted 1 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: B

A - It doesn't address the cost optimization for burst traffic. Rightsizing ensures you're not over-provisioning your baseline resources, but it doesn't provide a mechanism to dynamically scale up with cost-effective Spot instances during traffic spikes.

B - The application can scale up quickly with Fargate Spot tasks to handle the increased load, and if a Spot task is interrupted, the remaining Fargate tasks can continue to serve traffic (and the system can try to acquire new Spot capacity).

C - Fargate Spot is not recommended for steady-state workloads because it can be interrupted. This would compromise the application's 24/7 availability.

D - Like A, it fails to leverage Fargate Spot for cost savings.

Combining Fargate for steady state with Fargate Spot for burst traffic, the 24/7 HA is actually guaranteed.

upvoted 3 times

Liongeek 1 year, 5 months ago

Selected Answer: D

It's D.

Compute Optimizer supports recommendations for Fargate Tasks

<https://aws.amazon.com/about-aws/whats-new/2022/12/aws-compute-optimizer-amazon-ecs-services-aws-fargate/>

Option B and C recommend Spot, but that's not an option since the the question asks for 24x7 availability.

upvoted 3 times

thiahthura 1 year, 5 months ago

Selected Answer: D

D is right answer. Company desire is $24 * 7$. So I think spot should be.

upvoted 2 times

EllenLiu 1 year, 5 months ago

Selected Answer: D

available for $24 * 7$, spot is out as it can be interrupted any time.

upvoted 4 times

spoved 1 year, 8 months ago

Selected Answer: D

<https://aws.amazon.com/blogs/aws-cloud-financial-management/how-to-take-advantage-of-rightsizing-recommendation-preferences-in-compute-optimizer/>

be available 24 hours a day, 7 days a week

must be highly available

=> D

upvoted 4 times

FlyingHawk 1 year, 4 months ago

Rightsizing is important for cost optimization, but it does not address the need for handling burst traffic.

upvoted 1 times

dhewa 1 year, 9 months ago

Selected Answer: B

This combination leverages the cost benefits of Fargate Spot for burst traffic while ensuring steady performance with regular Fargate instances.

upvoted 2 times

pujithacg8 1 year, 10 months ago

B is the right answer

upvoted 3 times

swati1508 1 year, 10 months ago

B- for short work use spot

upvoted 3 times

A company is building an application in the AWS Cloud. The application is hosted on Amazon EC2 instances behind an Application Load Balancer (ALB). The company uses Amazon Route 53 for the DNS.

The company needs a managed solution with proactive engagement to detect against DDoS attacks.

Which solution will meet these requirements?

- A. Enable AWS Config. Configure an AWS Config managed rule that detects DDoS attacks.
- B. Enable AWS WAF on the ALB. Create an AWS WAF web ACL with rules to detect and prevent DDoS attacks. Associate the web ACL with the ALB.
- C. Store the ALB access logs in an Amazon S3 bucket. Configure Amazon GuardDuty to detect and take automated preventative actions for DDoS attacks.
- D. Subscribe to AWS Shield Advanced. Configure hosted zones in Route 53. Add ALB resources as protected resources.

Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

dhewa **Highly Voted** 1 year, 9 months ago

Selected Answer: D

Tip: DDoS = Shield, SQL injection = WAF

upvoted 6 times

LeonSauveterre **Most Recent** 1 year, 5 months ago

Selected Answer: D

For your references:

1. AWS WAF (Web Application Firewall) - SQL injections, cross-site scripting (XSS), and other common web exploits.
2. Shield Standard - common DDoS attacks; Shield Advanced - Provides advanced DDoS protection with 24/7 support, cost protection for scaling due to attacks, and more detailed attack analytics.
3. AWS Network Firewall - Blocks unauthorized access and inspect traffic for VPCs.
4. Amazon GuardDuty - Detects threats by analyzing AWS logs like CloudTrail, DNS, and VPC Flow Logs. Identifies potential malicious activities like brute force attacks and data exfiltration.
5. AWS Macie - Helps identify and protect sensitive data like PII stored in S3 buckets.

upvoted 3 times

LeonSauveterre 1 year, 5 months ago

6. AWS Detective - Analyzes and visualizes security-related data to identify root causes of potential security issues.
7. AWS Config - Monitors and evaluates configurations of AWS resources to ensure compliance with security best practices.
8. AWS Inspector - Automatically scans EC2 instances and container-based workloads for known vulnerabilities.
9. AWS CloudTrail - Tracks API calls and user activity for auditing and compliance.

upvoted 2 times

AMEJack 1 year, 6 months ago

Selected Answer: D

Shield Advanced signals a DDoS detection and starts analyzing the traffic for an attack signature. If a signature is found, it is first tested on past traffic to reduce the risk of false positive, then if it's safe to use it, a corresponding WAF rule is placed in the previously created rule group. After a certain time, when the attack is stopped, the rule is automatically removed from the rule group. When successful, this process takes several minutes.

upvoted 2 times

komorebi 1 year, 10 months ago

Selected Answer: D

Answer is D

upvoted 2 times

swati1508 1 year, 10 months ago

D for DDOS shield advance

upvoted 2 times

A company hosts a video streaming web application in a VPC. The company uses a Network Load Balancer (NLB) to handle TCP traffic for real-time data processing. There have been unauthorized attempts to access the application.

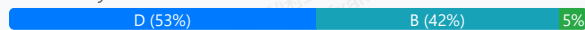
The company wants to improve application security with minimal architectural change to prevent unauthorized attempts to access the application.

Which solution will meet these requirements?

- A. Implement a series of AWS WAF rules directly on the NLB to filter out unauthorized traffic.
- B. Recreate the NLB with a security group to allow only trusted IP addresses.
- C. Deploy a second NLB in parallel with the existing NLB configured with a strict IP address allow list.
- D. Use AWS Shield Advanced to provide enhanced DDoS protection and prevent unauthorized access attempts. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

dragosky **Highly Voted** 1 year, 5 months ago

Selected Answer: D

Use AWS Shield Advanced to provide enhanced DDoS protection and prevent unauthorized access attempts.

upvoted 5 times

In2718 **Most Recent** 10 months ago

Selected Answer: A

WAF -ACL block IP addresses

upvoted 1 times

Clouddon 11 months ago

Selected Answer: D

✓ Why Option D is correct:
AWS Shield Advanced provides:

Enhanced DDoS protection for NLBs, Route 53, CloudFront, and EC2.

Proactive monitoring and automatic mitigation of unauthorized or suspicious activity.

24/7 DDoS Response Team (DRT) support.

Minimal or no architectural change required — simply enable on the NLB and related resources.

While Shield Advanced primarily focuses on DDoS protection, it also integrates with AWS Firewall Manager for centralized security management and allows defining custom rules.

upvoted 2 times

bora4motion 1 year ago

Selected Answer: D

It can't be B. You are "netflix" lets say. You don't know which IP addresses are trusted to use them in SG. It's stupid.

upvoted 2 times

Steve122 8 months, 3 weeks ago

Correct. Forget IPs and focus on higher OSI Layer protection.

upvoted 1 times

GOTJ 1 year, 4 months ago

Selected Answer: B

Several links below prove that option "B" is correct, regardless of the "NLB recreation" drawback.

It's time to kick option C out (it was my first guess, I must confess)

DDoS prevention, even though useful, is not required by the company. The company is interested SOLELY in prevent "unauthorized access" to the application. In the absence of more specific information, I came to the conclusion that the company detected some "weird ips" trying to access the application. After some investigation, I came to the conclusion that AWS Shield Advanced doesn't provide this feature for ips if they weren't previously involved in a DDoS attack.

In an on-premises environment, I would setup a firewall rule with an allow list to accomplish this requirement, which is precisely what option "B" offers.

upvoted 3 times

FlyingHawk 1 year, 4 months ago

Selected Answer: B

<https://aws.amazon.com/blogs/containers/network-load-balancers-now-support-security-groups/>

upvoted 4 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: D

A - AWS WAF works at Layer 7 (application layer) and is designed for HTTP/HTTPS traffic, so WAF works only with Application Load Balancers (ALB), API Gateway, and CloudFront.

C - Works but comes with unnecessary complexity and introduces architectural changes.

So A & C are out. I'm actually torn between B and D.

If the primary issue is unauthorized IP access and not large-scale DDoS attacks, then B might be the answer, but NLBs do not directly associate with security groups. Security groups are applied to the targets of the NLB (EC2 instances, IP addrs, or ALBs). Also, this is an architectural change.

On the other hand, if the primary concern includes DDoS attacks, option D (AWS Shield Advanced) becomes more relevant but this is so much more expensive and may still be overkill for simple IP-based access control. I'm gonna go with D if it shows up in my exam.

upvoted 4 times

Salilgen 1 year, 5 months ago

<https://docs.aws.amazon.com/elasticloadbalancing/latest/network/load-balancer-security-groups.html>

upvoted 2 times

FlyingHawk 1 year, 4 months ago

AWS NLB supports the security group now: <https://aws.amazon.com/blogs/containers/network-load-balancers-now-support-security-groups/>

upvoted 4 times

Denise123 1 year, 5 months ago

Selected Answer: B

Tricky one. About option A > AWS WAF does not support Network Load Balancers (NLBs) directly. NLBs operate at the transport layer (Layer 4), while AWS WAF is designed to work with Application Load Balancers (ALBs) at the application layer (Layer 7).....Given that the requirement is to improve application security for a Network Load Balancer with minimal architectural changes, the most appropriate solution would be Option B

upvoted 2 times

EllenLiu 1 year, 5 months ago

Selected Answer: B

shield is only for DDos attacks protection

upvoted 1 times

Anyio 1 year, 5 months ago

Selected Answer: D

The correct answer is D. Use AWS Shield Advanced to provide enhanced DDoS protection and prevent unauthorized access attempts.

Explanation:

Option D: Correct. AWS Shield Advanced is designed to protect against DDoS attacks, which can be a source of unauthorized access attempts. It provides enhanced protection features for applications behind a Network Load Balancer, offering additional security measures without requiring significant architectural changes. By leveraging AWS Shield Advanced, the company can gain comprehensive DDoS protection tailored for use with their existing NLB setup.

Option B: Incorrect. NLBs do not support security groups which are applicable to instances, not to the NLB itself. In addition, recreating the NLB to deal with unauthorized access attempts does not align with the requirement for minimal architectural change.

upvoted 3 times

ckhemani 1 year, 6 months ago

A healthcare company is developing an AWS Lambda function that publishes notifications to an encrypted Amazon Simple Notification Service (Amazon SNS) topic. The notifications contain protected health information (PHI).

The SNS topic uses AWS Key Management Service (AWS KMS) customer managed keys for encryption. The company must ensure that the application has the necessary permissions to publish messages securely to the SNS topic.

Which combination of steps will meet these requirements? (Choose three.)

A. Create a resource policy for the SNS topic that allows the Lambda function to publish messages to the topic. **Most Voted**

B. Use server-side encryption with AWS KMS keys (SSE-KMS) for the SNS topic instead of customer managed keys.

C. Create a resource policy for the encryption key that the SNS topic uses that has the necessary AWS KMS permissions. **Most Voted**

D. Specify the Lambda function's Amazon Resource Name (ARN) in the SNS topic's resource policy.

E. Associate an Amazon API Gateway HTTP API with the SNS topic to control access to the topic by using API Gateway resource policies.

F. Configure a Lambda execution role that has the necessary IAM permissions to use a customer managed key in AWS KMS. **Most Voted**

Correct Answer: ACF

Community vote distribution

ACF (71%)

ADF (29%)

Comments

Abbas_Abi_AWS **Highly Voted** 1 year, 10 months ago

Selected Answer: ACF

A C F is corrcet
upvoted 8 times

In2718 **Most Recent** 9 months, 2 weeks ago

Selected Answer: ACF

When an encrypted SNS topic (with a customer managed KMS key) is used:

Publishing permissions to the SNS topic
The Lambda function needs explicit permission to publish messages to the SNS topic.
This is done with an SNS topic resource policy.
A. Create a resource policy for the SNS topic

KMS key permissions
Since the SNS topic is encrypted with a customer managed KMS key, the Lambda execution role must be allowed to use that key (kms:Encrypt, kms:GenerateDataKey*).

This requires:
IAM permissions on the Lambda role (so it can call KMS APIs).

A KMS key policy allowing the Lambda role to use the key.
C. Resource policy for the encryption key
F. IAM execution role with KMS permissions
upvoted 1 times

GOTJ 1 year, 4 months ago

Selected Answer: ADF

Does a SNS topic need to decrypt an encrypted message in order to deliver it to its subscribers? If so, option "C" should be in. If not, it should be out. And I think is out, unless SNS topic needs to apply certain logic to the message prior to subscription deliver (filters).

This question requires a CSE-KMS approach, which force the Lambda function to encrypt the message before publishing, so option "F" is a must (I think everybody here is on the same page with this one).

Also, option "A" should be obvious by now. And even though option "D" is redundant, it's technically correct, so it's a better choice.

upvoted 2 times

FlyingHawk 1 year, 4 months ago

Selected Answer: ACF

For C - <https://docs.aws.amazon.com/sns/latest/dg/sns-enable-encryption-for-topic.html>

Permissions for custom KMS keys – If using a custom KMS key, include the following in the key policy to allow Amazon SNS to encrypt and decrypt messages:

upvoted 1 times

FlyingHawk 1 year, 4 months ago

For A: <https://repost.aws/knowledge-center/sns-topic-lambda> and <https://repost.aws/knowledge-center/sns-topics-iam-errors-subscriber>

For F: <https://repost.aws/knowledge-center/lambda-kmsaccessdeniedexception-errors>

upvoted 1 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: ACF

A - This is a must-do.

B - As question stated, we have no reason to replace the keys.

C - This ensures the function can encrypt messages when publishing.

D - While specifying the Lambda function's ARN in the SNS topic's resource policy is part of the solution (it's how you grant the "sns:Publish" permission), it's not sufficient on its own. You also need the KMS key policy and the Lambda execution role permissions. This option is a part of option A, not a standalone answer.

E - The Lambda function interacts directly with the SNS topic, and API Gateway is not relevant.

F - Yes, so that this role will have policies that allow the "kms:Encrypt", "kms:GenerateDataKey," and "kms:Decrypt" actions on the specific KMS key used to encrypt the topic.

upvoted 3 times

EllenLiu 1 year, 5 months ago

Selected Answer: ACF

I don't understand why lambda needs KMS to encrypt message.

In this scenario, the Lambda function does not need direct permissions to use the KMS key because the encryption and decryption are fully managed by Amazon SNS as part of the SSE-KMS feature.

SNS is the only service lambda will talk to, The Lambda interacts with SNS using HTTPS and does not directly deal with the encrypted data or the KMS key. so Lambda only needs the permission to publish messages to the SNS topic.

I would like to choose F only if #F answer can be updated as below:

F. Configure a Lambda execution role that has the necessary IAM permissions to publish to the SNS topic.

A: Create a resource policy for the SNS topic => grants lambda the ability to publish messages to the topic

C: The KMS key resource policy must allow SNS to use the key for encryption and decryption.

F: Configure the Lambda Execution Role => SNS permissions to publish to the topic

upvoted 2 times

JA2018 1 year, 6 months ago

Selected Answer: ADF

#A: This is essential because the resource policy on the SNS topic will define which entities (like the Lambda function) are allowed to publish messages to it.

#D: By specifying the Lambda function's ARN in the SNS topic policy, you clearly grant access only to that specific Lambda function.

#F: Since the SNS topic uses a customer-managed KMS key, the Lambda execution role must have the necessary permissions to use that key for encryption/decryption when publishing messages.

upvoted 2 times

JA2018 1 year, 6 months ago

Why the other options are not correct:

#B: The question already states that the SNS topic is using customer-managed KMS keys, so there's no need to switch to server-side encryption with AWS managed keys.

#C: While technically you could create a resource policy on the encryption key itself, it's not the most secure approach as it would grant access to the key itself, not just the ability to use it for SNS encryption.

#E: Introducing an API Gateway layer is unnecessary complexity for this scenario, as you can directly control access to the SNS topic using the Lambda execution role and its permissions.

upvoted 1 times

JA2018 1 year, 6 months ago

Key takeaway:

- To ensure the Lambda function can securely publish messages to an encrypted SNS topic, you need to properly configure the SNS topic resource policy to allow the Lambda function access and make sure the Lambda execution role has the necessary KMS permissions to use the customer-managed encryption key.

upvoted 1 times

FlyingHawk 1 year, 4 months ago

C is correct since the SNS topic uses a customer managed key for encryption. The key's resource policy must allow the SNS service to use the key for encryption and decryption.

upvoted 1 times

FlyingHawk 1 year, 4 months ago

D is redundant because Option A already covers the need to allow the Lambda function to publish to the SNS topic. The ARN of the Lambda function would be included in the resource policy created in Option A.

upvoted 1 times

elmyth 1 year, 7 months ago

Selected Answer: ACF

D is correct too and C is not clear, but seems like it is about KMS policy and adding permissions for sns service which has to be added in case of CMK

upvoted 4 times

agbor_tambe 1 year, 8 months ago

Selected Answer: ADF

my answer

upvoted 1 times

Sergantus 1 year, 6 months ago

D is like a part of A, so it makes no sense to pick both, it should be A C F.

upvoted 1 times

progounick 1 year, 9 months ago

Selected Answer: ACF

ChatGPT agrees with me

upvoted 3 times

komorebi 1 year, 10 months ago

Selected Answer: ADF

Answer is ADF

upvoted 4 times

A company has an employee web portal. Employees log in to the portal to view payroll details. The company is developing a new system to give employees the ability to upload scanned documents for reimbursement. The company runs a program to extract text-based data from the documents and attach the extracted information to each employee's reimbursement IDs for processing.

The employee web portal requires 100% uptime. The document extract program runs infrequently throughout the day on an on-demand basis. The company wants to build a scalable and cost-effective new system that will require minimal changes to the existing web portal. The company does not want to make any code changes.

Which solution will meet these requirements with the LEAST implementation effort?

A. Run Amazon EC2 On-Demand Instances in an Auto Scaling group for the web portal. Use an AWS Lambda function to run the document extract program. Invoke the Lambda function when an employee uploads a new reimbursement document.

Most Voted

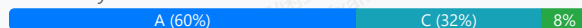
B. Run Amazon EC2 Spot Instances in an Auto Scaling group for the web portal. Run the document extract program on EC2 Spot Instances. Start document extract program instances when an employee uploads a new reimbursement document.

C. Purchase a Savings Plan to run the web portal and the document extract program. Run the web portal and the document extract program in an Auto Scaling group.

D. Create an Amazon S3 bucket to host the web portal. Use Amazon API Gateway and an AWS Lambda function for the existing functionalities. Use the Lambda function to run the document extract program. Invoke the Lambda function when the API that is associated with a new document upload is called.

Correct Answer: A

Community vote distribution



Comments

hpirnaj **Highly Voted** 1 year, 5 months ago

Selected Answer: C

i hate these types of questions . there is no perfect answer

A: on demand "not cost-effective" , also , how do you want to invoke Lambda . there is no API call

B: spot instance ("web portal requires 100% uptime")

D: s3 is for static website . it can work but payroll needs to update every 2 weeks and require extensive code change

C in my opinion is the only correct answer. it is scalable "ASG", no code change and cost effective "saving plan"

upvoted 6 times

Clouddon **Most Recent** 11 months ago

Selected Answer: A

It is not C. Purchase a Savings Plan to run both workloads

Savings Plans reduce cost, but:

You're still running document extraction code on EC2, which is less cost-effective and less scalable than Lambda.

Does not align with event-driven or on-demand execution needs.

Increases operational overhead.

upvoted 2 times

7dcef09 1 year, 1 month ago

Selected Answer: C

is this supposed to be a confusing question? C is scalable and most cost effective.

upvoted 1 times

Salilgen 1 year, 5 months ago

Selected Answer: C

Spot Instances are not suitable for workloads requiring 100% uptime then B is out.
D is the best solution but it requires some implementation effort.
Savings plan (that include Lambda too...) is cheaper than on-demand (option A).
Moreover, C doesn't require any code change or other AWS services.

upvoted 1 times

tch 1 year, 3 months ago

which saving plan to purchase ? Compute Savings Plans, EC2 Instance Savings Plans, and Amazon SageMaker Savings Plans???

upvoted 1 times

ckhemani 1 year, 6 months ago

Selected Answer: A

Answer: A

This solution offers the most scalable and cost-effective approach with minimal changes to the existing web portal and no code modifications.

Why Not Other Options?:

Option B (Spot Instances): Spot Instances are not suitable for workloads requiring 100% uptime, as they can be terminated by AWS with short notice.

Option C (Savings Plan): A Savings Plan could reduce costs but does not address the requirement for running the document extraction program efficiently or without code changes.

Option D (S3 with API Gateway and Lambda): This would require significant changes to the existing web portal setup, including moving the portal to S3 and reconfiguring its architecture, which contradicts the requirement of minimal implementation effort and no code change.

upvoted 4 times

XXXXXINN 1 year, 8 months ago

D makes more sense if the company asks to redesign the whole thing to achieve better operational management, performance, cost effective, etc. However, it requires us to provide solution with MINIMUM change... thus A it is I guess.

upvoted 3 times

JoeTromundo 1 year, 8 months ago

Selected Answer: A

"The company does not want to make any code changes."

Option D requires a complete re-architecture of the web portal to be hosted on Amazon S3 and API Gateway, which involves significant changes to the existing system. This does not align with the requirement of minimal changes to the current setup.

upvoted 4 times

rpmaws 1 year, 8 months ago

Selected Answer: A

because they don't want any change in code so A is correct.

upvoted 3 times

progounick 1 year, 9 months ago

Selected Answer: A

A ChatGPT agrees with me

upvoted 2 times

progounick 1 year, 9 months ago

Selected Answer: D

I think D is better choice. Even though A makes sense too, D seems the correct one

upvoted 1 times

RealPro111 1 year, 9 months ago

Selected Answer: D

Least effort, webportal is simply an interface: D

upvoted 1 times

744fdad 1 year, 10 months ago

i think D

upvoted 2 times

komorebi 1 year, 10 months ago

Answer is A

upvoted 1 times

swati1508 1 year, 10 months ago

I thinks it's D

upvoted 3 times

A media company has a multi-account AWS environment in the us-east-1 Region. The company has an Amazon Simple Notification Service (Amazon SNS) topic in a production account that publishes performance metrics. The company has an AWS Lambda function in an administrator account to process and analyze log data.

The Lambda function that is in the administrator account must be invoked by messages from the SNS topic that is in the production account when significant metrics are reported.

Which combination of steps will meet these requirements? (Choose two.)

- A. Create an IAM resource policy for the Lambda function that allows Amazon SNS to invoke the function. **Most Voted**
- B. Implement an Amazon Simple Queue Service (Amazon SQS) queue in the administrator account to buffer messages from the SNS topic that is in the production account. Configure the SQS queue to invoke the Lambda function.
- C. Create an IAM policy for the SNS topic that allows the Lambda function to subscribe to the topic. **Most Voted**
- D. Use an Amazon EventBridge rule in the production account to capture the SNS topic notifications. Configure the EventBridge rule to forward notifications to the Lambda function that is in the administrator account.
- E. Store performance metrics in an Amazon S3 bucket in the production account. Use Amazon Athena to analyze the metrics from the administrator account.

Correct Answer: AC

Community vote distribution

AC (48%)

AB (40%)

8%

Comments

Anyio **Highly Voted** 1 year, 5 months ago

Selected Answer: AB

The correct answers are A,B.

Explanation:

Option A: Correct. Creating an IAM resource policy for the Lambda function that allows Amazon SNS to invoke the function is necessary for SNS to have permission to trigger Lambda. This policy ensures that the Lambda function can be invoked by a service principal from the SNS service.

Option B: Correct. Using an Amazon Simple Queue Service (Amazon SQS) queue as an intermediary buffer allows for decoupling the SNS topic from the Lambda function, providing more reliability and handling burst traffic effectively. In this setup, the SNS topic can publish to the SQS queue, and the queue can then trigger the Lambda function to process the messages.

upvoted 5 times

Anyio 1 year, 5 months ago

Sorry this is the wrong answer (or second best answer) :). The Correct answer is A,C.

Using the Amazon SNS console, add a cross-account AWS Lambda subscription to an Amazon SNS topic.

the Lambda function resource policy allows SNS to invoke the function.

the SNS topic access policy allows Lambda to subscribe to the topic.

Note: The SNS topic resides in account A and the Lambda function resides in account B.

upvoted 4 times

siheom **Highly Voted** 1 year, 10 months ago

Selected Answer: AB

VOTE A,B

upvoted 5 times

Anyio **Most Recent** 1 year, 5 months ago

Selected Answer: AC

The Correct answer is A,C.

Using the Amazon SNS console, add a cross-account AWS Lambda subscription to an Amazon SNS topic.

the Lambda function resource policy allows SNS to invoke the function.

the SNS topic access policy allows Lambda to subscribe to the topic.

Note: The SNS topic resides in account A and the Lambda function resides in account B.

upvoted 2 times

EllenLiu 1 year, 5 months ago

Selected Answer: AC

A: resource-policy for Lambda: should grant SNS to access lambda permission

C: resource-policy for SNS: should specify who can subscribe SNS topic

upvoted 3 times

JA2018 1 year, 6 months ago

Selected Answer: AD

#A: This is the most direct way to allow the SNS topic in the production account to trigger the Lambda function in the administrator account. By creating an IAM policy on the Lambda function that grants SNS permission to invoke it, you establish the necessary access control.

#D: Using an EventBridge rule in the production account allows you to filter and route the SNS notifications specifically to the Lambda function in the administrator account, providing greater control and flexibility over the event delivery .

upvoted 1 times

agbor_tambe 1 year, 8 months ago

Selected Answer: AC

most reasonable

upvoted 1 times

moondooo 1 year, 8 months ago

Selected Answer: AC

Probably A and C

<https://repost.aws/knowledge-center/sns-with-crossaccount-lambda-subscription>

upvoted 3 times

progounick 1 year, 9 months ago

Selected Answer: AC

A and C seem to be the best answer

upvoted 1 times

dhewa 1 year, 9 months ago

Selected Answer: AC

No need to complicate stuff, AWS services already exist only permissions are missing. A&C will set up the necessary permissions and subscriptions for cross-account invocation of the Lambda function by the SNS topic.

upvoted 2 times

523db89 1 year, 9 months ago

A,C correct - While using SQS could be a solution for buffering messages, it introduces additional complexity

upvoted 1 times

jamesukae 1 year, 9 months ago

Selected Answer: BE

For me AB is contradict , why we invoke lambda function by both SNS and SQS?

I think BE is correct answer because question also need solution to analyze data.

upvoted 2 times

nebajp 1 year, 10 months ago

correct answer is AD

upvoted 4 times

JA2018 1 year, 6 months ago

- #A: This is the most direct way to allow the SNS topic in the production account to trigger the Lambda function in the administrator account. By creating an IAM policy on the Lambda function that grants SNS permission to invoke it, you establish the necessary access control [A, D].

- #D: Using an EventBridge rule in the production account allows you to filter and route the SNS notifications specifically to the Lambda function in the administrator account, providing greater control and flexibility over the event delivery [D].

upvoted 2 times

GOTJ 1 year, 4 months ago

I like your reasoning for option "D". However, with this setup, EventBridge should be the service that invoke the Lambda function:

Perf Metrics --> SNS --> EventBridge (for filtering and routing "the significant metrics") --> Lambda

Since you've detached Lambda from SNS adding a service in between, SNS should no longer invoke the Lambda Function and option "A" would be wrong, isn't it?

upvoted 1 times

A company is migrating an application from an on-premises location to Amazon Elastic Kubernetes Service (Amazon EKS). The company must use a custom subnet for pods that are in the company's VPC to comply with requirements. The company also needs to ensure that the pods can communicate securely within the pods' VPC.

Which solution will meet these requirements?

- A. Configure AWS Transit Gateway to directly manage custom subnet configurations for the pods in Amazon EKS.
- B. Create an AWS Direct Connect connection from the company's on-premises IP address ranges to the EKS pods.
- C. Use the Amazon VPC CNI plugin for Kubernetes. Define custom subnets in the VPC cluster for the pods to use. **Most Voted**
- D. Implement a Kubernetes network policy that has pod anti-affinity rules to restrict pod placement to specific nodes that are within custom subnets.

Correct Answer: C

Community vote distribution

C (100%)

Comments

officedepotadmin **Highly Voted** 1 year, 10 months ago

Selected Answer: C

The Amazon VPC Container Network Interface (CNI) plugin is the default network plugin for Amazon EKS. It allows Kubernetes pods to receive IP addresses from a VPC's subnet and enables pods to communicate securely within the VPC as if they were native VPC resources.

upvoted 8 times

Sarayounisaldossary **Most Recent** 1 year, 2 months ago

Selected Answer: C

The Amazon VPC CNI plugin is a Kubernetes networking plugin for EKS.

where it provides: a unique VPC IP address for each pod and native integration of Kubernetes networking into AWS VPC

- Pods can communicate like any EC2 instance on the same subnet.

upvoted 1 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: C

A - AWS Transit Gateway is used to manage connectivity between multiple VPCs, on-premises networks, or other AWS services. It does not manage pod-level subnet configurations in Amazon EKS.

B - This can't address the specific requirement to configure custom subnets for pods within the VPC.

C - This is the recommended approach.

D - Pod anti-affinity rules are used for scheduling constraints, not for subnet allocation or secure communication within the VPC.

upvoted 1 times

moondooo 1 year, 8 months ago

Selected Answer: C

Probably C

<https://repost.aws/knowledge-center/eks-custom-subnet-for-pod>

upvoted 3 times

dhewa 1 year, 9 months ago

Selected Answer: C

C all the way.

upvoted 2 times

A company hosts an ecommerce application that stores all data in a single Amazon RDS for MySQL DB instance that is fully managed by AWS. The company needs to mitigate the risk of a single point of failure.

Which solution will meet these requirements with the LEAST implementation effort?

A. Modify the RDS DB instance to use a Multi-AZ deployment. Apply the changes during the next maintenance window.

Most Voted

B. Migrate the current database to a new Amazon DynamoDB Multi-AZ deployment. Use AWS Database Migration Service (AWS DMS) with a heterogeneous migration strategy to migrate the current RDS DB instance to DynamoDB tables.

C. Create a new RDS DB instance in a Multi-AZ deployment. Manually restore the data from the existing RDS DB instance from the most recent snapshot.

D. Configure the DB instance in an Amazon EC2 Auto Scaling group with a minimum group size of three. Use Amazon Route 53 simple routing to distribute requests to all DB instances.

Correct Answer: A

Community vote distribution

A (100%)

Comments

example_ **Highly Voted** 1 year, 10 months ago

Selected Answer: A

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_UpgradeDBInstance.Maintenance.html

upvoted 5 times

2f2127f **Most Recent** 1 year ago

Selected Answer: A

It's because I said so

upvoted 1 times

Changwha 1 year, 6 months ago

A: This solution provides automated failover and high availability with minimal implementation effort.

upvoted 2 times

komorebi 1 year, 10 months ago

Selected Answer: A

Answer is A

upvoted 2 times

A company has multiple Microsoft Windows SMB file servers and Linux NFS file servers for file sharing in an on-premises environment. As part of the company's AWS migration plan, the company wants to consolidate the file servers in the AWS Cloud.

The company needs a managed AWS storage service that supports both NFS and SMB access. The solution must be able to share between protocols. The solution must have redundancy at the Availability Zone level.

Which solution will meet these requirements?

- A. Use Amazon FSx for NetApp ONTAP for storage. Configure multi-protocol access. **Most Voted**
- B. Create two Amazon EC2 instances. Use one EC2 instance for Windows SMB file server access and one EC2 instance for Linux NFS file server access.
- C. Use Amazon FSx for NetApp ONTAP for SMB access. Use Amazon FSx for Lustre for NFS access.
- D. Use Amazon S3 storage. Access Amazon S3 through an Amazon S3 File Gateway.

Correct Answer: A

Community vote distribution

A (100%)

Comments

FlyingHawk 1 year, 4 months ago

Selected Answer: A

<https://aws.amazon.com/fsx/netapp-ontap/features/>

Key Features of Amazon FSx for NetApp ONTAP:

Multi-Protocol Support: Simultaneously supports SMB, NFS, and iSCSI protocols.

Multi-AZ Redundancy: Provides high availability and automatic failover across Availability Zones.

Scalability: Scales storage capacity and performance independently.

Data Management: Includes advanced features like snapshots, cloning, and data deduplication.

Integration: Seamlessly integrates with AWS services and on-premises environments.

upvoted 2 times

Jeyaluxshan 1 year, 9 months ago

Amazon FSx for NetApp ONTAP uses Single-AZ and Multi-AZ deployment types. You can choose from four options: Single-AZ 1, Single-AZ 2, Multi-AZ 1, and Multi-AZ 2

upvoted 2 times

ccceb01 1 year, 9 months ago

Selected Answer: A

<https://aws.amazon.com/fsx/netapp-ontap/>

upvoted 3 times

komorebi 1 year, 10 months ago

Selected Answer: A

Answer is A

upvoted 2 times

example_ 1 year, 10 months ago

Selected Answer: A

<https://aws.amazon.com/blogs/storage/enabling-multiprotocol-workloads-with-amazon-fsx-for-netapp-ontap/>

upvoted 2 times

A software company needs to upgrade a critical web application. The application currently runs on a single Amazon EC2 instance that the company hosts in a public subnet. The EC2 instance runs a MySQL database. The application's DNS records are published in an Amazon Route 53 zone.

A solutions architect must reconfigure the application to be scalable and highly available. The solutions architect must also reduce MySQL read latency.

Which combination of solutions will meet these requirements? (Choose two.)

A. Launch a second EC2 instance in a second AWS Region. Use a Route 53 failover routing policy to redirect the traffic to the second EC2 instance.

B. Create and configure an Auto Scaling group to launch private EC2 instances in multiple Availability Zones. Add the instances to a target group behind a new Application Load Balancer. **Most Voted**

C. Migrate the database to an Amazon Aurora MySQL cluster. Create the primary DB instance and reader DB instance in separate Availability Zones. **Most Voted**

D. Create and configure an Auto Scaling group to launch private EC2 instances in multiple AWS Regions. Add the instances to a target group behind a new Application Load Balancer.

E. Migrate the database to an Amazon Aurora MySQL cluster with cross-Region read replicas.

Correct Answer: BC

Community vote distribution

BC (95%)

5%

Comments

example_ **Highly Voted** 1 year, 10 months ago

Selected Answer: BC

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Concepts.AuroraHighAvailability.html>

upvoted 6 times

Sarayounisaldossary **Most Recent** 1 year, 2 months ago

Selected Answer: BC

B. Auto Scaling + Load Balancer across AZs

Launch multiple EC2s across multiple AZs

Add to Application Load Balancer

Gives:

HA and Elastic scalability

C. Migrate to Aurora MySQL with a reader in a separate AZ

Aurora is high performance + replication built-in

Reader in another AZ → lower read latency

Built for fault tolerance

NOT A,D,E

A. EC2 in another Region + Route 53 Failover: Disaster recovery, not real-time scaling or latency reduction.

D. Auto Scaling across Regions: Not supported.

E. Aurora with cross-Region replica: Overkill and not needed here.

upvoted 1 times

Steve122 8 months, 3 weeks ago

If you see the word 'Region' among the answers -> DR! (No DR req. mentioned in the Q.)

upvoted 1 times

progounick 1 year, 9 months ago

Selected Answer: BC

It is obvious

upvoted 2 times

dhewa 1 year, 9 months ago

Selected Answer: BC

These simple options will help you achieve a robust, scalable, and highly available architecture for your web application

upvoted 3 times

nebajp 1 year, 10 months ago

Selected Answer: BC

correct Answer is B& C as it talks about high availability and scalability and rest of the options are for Disaster Recovery.

Right answer is B & C.

upvoted 3 times

nebajp 1 year, 10 months ago

correct Answer is B& C as it talks about high availability and scalability and rest of the options are for Disaster Recovery.

Right answer is B & C.

upvoted 2 times

komorebi 1 year, 10 months ago

Selected Answer: DE

Answer is DE

upvoted 1 times

JunsK1e 1 year, 10 months ago

Selected Answer: BC

B is correct and C, D is incorrect since the auto scaling is working within Region not Multi region. E incorrect also because the question is asking a high availability C is the best answer that E.

upvoted 4 times

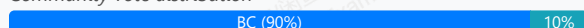
A company runs thousands of AWS Lambda functions. The company needs a solution to securely store sensitive information that all the Lambda functions use. The solution must also manage the automatic rotation of the sensitive information.

Which combination of steps will meet these requirements with the LEAST operational overhead? (Choose two.)

- A. Create HTTP security headers by using Lambda@Edge to retrieve and create sensitive information
- B. Create a Lambda layer that retrieves sensitive information **Most Voted**
- C. Store sensitive information in AWS Secrets Manager **Most Voted**
- D. Store sensitive information in AWS Systems Manager Parameter Store
- E. Create a Lambda consumer with dedicated throughput to retrieve sensitive information and create environmental variables

Correct Answer: BC

Community vote distribution



Comments

[Removed] **Highly Voted** 1 year, 9 months ago

Selected Answer: BC

C. Store sensitive information in AWS Secrets Manager.

AWS Secrets Manager securely stores sensitive information and provides automatic rotation of secrets, reducing the need for manual management.
B. Create a Lambda layer that retrieves sensitive information.

Using a Lambda layer allows multiple Lambda functions to access the sensitive information stored in Secrets Manager without needing to duplicate retrieval logic in each function. This approach centralizes the retrieval process and reduces operational complexity.

upvoted 9 times

awisaw 1 year, 9 months ago

chatgpt answer is now C and D

upvoted 1 times

sOI852POL 1 year, 7 months ago

AWS public documentation and other professional forums instead, kindly!

upvoted 3 times

pujithacg8 **Highly Voted** 1 year, 10 months ago

D doesn't provide automatic rotation

Answer will be B and C

upvoted 8 times

c3c7e62 **Most Recent** 1 year, 1 month ago

Selected Answer: BC

A - not relate, as question does not mention the use of HTTP.

D - technical works, but does not support automatic rotation.

E - not secure to use the environmental variables.

Question and Concern - for the exam, have to pick option B & C. However, really wonder if the approach is practical in real world for secure solution.

1) Secrets Manager encrypt the sensitive info. Lambda Layer saves those sensitive information into /opt layer (in zip format etc) and shares with all other. However, I cannot see there is description the information is encrypted for certain format.

2) After Secret Manager rotation, how can those saved content in /opt folder be refreshed ?

Open for discussion. Thanks.

upvoted 1 times

c3c7e62 1 year, 1 month ago

<https://docs.aws.amazon.com/lambda/latest/dg/chapter-layers.html>

upvoted 1 times

Sarayounisaldossary 1 year, 2 months ago

Selected Answer: BC

A. Lambda@Edge: Not relevant for secrets; Used for CDN/CloudFront.

D. Parameter Store: Doesn't support auto-rotation as easily as Secrets Manager.

E. Env Variables: Not secure and needs manual management.

upvoted 1 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: BC

A - Lambda@Edge is primarily used for managing HTTP request/response behavior in a CDN environment with Amazon CloudFront. It is not designed for securely storing or rotating sensitive information.

B - By creating a layer that securely retrieves sensitive information (for example, from AWS Secrets Manager), we will reduce duplication and operational overhead. The layer can also handle the logic for retrieving updated information if needed.

C - Automatic rotation is managed.

D - Automatic rotation is NOT managed.

E - Setting environment variables dynamically at runtime is not an efficient or scalable practice for "thousands of Lambda functions". Too much operational overhead.

upvoted 5 times

agbor_tambe 1 year, 8 months ago

Selected Answer: BC

BC is correct

upvoted 2 times

progounick 1 year, 9 months ago

Selected Answer: BC

B,C ChatGPT agrees with me

upvoted 1 times

komorebi 1 year, 10 months ago

Selected Answer: CD

Answer is CD

upvoted 1 times

example_ 1 year, 10 months ago

Selected Answer: CD

<https://docs.aws.amazon.com/systems-manager/latest/userguide/ps-integration-lambda-extensions.html>

<https://www.trustwave.com/en-us/resources/blogs/spiderlabs-blog/using-aws-secrets-manager-and-lambda-function-to-store-rotate-and-secure-keys/>

upvoted 1 times

A company has an internal application that runs on Amazon EC2 instances in an Auto Scaling group. The EC2 instances are compute optimized and use Amazon Elastic Block Store (Amazon EBS) volumes.

The company wants to identify cost optimizations across the EC2 instances, the Auto Scaling group, and the EBS volumes.

Which solution will meet these requirements with the MOST operational efficiency?

- A. Create a new AWS Cost and Usage Report. Search the report for cost recommendations for the EC2 instances the Auto Scaling group, and the EBS volumes.
- B. Create new Amazon CloudWatch billing alerts. Check the alert statuses for cost recommendations for the EC2 instances, the Auto Scaling group, and the EBS volumes.
- C. Configure AWS Compute Optimizer for cost recommendations for the EC2 instances, the Auto Scaling group and the EBS volumes. **Most Voted**
- D. Configure AWS Compute Optimizer for cost recommendations for the EC2 instances. Create a new AWS Cost and Usage Report. Search the report for cost recommendations for the Auto Scaling group and the EBS volumes.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Sarayounisaldossary 1 year, 2 months ago

Selected Answer: C

the question here focuses on (COST Optimization)

- AWS Cost and Usage Report is the most detailed report that shows your AWS usage and charges.

- cloudwatch is only for setting alerts based on setting policies-

- would go right-away with AWS compute Optimizer because this service gives recommendations to optimize your compute resources (like EC2, Lambda, etc.) for better performance and lower cost.

which is C and D

BUT when reading D choice there's nothing wrong with AWS Cost and Usage report but why would i make it more complex to get the cost optimization? AWS compute Optimizer is more than enough which is C option

upvoted 1 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: C

Go with Option C for the most efficient and straightforward solution for sure. Just one more thing: AWS Compute Optimizer doesn't provide cost data directly (such as detailed dollar amounts), so supplementing it with AWS Cost Explorer or Trusted Advisor is a good practice for deeper cost analysis.

upvoted 2 times

komorebi 1 year, 10 months ago

Selected Answer: C

Answer is C

upvoted 2 times

example_ 1 year, 10 months ago

Selected Answer: C

<https://aws.amazon.com/compute-optimizer/>

<https://docs.aws.amazon.com/compute-optimizer/latest/ug/what-is-compute-optimizer.html>

upvoted 4 times

A company is running a media store across multiple Amazon EC2 instances distributed across multiple Availability Zones in a single VPC. The company wants a high-performing solution to share data between all the EC2 instances, and prefers to keep the data within the VPC only.

What should a solutions architect recommend?

- A. Create an Amazon S3 bucket and call the service APIs from each instance's application
- B. Create an Amazon S3 bucket and configure all instances to access it as a mounted volume
- C. Configure an Amazon Elastic Block Store (Amazon EBS) volume and mount it across all instances
- D. Configure an Amazon Elastic File System (Amazon EFS) file system and mount it across all instances **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

Sarayounisaldossary 1 year, 2 months ago

Selected Answer: D

Why is EFS the correct answer?

Amazon EFS is a fully managed, shared file system.

It can be mounted on multiple EC2 instances at the same time.

It works across multiple Availability Zones.

High performance, and keeps all data within the VPC.

Perfect fit for the use case: "Share data between EC2 instances."

Other options:

S3 is object storage, not a shared file system.

EBS can be attached to only one instance at a time.

Mounting S3 isn't officially supported or reliable.

upvoted 1 times

aragon_saa 1 year, 10 months ago

Selected Answer: D

Answer is D

upvoted 4 times

muhammadahmer36 1 year, 10 months ago

Selected Answer: D

D is the right answer

upvoted 3 times

A company uses an Amazon RDS for MySQL instance. To prepare for end-of-year processing, the company added a read replica to accommodate extra read-only queries from the company's reporting tool. The read replica CPU usage was 60% and the primary instance CPU usage was 60%.

After end-of-year activities are complete, the read replica has a constant 25% CPU usage. The primary instance still has a constant 60% CPU usage. The company wants to rightsize the database and still provide enough performance for future growth.

Which solution will meet these requirements?

- A. Delete the read replica Do not make changes to the primary instance
- B. Resize the read replica to a smaller instance size Do not make changes to the primary instance **Most Voted**
- C. Resize the read replica to a larger instance size Resize the primary instance to a smaller instance size
- D. Delete the read replica Resize the primary instance to a larger instance

Correct Answer: B

Community vote distribution

B (56%)

D (44%)

Comments

Oghare **Highly Voted** 1 year, 8 months ago

Answer is B

Since the read replica is now underutilized with only 25% CPU usage, it can be resized to a smaller instance to save costs while still handling the reduced read queries. No changes to the primary instance are needed, as it is consistently running at 60% CPU usage, which is manageable, and the read replica will still offload read queries in the future.

upvoted 7 times

In2718 **Most Recent** 10 months ago

Selected Answer: B

A follow-up and you are wasting performance and money buy using option D. this looks good on paper but reality is it not. Terrible question because they try to bait you into a bad response when in reality you would never do such.

upvoted 2 times

In2718 10 months ago

Selected Answer: B

and still provide enough performance for future growth

upvoted 1 times

7dcef09 1 year, 1 month ago

Selected Answer: D

Delete the read replica and resize the primary instance seems to be the right choice for this question

upvoted 1 times

Sarayounisaldossary 1 year, 2 months ago

Selected Answer: D

You should remove an underused resource (read replica).

Use that budget to increase primary capacity, solving the CPU load.

Even though Option B is "safe" and cost-conscious, it ignores the performance needs of the primary, which is where the real issue lies. Option D is more aligned with AWS best practices and the question's goal of performance + future growth.

upvoted 2 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: B

Not A or D because they risk future scalability issues during traffic spikes. Moreover, when the end of next year comes, you have to bring back the deleted instance and configure it all over again. As a solutions architect, I wouldn't recommend things like that (personally).

upvoted 3 times

Ghoneam 1 year, 5 months ago

Selected Answer: D

D. Delete the read replica Resize the primary instance to a larger instance because

no need for read replica after year end

still provide enough performance for future growth.

upvoted 4 times

Abdullah2004 1 year, 9 months ago

No clear explanation here and everyone agreed with default answer

upvoted 1 times

aragon_saa 1 year, 10 months ago

Selected Answer: B

Answer is B

upvoted 1 times

muhammadahmer36 1 year, 10 months ago

Selected Answer: B

Resize the read replica to a smaller instance size Do not make changes to the primary instance

upvoted 2 times

A company is migrating its databases to Amazon RDS for PostgreSQL. The company is migrating its applications to Amazon EC2 instances. The company wants to optimize costs for long-running workloads.

Which solution will meet this requirement MOST cost-effectively?

- A. Use On-Demand Instances for the Amazon RDS for PostgreSQL workloads. Purchase a 1 year Compute Savings Plan with the No Upfront option for the EC2 instances.
- B. Purchase Reserved Instances for a 1 year term with the No Upfront option for the Amazon RDS for PostgreSQL workloads. Purchase a 1 year EC2 Instance Savings Plan with the No Upfront option for the EC2 instances.
- C. Purchase Reserved Instances for a 1 year term with the Partial Upfront option for the Amazon RDS for PostgreSQL workloads. Purchase a 1 year EC2 Instance Savings Plan with the Partial Upfront option for the EC2 instances.
- D. Purchase Reserved Instances for a 3 year term with the All Upfront option for the Amazon RDS for PostgreSQL workloads. Purchase a 3 year EC2 Instance Savings Plan with the All Upfront option for the EC2 instances. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

LeonSauveterre **Highly Voted** 1 year, 5 months ago

Selected Answer: D

What you should know:

1. No Upfront - Pay nothing at the time of purchase. Payments are spread out monthly over the term (1 or 3 years). Lowest savings with high flexibility.
2. Partial Upfront - Pay a portion of the cost upfront, with reduced monthly payments over the term. Moderate savings with moderate flexibility.
3. All Upfront - Pay the full cost upfront at the time of purchase. No additional payments required during the term. Highest savings with the least flexibility.

Merely by the question, option D would be the answer because it's best for companies with a stable workload, long-term commitment, and sufficient budget for upfront payments.

Just to add something, generally option C would be the best in real world cases because it strikes a good balance between savings and upfront investment, and suitable for mid-term commitments with moderate budget flexibility.

upvoted 6 times

LeonSauveterre 1 year, 5 months ago

uitable => suitable sorry

upvoted 1 times

744fdad **Most Recent** 1 year, 10 months ago

the longer the reservation the more the savings too for reserved

upvoted 3 times

muhammadahmer36 1 year, 10 months ago

Selected Answer: D

D is right

upvoted 1 times

nebajp 1 year, 10 months ago

Selected Answer: D

Correct Answer - D,

All upfront is cheaper than partial and no upfront.

upvoted 1 times

swati1508 1 year, 10 months ago

D is correct

upvoted 2 times

JunsK1e 1 year, 10 months ago

Selected Answer: D

letter D is the correct, It gives you big discount if you purchase with all upfront

upvoted 3 times

A company is using an Amazon Elastic Kubernetes Service (Amazon EKS) cluster. The company must ensure that Kubernetes service accounts in the EKS cluster have secure and granular access to specific AWS resources by using IAM roles for service accounts (IRSA).

Which combination of solutions will meet these requirements? (Choose two.)

- A. Create an IAM policy that defines the required permissions. Attach the policy directly to the IAM role of the EKS nodes.
- B. Implement network policies within the EKS cluster to prevent Kubernetes service accounts from accessing specific AWS services.
- C. Modify the EKS cluster's IAM role to include permissions for each Kubernetes service account. Ensure a one-to-one mapping between IAM roles and Kubernetes roles.
- D. Define an IAM role that includes the necessary permissions. Annotate the Kubernetes service accounts with the Amazon ResourceName (ARN) of the IAM role. **Most Voted**
- E. Set up a trust relationship between the IAM roles for the service accounts and an OpenID Connect (OIDC) identity provider. **Most Voted**

Correct Answer: DE

Community vote distribution

DE (100%)

Comments

Dantecito **Highly Voted** 1 year, 3 months ago

Selected Answer: DE

This was in my Exam today. Good Luck!!!!!!!!!!!!!!
upvoted 7 times

LeonSauveterre **Most Recent** 1 year, 5 months ago

Selected Answer: DE

A - Attaching policies to the IAM role of the EKS nodes provides broad permissions to all pods running on the nodes. IRSA is designed to eliminate such over-permissive configurations.

B - AWS resource access is managed by IAM, not Kubernetes network policies.

C - EKS cluster's IAM role is not meant to handle granular access for service accounts.

D - Ensures fine-grained access control at the service account level.

E - Allows Kubernetes service accounts to assume IAM roles securely.

upvoted 2 times

spoved 1 year, 8 months ago

<https://docs.aws.amazon.com/eks/latest/userguide/iam-roles-for-service-accounts.html>

<https://docs.aws.amazon.com/eks/latest/userguide/associate-service-account-role.html>

<https://docs.aws.amazon.com/eks/latest/userguide/enable-iam-roles-for-service-accounts.html>

=> DE

upvoted 4 times

moondooo 1 year, 8 months ago

Selected Answer: DE

probably D and E

<https://docs.aws.amazon.com/emr/latest/EMR-on-EKS-DevelopmentGuide/setting-up-enable-IAM.html>

upvoted 2 times

744fdad 1 year, 10 months ago

my educated guess C, E
upvoted 1 times

A company regularly uploads confidential data to Amazon S3 buckets for analysis.

The company's security policies mandate that the objects must be encrypted at rest. The company must automatically rotate the encryption key every year. The company must be able to track key rotation by using AWS CloudTrail. The company also must minimize costs for the encryption key.

Which solution will meet these requirements?

- A. Use server-side encryption with customer-provided keys (SSE-C)
- B. Use server-side encryption with Amazon S3 managed keys (SSE-S3)
- C. Use server-side encryption with AWS KMS keys (SSE-KMS) **Most Voted**
- D. Use server-side encryption with customer managed AWS KMS keys

Correct Answer: C

Community vote distribution

C (56%)

D (44%)

Comments

nebajp **Highly Voted** 1 year, 10 months ago

Selected Answer: C

SSE keys provided usage fee application and there is no monthly charges, hence its a correct option.
D is highly cost option with monthly and usage fee. which is incorrect.

upvoted 10 times

e10c971 **Most Recent** 6 months ago

Selected Answer: D

from AI :SSE-S3 (Amazon S3 managed keys): Rotation is automatic, but you cannot track rotation in CloudTrail because S3-managed keys are not customer-controlled.

SSE-C (customer-provided keys): No rotation support; you must manage keys manually.

SSE-KMS (AWS managed key): AWS rotates keys automatically every year, but you cannot control or track rotation events in CloudTrail.

Customer managed KMS key: You can enable automatic annual rotation, and CloudTrail logs all KMS API calls, including rotation events.

upvoted 1 times

In2718 10 months ago

Selected Answer: D

for all the reasons stated...

upvoted 2 times

Clouddon 10 months, 4 weeks ago

Selected Answer: D

✗ Why Other Options Are Incorrect:

A. SSE-C (Customer-provided keys)

You must provide and manage keys yourself — AWS does not store or rotate them ✗

No CloudTrail logging for encryption operations ✗

No automatic rotation ✗

B. SSE-S3 (S3-managed keys)

Keys are managed entirely by AWS, with no visibility or control ✗

No CloudTrail logging for key usage ✗

No automatic key rotation under your control ❌

C. SSE-KMS (AWS-managed KMS keys)

Uses AWS-managed KMS keys, which do not support automatic key rotation ❌

Less expensive than customer managed keys, but lacks key rotation and policy control ❌

upvoted 2 times

tch 1 year, 3 months ago

Selected Answer: C

should put full name as "Use server-side encryption with AWS KMS managed keys"

answer D does not minimize the cost

upvoted 1 times

yangbo 1 year, 3 months ago

Selected Answer: D

"SSE-KMS (AWS managed KMS keys): Automatically rotated, but the rotation records are not visible. Users cannot view the RotateKey event in CloudTrail."

upvoted 2 times

Dantecito 1 year, 3 months ago

Selected Answer: D

C. since May 2022 managed SSE-KMS can rotate every year <https://docs.aws.amazon.com/kms/latest/developerguide/rotate-keys.html>

D. I still prefer D because the question says "The company must automatically rotate the encryption key every year" it does not say "AWS must automatically..."

upvoted 2 times

tch 1 year, 3 months ago

customer managed keys has monthly fee for existence of keys (pro-rated hourly). Also charged for key usage.... not an good idea

upvoted 1 times

GOTJ 1 year, 4 months ago

Selected Answer: C

With all due respect, I think option "C" is poorly written, and it reflects on the lack of consensus with this one. A "server-side encryption with customer managed AWS KMS keys" (option "D") is nothing but an implementation of a "server-side encryption with AWS KMS keys" (option "C"), and both are SSE-KMS. What I think op tried to express with option C is: "Use server-side encryption with AWS-MANAGED AWS KMS keys (SSE-KMS). If my guess is correct, option "C" is the right answer, according to the table shown in this link:

<https://docs.aws.amazon.com/kms/latest/developerguide/concepts.html#aws-managed-cmk>

Option C and Option D satisfied all the technical requirements of the company, but option C is \$1/month cheaper than option D

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Selected Answer: C

AWS-managed keys are automatically rotated every year (approximately 365 days) by AWS.

Key rotation cannot be enabled or disabled for AWS-managed keys, ensuring compliance with the rotation requirement.

When AWS KMS rotates the key material for an AWS-managed key, it writes: A KMS CMK Rotation event to Amazon EventBridge, A RotateKey event to AWS CloudTrail.

This allows the company to track key rotation events for auditing purposes.

Cost Minimization: AWS-managed keys are free to use (no additional cost beyond the base AWS KMS pricing for API calls). This helps minimize costs.

<https://docs.aws.amazon.com/kms/latest/developerguide/rotate-keys.html>

upvoted 2 times

FlyingHawk 1 year, 4 months ago

You can enable automatic key rotation on the customer managed keys that you use for server-side encryption in AWS services. if disabled, then won't meet the requirement "The company must automatically rotate the encryption key every year"

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Why not D?

While customer-managed KMS keys also support automatic rotation and CloudTrail logging, they incur additional costs:

Customer-managed keys have a monthly fee (e.g., \$1 per key per month).

This would increase costs compared to AWS-managed keys, which are free.

Since the requirement is to minimize costs, AWS-managed keys are the better choice.

upvoted 1 times

joanna91 1 year, 4 months ago

Selected Answer: C

"Automatically" rotate, then make it handled by AWS service, not by customer -> C, not D

upvoted 2 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: D

"automatically rotate the encryption key" => C or D (because of KMS), then

"able to track key rotation" => Just D.

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Monitoring key rotation

When AWS KMS rotates the key material for an AWS managed key or customer managed key, it writes a KMS CMK Rotation event to Amazon EventBridge and a RotateKey event to your AWS CloudTrail log. You can use these records to verify that the KMS key was rotated.

upvoted 1 times

Anyio 1 year, 5 months ago

Selected Answer: D

Option C: Incorrect. Though server-side encryption with AWS KMS keys (SSE-KMS) would allow AWS to manage keys and enable logging via AWS CloudTrail, this option uses AWS-managed keys instead of customer-managed keys, limiting control over key rotations. Additionally, there can be more costs involved in using AWS-managed KMS keys compared to the customer managing their own.

Option D: Correct. Using server-side encryption with customer-managed AWS KMS keys allows the company to have full control over the encryption keys, including managing and ensuring automatic rotation every year. Moreover, AWS CloudTrail can be employed to log events associated with AWS KMS, enabling the tracking of when keys are rotated. This option balances cost-effectiveness with the operational requirements specified, as it provides the necessary control without unnecessary expenses from more specialized AWS services.

upvoted 1 times

JA2018 1 year, 6 months ago

Selected Answer: D

I will choose Option D for the following reasons:

#1 Automatic key rotation: AWS KMS allows you to set up automatic key rotation for customer managed keys, which fulfills the requirement to rotate encryption keys yearly.

2 CloudTrail tracking: All KMS key operations are logged in CloudTrail, enabling you to track key rotation activity.

#3 Lowest cost: While using customer-provided keys (SSE-C) might seem cost-effective at first glance, managing your own keys adds complexity and can be more expensive in the long run.

#\$ Compliance with security policies: Using customer managed KMS keys ensures that the company has full control over the encryption keys, meeting the stringent security requirements

upvoted 1 times

FlyingHawk 1 year, 4 months ago

There is no requirement for the company has full control over the encryption keys. the requirement is must auto-rotate the keys every year, must be able to track rotation via cloudtrail, and minimise cost. C meets requirement as AWS KMS with AWS Managed Key auto-rotate yearly too, you cannot change the rotation period or disable the rotation, it also logs the rotation events and no cost for storing it. you only need the customer key if you want the flexibility to delete it or change rotation period or disable it... etc but this is not required in the problem.

upvoted 1 times

XXXXXINN 1 year, 7 months ago

D

auto-rotation feature > customer managed key

upvoted 1 times

XXXXXINN 1 year, 8 months ago

D. customer needs to see the logs from Cloudtrail!

upvoted 1 times

sOI852POL 1 year, 7 months ago

Even with AWS KMS keys, rotation is logged on ctrail. Answer is D.

<https://docs.aws.amazon.com/kms/latest/developerguide/rotate-keys.html#:~:text=Monitoring%20key%20rotation,key%20was%20rotated.>

upvoted 1 times

sOI852POL 1 year, 9 months ago

Selected Answer: C

Answer is C.

There is no monthly fee for AWS managed keys

<https://docs.aws.amazon.com/kms/latest/developerguide/concepts.html#aws-managed-cmk>

upvoted 3 times

elmyth 1 year, 9 months ago

Selected Answer: C

Customer managed key: Monthly fee (pro-rated hourly) + Per-use fee + rotation and cloudtrail

AWS managed key: No monthly fee + Per-use fee (some AWS services pay this fee for you)+ rotation and cloudtrail

upvoted 4 times

A company has migrated several applications to AWS in the past 3 months. The company wants to know the breakdown of costs for each of these applications. The company wants to receive a regular report that includes this information.

Which solution will meet these requirements MOST cost-effectively?

- A. Use AWS Budgets to download data for the past 3 months into a .csv file. Look up the desired information.
- B. Load AWS Cost and Usage Reports into an Amazon RDS DB instance. Run SQL queries to get the desired information.
- C. Tag all the AWS resources with a key for cost and a value of the application's name. Activate cost allocation tags. Use Cost Explorer to get the desired information. **Most Voted**
- D. Tag all the AWS resources with a key for cost and a value of the application's name. Use the AWS Billing and Cost Management console to download bills for the past 3 months. Look up the desired information.

Correct Answer: C

Community vote distribution

C (67%)

D (33%)

Comments

spoved **Highly Voted** 1 year, 8 months ago

Selected Answer: C

- Organizing and tracking costs using AWS cost allocation tags

<https://docs.aws.amazon.com/awsaccountbilling/latest/aboutv2/cost-alloc-tags.html>

- Cost Explorer uses the same dataset that is used to generate the AWS Cost and Usage Reports and the detailed billing reports. For a comprehensive review of the data, you can download it into a comma-separated value (CSV) file.

<https://docs.aws.amazon.com/cost-management/latest/userguide/ce-what-is.html>

upvoted 5 times

rpmaws **Most Recent** 1 year, 8 months ago

Selected Answer: C

cost explorer allows cost analysis of up to 13 months.

upvoted 3 times

ccceb01 1 year, 9 months ago

Selected Answer: D

<https://docs.aws.amazon.com/awsaccountbilling/latest/aboutv2/cost-alloc-tags.html>

upvoted 4 times

An ecommerce company is preparing to deploy a web application on AWS to ensure continuous service for customers. The architecture includes a web application that the company hosts on Amazon EC2 instances, a relational database in Amazon RDS, and static assets that the company stores in Amazon S3.

The company wants to design a robust and resilient architecture for the application.

Which solution will meet these requirements?

- A. Deploy Amazon EC2 instances in a single Availability Zone. Deploy an RDS DB instance in the same Availability Zone. Use Amazon S3 with versioning enabled to store static assets.
- B. Deploy Amazon EC2 instances in an Auto Scaling group across multiple Availability Zones. Deploy a Multi-AZ RDS DB instance. Use Amazon CloudFront to distribute static assets. **Most Voted**
- C. Deploy Amazon EC2 instances in a single Availability Zone. Deploy an RDS DB instance in a second Availability Zone for cross-AZ redundancy. Serve static assets directly from the EC2 instances.
- D. Use AWS Lambda functions to serve the web application. Use Amazon Aurora Serverless v2 for the database. Store static assets in Amazon Elastic File System (Amazon EFS) One Zone-Infrequent Access (One Zone-IA).

Correct Answer: B

Community vote distribution

B (100%)

Comments

NH_User 6 months ago

Selected Answer: B

B is the answer
upvoted 1 times

mk168898 1 year, 7 months ago

keyword is multi-AZ
upvoted 3 times

Jeyaluxshan 1 year, 9 months ago

B is the answer
upvoted 2 times

pujithacg8 1 year, 10 months ago

B is correct
upvoted 2 times

An ecommerce company runs several internal applications in multiple AWS accounts. The company uses AWS Organizations to manage its AWS accounts.

A security appliance in the company's networking account must inspect interactions between applications across AWS accounts.

Which solution will meet these requirements?

- A. Deploy a Network Load Balancer (NLB) in the networking account to send traffic to the security appliance. Configure the application accounts to send traffic to the NLB by using an interface VPC endpoint in the application accounts.
- B. Deploy an Application Load Balancer (ALB) in the application accounts to send traffic directly to the security appliance.
- C. Deploy a Gateway Load Balancer (GWLB) in the networking account to send traffic to the security appliance. Configure the application accounts to send traffic to the GWLB by using an interface GWLB endpoint in the application accounts. **Most Voted**
- D. Deploy an interface VPC endpoint in the application accounts to send traffic directly to the security appliance.

Correct Answer: C

Community vote distribution

C (100%)

Comments

mk168898 **Highly Voted** 1 year, 7 months ago
security appliance, inspect interaction => GWLB
upvoted 5 times

poramesh **Most Recent** 1 year, 3 months ago
Finally a ques on GLB
upvoted 4 times

EllenLiu 1 year, 5 months ago
Selected Answer: C
<https://docs.aws.amazon.com/elasticloadbalancing/latest/gateway/getting-started.html>
upvoted 1 times

swati1508 1 year, 10 months ago
Inspect traffic for that use GWLB OPTION C
upvoted 3 times

komorebi 1 year, 10 months ago
Selected Answer: C
Answer is C
upvoted 2 times

example_ 1 year, 10 months ago
Selected Answer: C
<https://docs.aws.amazon.com/vpc/latest/privatelink/create-gateway-load-balancer-endpoint-service.html>
upvoted 3 times

A company runs its production workload on an Amazon Aurora MySQL DB cluster that includes six Aurora Replicas. The company wants near-real-time reporting queries from one of its departments to be automatically distributed across three of the Aurora Replicas. Those three replicas have a different compute and memory specification from the rest of the DB cluster.

Which solution meets these requirements?

- A. Create and use a custom endpoint for the workload **Most Voted**
- B. Create a three-node cluster clone and use the reader endpoint
- C. Use any of the instance endpoints for the selected three nodes
- D. Use the reader endpoint to automatically distribute the read-only workload

Correct Answer: A

Community vote distribution

A (100%)

Comments

FlyingHawk 1 year, 4 months ago

Selected Answer: A

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Endpoints.Custom.html>

A custom endpoint for an Aurora cluster represents a set of DB instances that you choose. When you connect to the endpoint, Aurora performs connection balancing and chooses one of the instances in the group to handle the connection. You define which instances this endpoint refers to, and you decide what purpose the endpoint serves.

upvoted 2 times

FlyingHawk 1 year, 4 months ago

custom endpoint for an Aurora cluster represents a set of DB instances that you choose. When you connect to the endpoint, Aurora performs connection balancing and chooses one of the instances in the group to handle the connection. You define which instances this endpoint refers to, and you decide what purpose the endpoint serves.

An Aurora DB cluster has no custom endpoints until you create one. You can create up to five custom endpoints for each provisioned Aurora cluster or Aurora Serverless v2 cluster. You can't use custom endpoints for Aurora Serverless v1 clusters.

The custom endpoint provides balanced database connections based on criteria other than the read-only or read/write capability of the DB instances.

upvoted 2 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: A

A - You can assign the three replicas with different compute and memory specifications to the custom endpoint, ensuring that queries are distributed only among these replicas.

B - Expensive, cumbersome, and unnecessary. The clone has separate storage and compute resources.

C - You can't target just the three replicas you want.

D - The reader endpoint automatically distributes read queries across *ALL* Aurora Replicas in the cluster.

upvoted 2 times

JA2018 1 year, 6 months ago

Selected Answer: A

This is why I choose Option A:

Using a Custom endpoint: This allows you to specify a specific set of Aurora replicas be used for the reporting queries, enabling you to target the three replicas with different compute and memory specifications while distributing the workload across them.

upvoted 1 times

JA2018 1 year, 6 months ago

Why the other options are not ideal:

- #B. Create a three-node cluster clone and use the reader endpoint: While you can create a separate cluster, using the "reader endpoint" would distribute queries across all replicas in that cluster, not just the three you want to target with different specs.
- #C. Use any of the instance endpoints for the selected three nodes: This approach would require manual connection management to each specific replica, not providing automatic load balancing across the chosen three nodes.
- #D. Use the reader endpoint to automatically distribute the read-only workload: The standard "reader endpoint" distributes reads across all Aurora replicas in the cluster, not just the selected three with specific configurations.

upvoted 1 times

A company runs a Node.js function on a server in its on-premises data center. The data center stores data in a PostgreSQL database. The company stores the credentials in a connection string in an environment variable on the server. The company wants to migrate its application to AWS and to replace the Node.js application server with AWS Lambda. The company also wants to migrate to Amazon RDS for PostgreSQL and to ensure that the database credentials are securely managed.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Store the database credentials as a parameter in AWS Systems Manager Parameter Store. Configure Parameter Store to automatically rotate the secrets every 30 days. Update the Lambda function to retrieve the credentials from the parameter.
- B. Store the database credentials as a secret in AWS Secrets Manager. Configure Secrets Manager to automatically rotate the credentials every 30 days. Update the Lambda function to retrieve the credentials from the secret. **Most Voted**
- C. Store the database credentials as an encrypted Lambda environment variable. Write a custom Lambda function to rotate the credentials. Schedule the Lambda function to run every 30 days.
- D. Store the database credentials as a key in AWS Key Management Service (AWS KMS). Configure automatic rotation for the key. Update the Lambda function to retrieve the credentials from the KMS key.

Correct Answer: B

Community vote distribution

B (100%)

Comments

LeonSauveterre 1 year, 5 months ago

Selected Answer: B

A - Parameter Store doesn't support automatic rotation.

C - Don't write a custom function when you can use off-the-rack ones.

D - AWS KMS is a service for managing encryption keys, not for managing secrets like database credentials. Also it doesn't provide direct support for managing or rotating database credentials.

upvoted 2 times

aragon_saa 1 year, 9 months ago

Selected Answer: B

Answer is B

upvoted 1 times

A company wants to replicate existing and ongoing data changes from an on-premises Oracle database to Amazon RDS for Oracle. The amount of data to replicate varies throughout each day. The company wants to use AWS Database Migration Service (AWS DMS) for data replication. The solution must allocate only the capacity that the replication instance requires.

Which solution will meet these requirements?

A. Configure the AWS DMS replication instance with a Multi-AZ deployment to provision instances across multiple Availability Zones.

B. Create an AWS DMS Serverless replication task to analyze and replicate the data while provisioning the required capacity.

Most Voted

C. Use Amazon EC2 Auto Scaling to scale the size of the AWS DMS replication instance up or down based on the amount of data to replicate.

D. Provision AWS DMS replication capacity by using Amazon Elastic Container Service (Amazon ECS) with an AWS Fargate launch type to analyze and replicate the data while provisioning the required capacity.

Correct Answer: B

Community vote distribution

B (88%)

13%

Comments

JunsK1e **Highly Voted** 1 year, 10 months ago

Selected Answer: B

Correct answer is B since the question need to allocate only the capacity that the replication instance requires
upvoted 6 times

EllenLiu **Most Recent** 1 year, 5 months ago

Selected Answer: B

https://docs.aws.amazon.com/dms/latest/userguide/CHAP_Serverless.html
upvoted 1 times

pujithacg8 1 year, 10 months ago

correct answer could be "B"
upvoted 2 times

komorebi 1 year, 10 months ago

Selected Answer: A

Answer is A
upvoted 1 times

A company has a multi-tier web application. The application's internal service components are deployed on Amazon EC2 instances. The internal service components need to access third-party software as a service (SaaS) APIs that are hosted on AWS.

The company needs to provide secure and private connectivity from the application's internal services to the third-party SaaS application. The company needs to ensure that there is minimal public internet exposure.

Which solution will meet these requirements?

- A. Implement an AWS Site-to-Site VPN to establish a secure connection with the third-party SaaS provider.
- B. Deploy AWS Transit Gateway to manage and route traffic between the application's VPC and the third-party SaaS provider.
- C. Configure AWS PrivateLink to allow only outbound traffic from the VPC without enabling the third-party SaaS provider to establish.
- D. Use AWS PrivateLink to create a private connection between the application's VPC and the third-party SaaS provider.

Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

LeonSauveterre 1 year, 5 months ago

Selected Answer: D

C - PrivateLink exists because of its main benefit of bidirectional communication for secure access without public internet exposure. By doing this, you're restricting the SaaS provider's ability to establish sessions. Option C is misleading and just plain wrong. Option D alone is enough.

upvoted 4 times

c3c7e62 1 year, 1 month ago

D is correct, just want to clarify, PrivateLink setup is unidirectional.

"AWS PrivateLink — Use AWS PrivateLink when you have a client/server set up where you want to allow one or more consumer VPCs unidirectional access to a specific service or set of instances in the service provider VPC or certain AWS services. Only the clients with access in the consumer VPC can initiate a connection to the service in the service provider VPC or AWS service."

That means -

Inbound to service: Yes (from client to the endpoint service).

Outbound from service to client: No, unless the client explicitly exposes something .

refer to this -

<https://docs.aws.amazon.com/whitepapers/latest/building-scalable-secure-multi-vpc-network-infrastructure/aws-privatelink.html>

upvoted 1 times

spoved 1 year, 8 months ago

Selected Answer: D

<https://docs.aws.amazon.com/prescriptive-guidance/latest/integrate-third-party-services/architecture-1.html>

It is limited to only TCP traffic and unidirectional communication. The third-party workloads cannot initiate communication back to your account.

upvoted 4 times

56ce46c 1 year, 8 months ago

I think C is correct

2. Restrict Inbound Traffic via Security Groups:

To prevent the third-party SaaS provider from establishing inbound connections to your VPC, use Security Groups attached to the VPC Endpoint Interface.

Outbound Traffic Allowed: Ensure that your security groups allow outbound traffic to the SaaS provider's IP ranges or endpoints.

Restrict Inbound Traffic: You should block all inbound traffic on the VPC Endpoint Interface by configuring the security group rules. For example:

Inbound Rules: Block all traffic (or leave it empty).

Outbound Rules: Allow outbound connections to the IP addresses or ports specified by the SaaS provider.

upvoted 1 times

komorebi 1 year, 10 months ago

Selected Answer: D

Answer is D

upvoted 3 times

JunsK1e 1 year, 10 months ago

Selected Answer: D

D is correct

upvoted 2 times

A solutions architect needs to connect a company's corporate network to its VPC to allow on-premises access to its AWS resources. The solution must provide encryption of all traffic between the corporate network and the VPC at the network layer and the session layer. The solution also must provide security controls to prevent unrestricted access between AWS and the on-premises systems.

Which solution meets these requirements?

- A. Configure AWS Direct Connect to connect to the VPC. Configure the VPC route tables to allow and deny traffic between AWS and on premises as required.
- B. Create an IAM policy to allow access to the AWS Management Console only from a defined set of corporate IP addresses. Restrict user access based on job responsibility by using an IAM policy and roles.
- C. Configure AWS Site-to-Site VPN to connect to the VPC. Configure route table entries to direct traffic from on premises to the VPC. Configure instance security groups and network ACLs to allow only required traffic from on premises. **Most Voted**
- D. Configure AWS Transit Gateway to connect to the VPC. Configure route table entries to direct traffic from on premises to the VPC. Configure instance security groups and network ACLs to allow only required traffic from on premises.

Correct Answer: C

Community vote distribution

C (96%)

4%

Comments

Abbas_Abi_AWS **Highly Voted** 1 year, 10 months ago

Selected Answer: C

AWS Direct Connect does not provide encryption by itself; it is often used in conjunction with VPN for encrypted traffic. Direct Connect primarily offers a dedicated connection and does not inherently satisfy the encryption requirement.

upvoted 8 times

LeonSauveterre **Most Recent** 1 year, 5 months ago

Selected Answer: C

A - Direct Connect provides private connectivity but does not inherently include encryption.

B - IAM policies is irrelevant.

C - AWS Site-to-Site VPN uses IPsec (network layer) and SSL/TLS (session layer) to encrypt all traffic between the on-premises network and the AWS VPC.

D - Just like option A.

upvoted 3 times

Anyio 1 year, 5 months ago

Selected Answer: C

The correct answer is C. Configure AWS Site-to-Site VPN to connect to the VPC. Configure route table entries to direct traffic from on premises to the VPC. Configure instance security groups and network ACLs to allow only required traffic from on premises.

Explanation:

Option C: Correct. AWS Site-to-Site VPN offers encrypted network connections over the internet, providing encryption of all traffic at the network layer (IPsec) between the on-premises network and the VPC. Moreover, by using route tables, security groups, and network ACLs, you can carefully control the flow of traffic and restrict access, thereby meeting the requirement of preventing unrestricted access.

upvoted 3 times

blehbleh 1 year, 8 months ago

Selected Answer: C

This is C, but not for all the reasons everyone is posting. D, also encrypts traffic and works at the network layer and also has security controls to prevent unrestricted access between AWS and on-premises systems.

So, if you thought D like I did initially you were very close. The reason it is C, is because C works at both the network and session layer while doing all the other requirements as well. Whereas D only works at the network layer.

Happy studying!

upvoted 4 times

komorebi 1 year, 10 months ago

Selected Answer: D

Answer is D

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Inter-Region gateway peering uses the same network infrastructure as VPC peering. Therefore traffic is encrypted using AES-256 encryption at the virtual network layer as it travels between Regions. Traffic is also encrypted using AES-256 encryption at the physical layer when it traverses network links that are outside of the physical control of AWS. As a result, traffic is double encrypted on network links outside the physical control of AWS.

Within the same Region, traffic is encrypted at the physical layer only when it traverses network links that are outside of the physical control of AWS

upvoted 1 times

JunsK1e 1 year, 10 months ago

Selected Answer: C

C is correct question needs to access between on prem and AWS

upvoted 4 times

A company has a custom application with embedded credentials that retrieves information from a database in an Amazon RDS for MySQL DB cluster. The company needs to make the application more secure with minimal programming effort. The company has created credentials on the RDS for MySQL database for the application user.

Which solution will meet these requirements?

- A. Store the credentials in AWS Key Management Service (AWS KMS). Create keys in AWS KMS. Configure the application to load the database credentials from AWS KMS. Enable automatic key rotation
- B. Store the credentials in encrypted local storage. Configure the application to load the database credentials from the local storage. Set up a credentials rotation schedule by creating a cron job.
- C. Store the credentials in AWS Secrets Manager. Configure the application to load the database credentials from Secrets Manager. Set up a credentials rotation schedule by creating an AWS Lambda function for Secrets Manager. **Most Voted**
- D. Store the credentials in AWS Systems Manager Parameter Store. Configure the application to load the database credentials from Parameter Store. Set up a credentials rotation schedule in the RDS for MySQL database by using Parameter Store.

Correct Answer: C

Community vote distribution

C (100%)

Comments

LeonSauveterre 1 year, 5 months ago

Selected Answer: C

- A - AWS KMS is used for encryption and key management, not for storing and rotating credentials themselves.
- B - Why manually set up a cron job?
- C - Securely stores and manages sensitive information such as credentials, and provides native support for automatic rotation.
- D - Parameter Store doesn't provide automatic rotation.

upvoted 2 times

Anyio 1 year, 5 months ago

Selected Answer: C

The correct answer is C. Store the credentials in AWS Secrets Manager. Configure the application to load the database credentials from Secrets Manager. Set up a credentials rotation schedule by creating an AWS Lambda function for Secrets Manager.

Explanation:

Option C: Correct. AWS Secrets Manager is designed to securely store, manage, and retrieve database credentials. It supports automatic credentials rotation for Amazon RDS databases with minimal programming effort. Secrets Manager can automatically rotate the Amazon RDS database credentials using a built-in Lambda function, providing an integrated and secure solution for handling credentials.

upvoted 1 times

Omshanti 1 year, 8 months ago

Selected Answer: C

AWS Secret manager securely stores data base user id and passwords

upvoted 2 times

komorebi 1 year, 10 months ago

Selected Answer: C

Answer is C

upvoted 2 times

A company wants to move its application to a serverless solution. The serverless solution needs to analyze existing data and new data by using SQL. The company stores the data in an Amazon S3 bucket. The data must be encrypted at rest and replicated to a different AWS Region.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Create a new S3 bucket that uses server-side encryption with AWS KMS multi-Region keys (SSE-KMS). Configure Cross-Region Replication (CRR). Load the data into the new S3 bucket. Use Amazon Athena to query the data.
- B. Create a new S3 bucket that uses server-side encryption with Amazon S3 managed keys (SSE-S3). Configure Cross-Region Replication (CRR). Load the data into the new S3 bucket. Use Amazon RDS to query the data.
- C. Configure Cross-Region Replication (CRR) on the existing S3 bucket. Use server-side encryption with Amazon S3 managed keys (SSE-S3). Use Amazon Athena to query the data. **Most Voted**
- D. Configure S3 Cross-Region Replication (CRR) on the existing S3 bucket. Use server-side encryption with AWS KMS multi-Region keys (SSE-KMS). Use Amazon RDS to query the data.

Correct Answer: C

Community vote distribution



Comments

LeonSauveterre **Highly Voted** 1 year, 5 months ago

Selected Answer: C

- A - A new bucket + KMS multi-Region keys = Too much operational oversight.
- B - RDS is not a serverless solution
- C - By using the existing S3 bucket, you eliminate the need to create a new bucket and load data into it.
- D - Athena supports querying using SQL, so that rules out RDS.

upvoted 9 times

dhewa **Highly Voted** 1 year, 9 months ago

Selected Answer: A

A wins because it gives us encryption with AWS KMS multi-Region keys

upvoted 7 times

46a8566 **Most Recent** 4 months, 3 weeks ago

Selected Answer: A

S3 with SSE-KMS multi-Region keys + CRR + Athena meets all requirements with least overhead.

upvoted 1 times

Catlyn 5 months, 3 weeks ago

Selected Answer: C

correct

upvoted 1 times

In2718 9 months, 2 weeks ago

Selected Answer: C

Chat AI -

Answer C - Configure CRR on the existing S3 bucket with SSE-S3 and use Amazon Athena to query the data.

Serverless SQL: Athena is fully serverless and queries data directly in S3.

Encryption at rest: SSE-S3 (S3-managed keys) satisfies encryption with zero KMS key management.

Cross-Region Replication: CRR works seamlessly with SSE-S3.

Least overhead: No new bucket, no data reload/migration, no KMS key creation/policies, and no databases to manage (unlike RDS).

upvoted 2 times

dce7e4b 1 year ago

Selected Answer: A

I would go on option A.

Considering this article: <https://docs.aws.amazon.com/AmazonS3/latest/userguide/replication-config-for-kms-objects.html>

If we just activate the replication on the existing bucket, the unencrypted data will be replicated unencrypted.

If there was an option for creating a new bucket with crr, sse-s3 AND Athena I would have chosen this one.

But the nearest solution is Wright en with rds instead of Athena. So go for A

upvoted 2 times

ElI89 1 year, 2 months ago

Selected Answer: A

A is correct.

everyone. voting for c i think isn't paying attention to the part where you need the data replicated to a different region. how are you meant to replicate it without a new bucket?

everyone saying use the existing bucket.... urm what? how can you replicate into the same bucket that's in the same region and expect it to be in a different region?

clearly A.

upvoted 2 times

tch 1 year, 2 months ago

Selected Answer: C

we don't need to create a policy specifically for Amazon S3-managed keys (SSE-S3) to encrypt objects in your S3 bucket... should be more easy and LEAST operational overhead

upvoted 2 times

yangbo 1 year, 3 months ago

Selected Answer: C

The default encryption for S3 is SSE-S3, and cross-region replication can be enabled normally.

upvoted 2 times

FlyingHawk 1 year, 4 months ago

Selected Answer: A

Amazon S3 managed keys is region specific, for CRR, we must use KMS multi-region keys.

<https://docs.aws.amazon.com/kms/latest/developerguide/multi-region-keys-overview.html>

upvoted 5 times

FlyingHawk 1 year, 4 months ago

I checked AI, ChatGPT, Gemini, Claude AI and DeepSeek think A is correct, but meta AI thinks C is correct. SSE-S3 is the default encryption, it will be least operational effort for sure. but with SSE-S3 (server-side encryption with S3-managed keys), AWS manages the encryption keys, and each region will use its own region-specific encryption key.

There is no shared key between the source and destination buckets. AWS handles encryption and decryption seamlessly in each region. This approach is operationally simple but lacks control and consistency for encryption keys across regions.

upvoted 2 times

trinh_le 1 year, 5 months ago

Selected Answer: C

It's not require highly sensitive data or complexity gain permission. So it should use sse-s3 is suitable

upvoted 2 times

Anyio 1 year, 5 months ago

Selected Answer: C

The correct answer is C.

Explanation:

Option A: Incorrect. Although using AWS KMS multi-Region keys (SSE-KMS) and Amazon Athena to query the data meet the security and SQL querying requirements, creating a new S3 bucket and handling data migration increases operational overhead unnecessarily if the data already exists

Option C: Correct. Configuring Cross-Region Replication (CRR) on the existing S3 bucket makes efficient use of existing infrastructure and leverages server-side encryption with Amazon S3 managed keys (SSE-S3) to ensure data encryption at rest with lower operational complexity and costs compared to using KMS keys. Using Amazon Athena allows querying the data directly in S3, offering serverless and flexible SQL querying capabilities with minimal setup and operational overhead.

upvoted 2 times

JA2018 1 year, 6 months ago

Selected Answer: C

I will chose Option C for the following reasons:

#1: Least operational overhead: Choosing "SSE-S3" over "SSE-KMS" minimizes operational overhead as it automatically manages encryption keys within S3, eliminating the need for additional KMS key management.

#2: Existing S3 bucket: Reusing the existing bucket avoids the extra step of creating a new one and migrating data.

#3: Athena for querying: Athena is a serverless solution ideal for querying large datasets stored in S3, aligning with the requirement for a serverless architecture.

upvoted 2 times

JA2018 1 year, 6 months ago

For Option A: While using KMS multi-region keys provides more control, it adds extra management complexity compared to SSE-S3.

upvoted 2 times

Abdullah2004 1 year, 9 months ago

Selected Answer: A

A is correct

upvoted 3 times

komorebi 1 year, 10 months ago

Selected Answer: A

Answer is A

upvoted 4 times

JunsK1e 1 year, 10 months ago

Selected Answer: C

C is correct because it needs to replicate to different AWS region

upvoted 2 times

A company has a web application that has thousands of users. The application uses 8-10 user-uploaded images to generate AI images. Users can download the generated AI images once every 6 hours. The company also has a premium user option that gives users the ability to download the generated AI images anytime.

The company uses the user-uploaded images to run AI model training twice a year. The company needs a storage solution to store the images.

Which storage solution meets these requirements MOST cost-effectively?

- A. Move uploaded images to Amazon S3 Glacier Deep Archive. Move premium user-generated AI images to S3 Standard. Move non-premium user-generated AI images to S3 Standard-Infrequent Access (S3 Standard-IA). **Most Voted**
- B. Move uploaded images to Amazon S3 Glacier Deep Archive. Move all generated AI images to S3 Glacier Flexible Retrieval.
- C. Move uploaded images to Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA). Move premium user-generated AI images to S3 Standard. Move non-premium user-generated AI images to S3 Standard-Infrequent Access (S3 Standard-IA).
- D. Move uploaded images to Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA). Move all generated AI images to S3 Glacier Flexible Retrieval.

Correct Answer: A

Community vote distribution

A (61%)

C (32%)

7%

Comments

dhewa **Highly Voted** 1 year, 9 months ago

Selected Answer: A

The company uses the user-uploaded images to run AI model training twice a year. So for this Deep Archive will be necessary.
upvoted 7 times

JA2018 1 year, 6 months ago

But moving uploaded images to Glacier Deep Archive is not ideal as it has a very high retrieval cost and long retrieval time, which is not suitable for the occasional access needed for AI model training.

upvoted 2 times

blehbleh **Highly Voted** 1 year, 8 months ago

Selected Answer: A

This is A. We care about cost effectiveness. In regards to images being used twice a year "For images accessed only twice a year, S3 Glacier Deep Archive would be the more cost-effective option compared to S3 One Zone Infrequent Access, as it is designed for extremely infrequent access and offers the lowest storage cost within AWS S3 storage classes; while S3 One Zone Infrequent Access is cheaper than standard Infrequent Access, it still might be slightly more expensive for data accessed as rarely as twice a year." In regards to premium users keep them standard. Non premium users can be in the infrequent since they have 6 hrs.

Cost Effective!

upvoted 5 times

JA2018 1 year, 6 months ago

But moving uploaded images to Glacier Deep Archive is not ideal as it has a very high retrieval cost and long retrieval time, which is not suitable for the occasional access needed for AI model training.

upvoted 1 times

tch **Most Recent** 1 year, 3 months ago

Selected Answer: A

S3 Glacier Deep Archive is Amazon S3's lowest-cost storage class and supports long-term retention and digital preservation for data that may be accessed once or twice in a year

B& D are out.....

upvoted 2 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: C

A - Deep Archive is designed for long-term cold storage and can spend hours to retrieve each object. For AI model training, you'd hate it if you must spend hundreds/thousands of hours before you can actually start training it.

B - Premium users would be so mad if they had to wait for hours to get their generated images.

C - S3 One Zone-IA reduces costs, and it's great now that we don't need high durability across multiple availability zones. Premium users will experience low latency to get their generated images.

D - Glacier tiers are not designed for frequent access. This is bad practice.

One more thing about option C, since training happens extremely infrequently and the images can be re-uploaded if necessary, One Zone IA's lower redundancy is an acceptable trade-off for cost savings.

upvoted 3 times

Cpso 1 year, 6 months ago

Selected Answer: A

I choose A. But it not best answer. for standard-ia you get data immediately but pay for request which not make sense for requirement. The best fit of non-premium user is Glacier Flexible standard(1-5hour)

upvoted 3 times

JA2018 1 year, 6 months ago

But moving uploaded images to Glacier Deep Archive is not ideal as it has a very high retrieval cost and long retrieval time, which is not suitable for the occasional access needed for AI model training.

upvoted 1 times

Oghare 1 year, 8 months ago

this is A

S3 Glacier Deep Archive is the most cost-effective storage for long-term infrequently accessed data, ideal for the user-uploaded images used twice a year for AI training.

S3 Standard for premium user-generated AI images ensures fast access, which is needed for frequent downloads.

S3 Standard-IA for non-premium user-generated AI images is cost-effective for less frequent access, as it charges lower for storage but slightly more for retrieval, which fits the 6-hour download frequency.

Not C because, although S3 One Zone-IA is a lower-cost option as well it provides less durability because it stores data in only one Availability Zone. While it is cost-effective, it increases the risk of data loss for critical AI training data.

B and D is out

S3 Glacier Flexible Retrieval for all generated AI images would likely introduce unacceptable retrieval delays for premium users, as they require immediate access to download images

upvoted 3 times

JA2018 1 year, 6 months ago

But moving uploaded images to Glacier Deep Archive is not ideal as it has a very high retrieval cost and long retrieval time, which is not suitable for the occasional access needed for AI model training.

upvoted 1 times

kbgsgsgs 1 year, 8 months ago

Selected Answer: C

S3 Glacier Deep Archive does not meet the 6-hour download requirement because it takes time to access data. User can download even if you are not a premium user

upvoted 1 times

blehbleh 1 year, 8 months ago

it says "Users can download the generated AI images once every 6 hours." The upload images are used for training only 2 times a year which meets the s3 glacier deep dive. The standard users are put in infrequent access and premium users are put in standard. I don't understand how people are messing this up.

upvoted 3 times

[Removed] 1 year, 9 months ago

Selected Answer: B

B is correct

Explanation:

S3 Glacier Deep Archive is the most cost-effective storage option for data that is rarely accessed. Since the user-uploaded images are only used twice a year for AI model training, storing them in Glacier Deep Archive is ideal for minimizing costs. The longer retrieval time (up to 12 hours) is acceptable given the infrequent access.

S3 Glacier Flexible Retrieval is suitable for storing the generated AI images because it balances cost and retrieval time. Regular users can download images every 6 hours, which Glacier Flexible Retrieval can accommodate with its flexible retrieval options (ranging from minutes to hours). This

solution also works for premium users, who might need more frequent access. While S3 Standard or Standard-IA could be used, Glacier Flexible Retrieval offers significant cost savings while still meeting the access requirements.

upvoted 2 times

JA2018 1 year, 6 months ago

Option B is using Glacier Flexible Retrieval.

Storing all generated AI images in Glacier Flexible Retrieval is also not cost-effective as the premium users require faster access to their images.

upvoted 1 times

officedepotadmin 1 year, 9 months ago

Selected Answer: C

S3 One Zone-IA is a cost-effective storage option for images that are accessed infrequently but are still needed for AI model training twice a year. One Zone-IA stores data in a single Availability Zone, making it less expensive but still highly available within that zone. Premium users need frequent access to their AI-generated images so S3. Non-premium users access their AI-generated images less frequently (once every 6 hours) so S3 Standard-IA

upvoted 5 times

pujithacg8 1 year, 10 months ago

A is correct as Glacier deep archive provides the lowest-cost storage class.

upvoted 1 times

JA2018 1 year, 6 months ago

But moving uploaded images to Glacier Deep Archive is not ideal as it has a very high retrieval cost and long retrieval time, which is not suitable for the occasional access needed for AI model training.

upvoted 1 times

A company is developing machine learning (ML) models on AWS. The company is developing the ML models as independent microservices. The microservices fetch approximately 1 GB of model data from Amazon S3 at startup and load the data into memory. Users access the ML models through an asynchronous API. Users can send a request or a batch of requests.

The company provides the ML models to hundreds of users. The usage patterns for the models are irregular. Some models are not used for days or weeks. Other models receive batches of thousands of requests at a time.

Which solution will meet these requirements?

- A. Direct the requests from the API to a Network Load Balancer (NLB). Deploy the ML models as AWS Lambda functions that the NLB will invoke. Use auto scaling to scale the Lambda functions based on the traffic that the NLB receives.
- B. Direct the requests from the API to an Application Load Balancer (ALB). Deploy the ML models as Amazon Elastic Container Service (Amazon ECS) services that the ALB will invoke. Use auto scaling to scale the ECS cluster instances based on the traffic that the ALB receives.
- C. Direct the requests from the API into an Amazon Simple Queue Service (Amazon SQS) queue. Deploy the ML models as AWS Lambda functions that SQS events will invoke. Use auto scaling to increase the number of vCPUs for the Lambda functions based on the size of the SQS queue.
- D. Direct the requests from the API into an Amazon Simple Queue Service (Amazon SQS) queue. Deploy the ML models as Amazon Elastic Container Service (Amazon ECS) services that read from the queue. Use auto scaling for Amazon ECS to scale both the cluster capacity and number of the services based on the size of the SQS queue. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

FlyingHawk 1 year, 4 months ago

Selected Answer: D

Given that the problem statement does not specify the processing time, Option D (ECS) is the safer choice because it does not have the same limitations as Lambda (cold start and 15 minutes execution). However, if the processing time is short and cold starts are acceptable, Option C (Lambda) could also be a cost-effective solution.

upvoted 2 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: D

A - Lambda has a cold start latency. There are ways to optimize that (like the environment variable config which I actually did when I used Lambda once), but loading 1GB into memory is still too much.

B - Using ALB makes handling sudden bursts of traffic even more challenging.

C - Just like A if I'm correct.

D - Each ML model needs to load 1 GB of model data at startup ==> Containers in ECS are well-suited for this because they provide persistent execution environments that avoid the overhead of reloading the model data frequently.

upvoted 1 times

RamanAgarwal 1 year, 5 months ago

Selected Answer: C

Since AWS Lambda supports 1GB of memory and can be scaled seamlessly, the ans should be C. Question says a lot of APIs are not used frequently so keeping them in ECS will result in higher costs. Also these are async operations means the response is not time bound hence we can wait for lambda to startup and scale up based on the size of the SQS queue.

upvoted 2 times

kbgsgrs 1 year, 8 months ago

Selected Answer: B

Why should I use SQS in option D? Wouldn't ALB be enough?

upvoted 2 times

Sergantus 1 year, 6 months ago

It's talking about accessing models through an asynchronous API, so decoupling is needed (SQS)

upvoted 1 times

A company runs a web application on Amazon EC2 instances in an Auto Scaling group behind an Application Load Balancer (ALB). The application stores data in an Amazon Aurora MySQL DB cluster.

The company needs to create a disaster recovery (DR) solution. The acceptable recovery time for the DR solution is up to 30 minutes. The DR solution does not need to support customer usage when the primary infrastructure is healthy.

Which solution will meet these requirements?

A. Deploy the DR infrastructure in a second AWS Region with an ALB and an Auto Scaling group. Set the desired capacity and maximum capacity of the Auto Scaling group to a minimum value. Convert the Aurora MySQL DB cluster to an Aurora global database. Configure Amazon Route 53 for an active-passive failover with ALB endpoints. **Most Voted**

B. Deploy the DR infrastructure in a second AWS Region with an ALB. Update the Auto Scaling group to include EC2 instances from the second Region. Use Amazon Route 53 to configure active-active failover. Convert the Aurora MySQL DB cluster to an Aurora global database.

C. Back up the Aurora MySQL DB cluster data by using AWS Backup. Deploy the DR infrastructure in a second AWS Region with an ALB. Update the Auto Scaling group to include EC2 instances from the second Region. Use Amazon Route 53 to configure active-active failover. Create an Aurora MySQL DB cluster in the second Region. Restore the data from the backup.

D. Back up the infrastructure configuration by using AWS Backup. Use the backup to create the required infrastructure in a second AWS Region. Set the Auto Scaling group desired capacity to zero. Use Amazon Route 53 to configure active-passive failover. Convert the Aurora MySQL DB cluster to an Aurora global database.

Correct Answer: A

Community vote distribution

A (71%)

D (29%)

Comments

chwieobjom **Highly Voted** 1 year, 9 months ago

Selected Answer: A

RTO 30 minute, warm standby.

upvoted 13 times

spoved **Highly Voted** 1 year, 8 months ago

Selected Answer: D

<https://docs.aws.amazon.com/whitepapers/latest/disaster-recovery-workloads-on-aws/disaster-recovery-options-in-the-cloud.html>

The DR solution does not need to support customer usage when the primary infrastructure is healthy. -> Pilot Light

upvoted 7 times

jingen11 1 year, 7 months ago

Backup and restore RTO is up to hours

upvoted 4 times

In2718 **Most Recent** 10 months ago

Selected Answer: D

I would have gone with answer A if it were not for this one item - The acceptable recovery time for the DR solution is up to 30 minutes. That would be ideal for pilot light, which would be within the 30 minute window. AI - The recovery time objective (RTO) for an AWS Pilot Light disaster recovery strategy typically ranges from minutes to tens of minutes.

upvoted 1 times

zdi561 1 year, 3 months ago

Selected Answer: D

A is not right because the max and min are set to minimum which is zero. and the global table use resource.

upvoted 1 times

Salilgen 1 year, 5 months ago

Selected Answer: A

The question doesn't ask about the most cost effective solution. A is the option that ensures the lowest recovery time

upvoted 3 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: A

Well just a thought, I hate it when each option looks like a valid one... I think I'll go with A because it seems to me other options are somewhat flawed compared to A.

B - "The DR solution does not need to support customer usage when the primary infrastructure is healthy." So active-active failover is not necessary.

C - Restoring a database from AWS Backup can take hours to complete, let alone the 30-minute RTO. If I remember it correctly, it can take up to 7 days?

D - Just like C.

One more thing about option D, the desired capacity parameter in an Auto Scaling group determines the number of EC2 instances that should be running. By setting it to 0, the group will not launch any EC2 instances during normal operations, so costs are reduced when idle. That's a plus actually.

upvoted 6 times

Anyio 1 year, 5 months ago

Selected Answer: A

The correct answer is A.

Explanation:

Option A: Correct. This solution sets up disaster recovery infrastructure in a secondary region with minimal running costs by keeping the Auto Scaling group's desired and maximum capacity low. By converting the Aurora MySQL DB cluster to an Aurora global database, the company can provide rapid cross-region replication and promote the standby database to primary if needed, minimizing downtime. Configuring active-passive failover using Route 53 ensures the DR system is only activated when necessary, aligning with the requirement of not supporting customer usage when the primary infrastructure is healthy.

upvoted 3 times

Liongeek 1 year, 5 months ago

Selected Answer: A

RTO 30 min has to be warm standby.

AWS Backup DOES NOT back up configuration, it backs up DATA

upvoted 5 times

EllenLiu 1 year, 5 months ago

Selected Answer: D

pilot light, set auto scaling group desired capacity to 0

upvoted 1 times

Penjerla 1 year, 6 months ago

Selected Answer: A

Restoring Database from AWS back up could take from a few min to several hours. So not D

upvoted 3 times

JA2018 1 year, 6 months ago

Selected Answer: A

I will choose Option A for the following options:

#1 - Active-passive failover: This aligns with the requirement of a DR solution that doesn't need to support customer usage during normal operations, meaning the secondary region only activates when a primary failure occurs.

2 - Minimal Auto Scaling group capacity: Setting the desired and maximum capacity to a minimum value ensures minimal cost when the DR infrastructure is not actively used.

3 - Aurora Global Database: Converting the Aurora cluster to a global database allows for fast failover to the secondary region, meeting the 30-minute recovery time objective.

#4 - Separate AWS Region: Deploying the DR infrastructure in a different region provides geographic redundancy, crucial for disaster recovery.

upvoted 3 times

Atdotcom 1 year, 7 months ago

Selected Answer: D

D is pilot-light because desired min capacity =0 so ressources are live , services idle, RTO 10s.Ressources are provisionned and scale after even.
upvoted 2 times

elmyth 1 year, 7 months ago

Selected Answer: D

It can be A, if minimum value = 0.

It Can be D, if this approach is implemented before the disaster.

And "Set the Auto Scaling group desired capacity to zero" means that it supposed to be done before the disaster.

upvoted 3 times

A company is migrating its data processing application to the AWS Cloud. The application processes several short-lived batch jobs that cannot be disrupted. Data is generated after each batch job is completed. The data is accessed for 30 days and retained for 2 years.

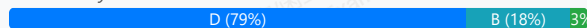
The company wants to keep the cost of running the application in the AWS Cloud as low as possible.

Which solution will meet these requirements?

- A. Migrate the data processing application to Amazon EC2 Spot Instances. Store the data in Amazon S3 Standard. Move the data to Amazon S3 Glacier Instant. Retrieval after 30 days. Set an expiration to delete the data after 2 years.
- B. Migrate the data processing application to Amazon EC2 On-Demand Instances. Store the data in Amazon S3 Glacier Instant Retrieval. Move the data to S3 Glacier Deep Archive after 30 days. Set an expiration to delete the data after 2 years.
- C. Deploy Amazon EC2 Spot Instances to run the batch jobs. Store the data in Amazon S3 Standard. Move the data to Amazon S3 Glacier Flexible Retrieval after 30 days. Set an expiration to delete the data after 2 years.
- D. Deploy Amazon EC2 On-Demand Instances to run the batch jobs. Store the data in Amazon S3 Standard. Move the data to Amazon S3 Glacier Deep Archive after 30 days. Set an expiration to delete the data after 2 years. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

nebajp **Highly Voted** 1 year, 10 months ago

Selected Answer: D

D is the correct answer
for 30 days - use Amazon S3 standard
2 years Retaining - Glacier Deep Archive
Can not be Disrupted - On-Demand Instances
upvoted 10 times

SR0312 **Highly Voted** 1 year, 10 months ago

Selected Answer: B

Job cannot be disrupted - On demand
upvoted 5 times

In2718 **Most Recent** 9 months, 2 weeks ago

Selected Answer: D

Answer D is cheaper.
Because:
Answer B locks you into 90 days minimum charge in Glacier Instant Retrieval, even though you only need 30 days of access.
Answer D pays S3 Standard rates for 30 days (slightly higher GB-month cost) but avoids the 60 days of wasted charges.
upvoted 1 times

FlyingHawk 1 year, 4 months ago

Selected Answer: D

1. The short-lived batch jobs cannot be disrupted -> on demand. A and C are out
2. The data is accessed for 30 days and retained for 2 years. means S3 Standard (30 days) -> S3 Glacier -> expire. S3 Glacier instant Retrieval has a minimum storage for 90 days, so B is out.
upvoted 3 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: D

B - Amazon S3 Glacier Instant Retrieval is an archive storage class that delivers the lowest-cost storage for long-lived data that is rarely accessed and requires retrieval in milliseconds. So this is wrong because "The data is accessed for 30 days".

upvoted 3 times

Anyio 1 year, 5 months ago

Selected Answer: D

The correct answer is D. Deploy Amazon EC2 On-Demand Instances to run the batch jobs. Store the data in Amazon S3 Standard. Move the data to Amazon S3 Glacier Deep Archive after 30 days. Set an expiration to delete the data after 2 years.

Explanation:

Option D: Correct. Since the batch jobs cannot be disrupted, using Amazon EC2 On-Demand Instances ensures the application runs without interruption. Initially storing data in Amazon S3 Standard allows for access during the 30-day period post-processing. Moving the data to Amazon S3 Glacier Deep Archive after 30 days optimizes storage costs for long-term retention of up to 2 years, the lowest cost for infrequent access. Setting an expiration ensures the data is deleted after 2 years, controlling costs and data lifecycle management.

upvoted 2 times

EllenLiu 1 year, 5 months ago

Selected Answer: D

I am always confused about the choice between s3 standard and s3 glacier instant...

short period of storage + frequently retrieve ==> s3 standard

long period of storage + infrequently retrieve ==> s3 glacier instant

s3 standard: \$0.023 per GB for storage / \$0.0004 get, select per 1000 request

s3 glacier instant: \$0.004 per GB for storage / \$0.01 get, select per 1000 request

<https://aws.amazon.com/s3/pricing/?nc=sn&loc=4>

upvoted 3 times

spoved 1 year, 8 months ago

Selected Answer: B

<https://aws.amazon.com/about-aws/whats-new/2021/11/amazon-s3-glacier-instant-retrieval-storage-class/>

- The easiest way to store data in S3 Glacier Instant Retrieval is to use the S3 PUT API to upload data directly, or use S3 Lifecycle to transition data from the S3 Standard and S3 Standard-IA storage classes.

- The company wants to keep the cost of running the application in the AWS Cloud as low as possible

=> B

upvoted 1 times

FlyingHawk 1 year, 4 months ago

S3 Glacier Instant Retrieval has a minimum storage duration of 90 days. If you move data to S3 Glacier Deep Archive before 90 days, you will be charged for the full 90 days of storage, even if the data is deleted or moved earlier. This makes it cost-ineffective for the company's requirement to move data after 30 days.

upvoted 1 times

pujithacg8 1 year, 10 months ago

D is correct

upvoted 2 times

flaviobrf 1 year, 10 months ago

Selected Answer: D

I understand that D is the right answer

upvoted 5 times

siheom 1 year, 10 months ago

Selected Answer: C

I VOTE C

upvoted 1 times

tch 1 year, 3 months ago

A& C are out!

Answer is D

S3 Glacier Deep Archive is Amazon S3's lowest-cost storage class and supports long-term retention and digital preservation for data that may be accessed once or twice in a year.

upvoted 1 times

officedepotadmin 1 year, 9 months ago

you voted wrong

upvoted 1 times

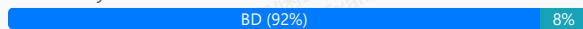
A company needs to design a hybrid network architecture. The company's workloads are currently stored in the AWS Cloud and in on-premises data centers. The workloads require single-digit latencies to communicate. The company uses an AWS Transit Gateway transit gateway to connect multiple VPCs.

Which combination of steps will meet these requirements MOST cost-effectively? (Choose two.)

- A. Establish an AWS Site-to-Site VPN connection to each VPC.
- B. Associate an AWS Direct Connect gateway with the transit gateway that is attached to the VPCs. **Most Voted**
- C. Establish an AWS Site-to-Site VPN connection to an AWS Direct Connect gateway.
- D. Establish an AWS Direct Connect connection. Create a transit virtual interface (VIF) to a Direct Connect gateway. **Most Voted**
- E. Associate AWS Site-to-Site VPN connections with the transit gateway that is attached to the VPCs.

Correct Answer: BD

Community vote distribution



Comments

LeonSauveterre **Highly Voted** 1 year, 5 months ago

Selected Answer: BD

A little memo for me. Hope it'll be helpful for you as well.

1. VPN Connection: For secure, encrypted traffic between on-premises and AWS over the public internet. ==> Relies on public internet (higher latency)
2. Direct Connect: For dedicated private network connectivity between on-premises and AWS (like data centers with large workloads). ==> Higher cost, longer setup time.
3. PrivateLink: For secure private connectivity to AWS services or third-party services (like Salesforce) within AWS. ==> Simple setup but limited to services configured with PrivateLink (not general connectivity).
4. VPC Peering: For direct connectivity between two AWS VPCs (like multi-account or region scenarios). ==> Does not support transitive connections not suitable for external services like Salesforce.

upvoted 6 times

MSAM16 **Most Recent** 1 year, 5 months ago

Selected Answer: BD

BD is correct

upvoted 2 times

jingen11 1 year, 7 months ago

Selected Answer: BE

Question asked about cheapest. D is not cheap

upvoted 1 times

Sergantus 1 year, 6 months ago

Single-digit latency is required

upvoted 3 times

muhammadahmer36 1 year, 10 months ago

Selected Answer: BD

BD is correct

upvoted 3 times

A global ecommerce company runs its critical workloads on AWS. The workloads use an Amazon RDS for PostgreSQL DB instance that is configured for a Multi-AZ deployment.

Customers have reported application timeouts when the company undergoes database failovers. The company needs a resilient solution to reduce failover time.

Which solution will meet these requirements?

- A. Create an Amazon RDS Proxy. Assign the proxy to the DB instance. **Most Voted**
- B. Create a read replica for the DB instance. Move the read traffic to the read replica.
- C. Enable Performance Insights. Monitor the CPU load to identify the timeouts.
- D. Take regular automatic snapshots. Copy the automatic snapshots to multiple AWS Regions.

Correct Answer: A

Community vote distribution

A (100%)

Comments

Clouddon 10 months, 4 weeks ago

Selected Answer: A

✗ Why other options are incorrect:

B. Create a read replica

✗ Read replicas do not help with failover time. They are used for read scaling and DR.

✗ Failover to a read replica is not automatic — requires manual promotion.

C. Enable Performance Insights

✗ Performance Insights helps with diagnostics, not failover handling or reduction.

✗ Does not reduce failover time — only helps analyze issues after they occur.

D. Take regular automatic snapshots

✗ Snapshots are for backup and restore, not for improving availability or reducing failover time.

✗ Provides disaster recovery, not active failover capability.

upvoted 1 times

aragon_saa 1 year, 9 months ago

Selected Answer: A

Answer is A

upvoted 2 times

A company has multiple Amazon RDS DB instances that run in a development AWS account. All the instances have tags to identify them as development resources. The company needs the development DB instances to run on a schedule only during business hours.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Create an Amazon CloudWatch alarm to identify RDS instances that need to be stopped. Create an AWS Lambda function to start and stop the RDS instances.
- B. Create an AWS Trusted Advisor report to identify RDS instances to be started and stopped. Create an AWS Lambda function to start and stop the RDS instances.
- C. Create AWS Systems Manager State Manager associations to start and stop the RDS instances.
- D. Create an Amazon EventBridge rule that invokes AWS Lambda functions to start and stop the RDS instances. **Most Voted**

Correct Answer: D

Community vote distribution

D (51%)

C (49%)

Comments

nebajp **Highly Voted** 1 year, 10 months ago

Selected Answer: C

Correct Answer us C - it allows you to define and automatically enforce desired configurations for EC2 and RDS.

upvoted 5 times

Teeshe **Most Recent** 2 weeks, 6 days ago

Selected Answer: D

State Manager manages instance state/configuration (like running SSM documents). It works with EC2 instances, not RDS DB instances. RDS is a managed service, you can't run SSM associations against it.

upvoted 1 times

Clouddon 10 months, 4 weeks ago

Selected Answer: D

✗ Why other options are incorrect:

A. Create a CloudWatch alarm + Lambda

✗ CloudWatch alarms are used to react to metric thresholds, not for scheduled events.

✗ Doesn't align with the need to run "on a schedule."

B. Trusted Advisor + Lambda

✗ Trusted Advisor is used for optimization and best practices, not automation.

✗ No scheduling capability or direct integration with actions like start/stop.

C. Systems Manager State Manager

✗ State Manager is meant for desired state configuration of instances, not for managing RDS start/stop schedules.

✗ It is not designed for time-based scheduling of RDS operations.

upvoted 1 times

ieffiong 1 year, 3 months ago

Selected Answer: D

When comparing the two:

Option C is less straightforward because State Manager isn't designed specifically for start/stop operations on RDS. It would likely require additional complexity (e.g., custom automation documents and potential workarounds) to achieve a scheduled start/stop functionality.

Option D leverages EventBridge's native scheduling capabilities to invoke Lambda functions, offering a more direct and streamlined approach with the least operational overhead.

upvoted 2 times

FlyingHawk 1 year, 4 months ago

Selected Answer: C

based on the AWS doc, you can use AWS Systems Manager State Manager associations to start and stop the RDS instances:

1. <https://docs.aws.amazon.com/systems-manager-automation-runbooks/latest/userguide/automation-ref-rds.html>

AWS-StartRdsInstance

AWS-StartStopAuroraCluster

AWS-StopRdsInstance

2. <https://docs.aws.amazon.com/systems-manager/latest/userguide/scheduling-automations-state-manager-associations.html>

D is also workable solution, but it will take more effort to implement.

upvoted 4 times

FlyingHawk 1 year, 4 months ago

If State Manager does not fully meet your needs, you can also use AWS Lambda with Amazon EventBridge to achieve similar functionality. This approach provides more flexibility but requires additional setup and management

<https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/automatically-stop-and-start-an-amazon-rds-db-instance-using-aws-systems-manager-maintenance-windows.html>

upvoted 1 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: C

A - CloudWatch alarms are designed to respond to metric thresholds (like CPU/memory utilization) rather than schedules.

B - Trusted Advisor provides cost optimization and resource usage recommendations but can't schedule anything.

D - Works but more complicated than option C. We don't wanna manually compose and then maintain lambda functions right?

upvoted 2 times

ad11934 8 months, 2 weeks ago

An Amazon EventBridge scheduled rule (cron) is the simplest, lowest-overhead way to run tasks on a fixed business-hours schedule.

upvoted 1 times

Liongeek 1 year, 5 months ago

Selected Answer: C

Setting State Manager with AWS managed Documents (StartEC2Instance, StopEC2Instance, StartRDSInstance and StopRDSInstance) it's easier and faster than creating your own Lambda functions and event bridge rules.

Letter C is the answ.

upvoted 1 times

bujuman 1 year, 6 months ago

Selected Answer: D

AWS Systems Manager State Manager has limitations regarding MSQ and the Db engine is not mentioned.

Check the this link to confirm D option : <https://aws.amazon.com/blogs/database/save-costs-by-automating-the-start-and-stop-of-amazon-rds-instances-with-aws-lambda-and-amazon-eventbridge/>

upvoted 4 times

ARV14 1 year, 6 months ago

It should be c as requirement is least operational overhead. Creating 4 lambda functions, event bridge rule, deploying through cloud formation works but adds operational overhead

upvoted 2 times

Salilgen 1 year, 5 months ago

You are right: AWS Systems Manager has limitations (<https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/automatically-stop-and-start-an-amazon-rds-db-instance-using-aws-systems-manager-maintenance-windows.html#automatically-stop-and-start-an-amazon-rds-db-instance-using-aws-systems-manager-maintenance-windows-prereqs>).

Anyway, the question doesn't mention any of them

upvoted 1 times

KaZimirovich 1 year, 7 months ago

Selected Answer: D

While AWS Systems Manager State Manager can be used to manage configuration states of AWS resources, it is generally more complex to set up for straightforward use cases like schedule-based starting and stopping of RDS instances compared to using a direct scheduling method through EventBridge.

upvoted 4 times

jingen11 1 year, 7 months ago

Selected Answer: C

Start, restart, or stop managed nodes and Amazon Relational Database Service (Amazon RDS) instances.

upvoted 2 times

hharbiordun85 1 year, 8 months ago

D. Amazon EventBridge allows you to create rules based on a schedule (using cron expressions) to automate tasks. You can set up rules to start the RDS instances at the beginning of business hours and stop them at the end of business hours.

By using AWS Lambda in conjunction with EventBridge, you can create functions that handle

upvoted 1 times

siheom 1 year, 8 months ago

Selected Answer: D

VOTE D

upvoted 2 times

AbhiBK 1 year, 9 months ago

To meet the requirement of running Amazon RDS DB instances only during business hours with the least operational overhead, the best solution would be:

D. Create an Amazon EventBridge rule that invokes AWS Lambda functions to start and stop the RDS instances. This approach allows you to automate the scheduling of start and stop actions using EventBridge rules, which can trigger Lambda functions based on a cron expression. This setup is straightforward and requires minimal ongoing management

upvoted 4 times

AbhiBK 1 year, 9 months ago

Option C, which involves using AWS Systems Manager State Manager to start and stop the RDS instances, is indeed a viable solution. It allows you to automate the process of keeping your RDS instances in a desired state, such as starting and stopping them on a schedule¹².

However, the reason Option D (using Amazon EventBridge with AWS Lambda) might be preferred for this scenario is due to its simplicity and flexibility. EventBridge rules can be easily configured with cron expressions to trigger Lambda functions, which can start and stop the RDS instances. This setup typically involves fewer steps and less configuration compared to setting up State Manager associations and IAM roles³.

Both options are valid, but Option D generally offers a more straightforward approach with potentially lower operational overhead

upvoted 2 times

dhewa 1 year, 9 months ago

Selected Answer: C

AWS Systems Manager State Manager allows you to automate the process of starting and stopping RDS instances based on a defined schedule.

upvoted 3 times

komorebi 1 year, 10 months ago

Selected Answer: D

Answer is D

upvoted 4 times

A consumer survey company has gathered data for several years from a specific geographic region. The company stores this data in an Amazon S3 bucket in an AWS Region.

The company has started to share this data with a marketing firm in a new geographic region. The company has granted the firm's AWS account access to the S3 bucket. The company wants to minimize the data transfer costs when the marketing firm requests data from the S3 bucket.

Which solution will meet these requirements?

- A. Configure the Requester Pays feature on the company's S3 bucket. **Most Voted**
- B. Configure S3 Cross-Region Replication (CRR) from the company's S3 bucket to one of the marketing firm's S3 buckets.
- C. Configure AWS Resource Access Manager to share the S3 bucket with the marketing firm AWS account.
- D. Configure the company's S3 bucket to use S3 Intelligent-Tiering Sync the S3 bucket to one of the marketing firm's S3 buckets.

Correct Answer: A

Community vote distribution

A (67%)

B (33%)

Comments

jhon9012227 2 months, 3 weeks ago

Selected Answer: B

transferring cost is not an effective solution for me
upvoted 1 times

asosa_adorati 11 months, 3 weeks ago

Selected Answer: B

Ok Guys... The solution is A i understand, but if another company has to pay for the money i am saving that is not a solution for me... For me B is the Answer, is the most cost effective solution...
upvoted 1 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: A

Read again - "The company wants to minimize the data transfer costs when the marketing firm requests data from the S3 bucket."

So as long as the company saves money, bravo. We don't care about the marketing firm. When enabling Requester Pays, the marketing firm (requester) pays for the costs of data requests and data transfer out of the S3 bucket, instead of the data owner (the consumer survey company).

B - CRR does not inherently reduce costs. In fact, replication incurs additional storage and transfer costs for the data owner, who would still bear the cost of replication, which contradicts the requirement to minimize their own data transfer costs.

C - RAM is used to share resources like S3 buckets, but it doesn't affect data transfer costs.

D - This helps reduce storage costs by moving objects between storage classes based on access patterns, but it doesn't reduce data transfer costs for requests made by the marketing firm.

upvoted 1 times

Anyio 1 year, 5 months ago

Selected Answer: B

The correct answer is B. Configure S3 Cross-Region Replication (CRR) from the company's S3 bucket to one of the marketing firm's S3 buckets.

Explanation:

Option A: Incorrect. The Requester Pays feature can help distribute the cost of data requests by billing the requestor, but it does not reduce the overall data transfer costs. It only shifts who pays for the data transfer, not the amount.

Option B: Correct. S3 Cross-Region Replication (CRR) replicates the data to a bucket in the marketing firm's region. This allows the marketing firm to access the data from a local region, minimizing cross-region data transfer charges each time data is accessed.

upvoted 1 times

ARV14 1 year, 6 months ago

Selected Answer: B

This is the same question as question 88. One difference is "it's data transfer costs" in question 88. Here it is "minimize the data transfer costs". A or B can be correct based on whether they are asking for reducing data transfer costs for survey company or both of them.

upvoted 1 times

AMEJack 1 year, 6 months ago

Selected Answer: A

Sorry that was for the next question.

upvoted 1 times

martinadurcakova1 1 year, 8 months ago

Selected Answer: A

A. Configure the Requester Pays feature on the company's S3 bucket.

Explanation:

A. Configuring the Requester Pays feature on the company's S3 bucket is the most appropriate solution. With Requester Pays, the marketing firm's AWS account will be responsible for the data transfer costs when accessing the data in the S3 bucket, minimizing the data transfer costs for the consumer survey company.

B. Configuring S3 Cross-Region Replication (CRR) from the company's S3 bucket to one of the marketing firm's S3 buckets would not be the most cost-effective solution, as the company would still be responsible for the data transfer costs.

upvoted 4 times

MatAlves 1 year, 8 months ago

The Requester Pays feature allows the bucket owner to offload the data transfer costs to the requester. When this feature is enabled, the marketing firm would pay for the data transfer when they access the data in the survey company's S3 bucket, which effectively minimizes costs for the survey company.

This means that the most cost-effective solution for the survey company, given that the marketing firm is accessing the data, is "A"

upvoted 2 times

[Removed] 1 year, 9 months ago

Answer B seems to be more logic, the question didn't mention which account will pay

upvoted 1 times

Abdullah2004 1 year, 9 months ago

question ask for method to reduce cost in survey company, so A will do it

upvoted 1 times

JunsK1e 1 year, 10 months ago

Selected Answer: A

Letter A

upvoted 2 times

pujithacg8 1 year, 10 months ago

A is the correct answer

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/RequesterPaysBuckets.html>

upvoted 4 times

A company uses AWS to host its public ecommerce website. The website uses an AWS Global Accelerator accelerator for traffic from the internet. The Global Accelerator accelerator forwards the traffic to an Application Load Balancer (ALB) that is the entry point for an Auto Scaling group.

The company recently identified a DDoS attack on the website. The company needs a solution to mitigate future attacks.

Which solution will meet these requirements with the LEAST implementation effort?

- A. Configure an AWS WAF web ACL for the Global Accelerator accelerator to block traffic by using rate-based rules
- B. Configure an AWS Lambda function to read the ALB metrics to block attacks by updating a VPC network ACL
- C. Configure an AWS WAF web ACL on the ALB to block traffic by using rate-based rules **Most Voted**
- D. Configure an Amazon CloudFront distribution in front of the Global Accelerator accelerator

Correct Answer: C

Community vote distribution

C (83%)

A (17%)

Comments

[Removed] **Highly Voted** 1 year, 9 months ago

Selected Answer: C

C as AWS WAF cannot be used directly on global accelerator
upvoted 9 times

dragosky 1 year, 5 months ago

it can be used, but to block HTTP/HTTPS requests.
upvoted 1 times

nebajp **Highly Voted** 1 year, 10 months ago

Selected Answer: C

Correct answer C.
Global Accelerator does not work with WAF as it is suitable for TCP/UDP where as WAF integrates with Application Load Balancer which is on Layer 7 on OSI model, suitable for Web app (Http/Https)
upvoted 6 times

rpmaws **Most Recent** 1 year, 9 months ago

Selected Answer: C

WAF can be applied on ALB, API gateway or cloud front.
upvoted 5 times

sOI852POL 1 year, 9 months ago

Selected Answer: C

Answer is C.
Note: AWS Global Accelerator itself doesn't support AWS WAF.

<https://repost.aws/knowledge-center/globalaccelerator-aws-waf-filter-layer7-traffic>
upvoted 5 times

dhewa 1 year, 9 months ago

Selected Answer: A

Configuring the WAF directly on the Global Accelerator ensures that malicious traffic is blocked before it reaches the Application Load Balancer, providing an additional layer of protection.
upvoted 2 times

officedepotadmin 1 year, 9 months ago

Selected Answer: A

Global Accelerator can be integrated with AWS WAF to provide protection at the edge, meaning malicious traffic can be blocked before it reaches your Application Load Balancer (ALB) or other resources in your AWS environment.

upvoted 2 times

AWS_Debu 1 year, 9 months ago

Answer is A

AWS Global Accelerator (GA) can be used with AWS Web Application Firewall (WAF) to protect applications from web exploits and DDoS attacks:

Block HTTP method and header attacks

GA, WAF, and the Application Load Balancer can block access to Layer 7 HTTP method and headers. WAF uses web access control list (web ACL) rules with the load balancer to evaluate incoming traffic and only forward requests that comply with the rules to the endpoint.

Detect and mitigate web application layer request floods

GA can protect web applications running on Application Load Balancer, and when used with WAF, it can also detect and mitigate web application layer request floods.

Prevent DDoS attacks

upvoted 1 times

GOTJ 1 year, 4 months ago

Unfortunately, AWS WAF cannot protect such a resource, according to this link: <https://docs.aws.amazon.com/waf/latest/developerguide/waf-chapter.html>.

Currently supported resources are:

- o Amazon CloudFront distribution
- o Amazon API Gateway REST API
- o Application Load Balancer
- o AWS AppSync GraphQL API
- o Amazon Cognito user pool
- o AWS App Runner service
- o AWS Verified Access instance

upvoted 1 times

komorebi 1 year, 10 months ago

Selected Answer: A

Answer is A

upvoted 1 times

A company uses an Amazon DynamoDB table to store data that the company receives from devices. The DynamoDB table supports a customer-facing website to display recent activity on customer devices. The company configured the table with provisioned throughput for writes and reads.

The company wants to calculate performance metrics for customer device data on a daily basis. The solution must have minimal effect on the table's provisioned read and write capacity.

Which solution will meet these requirements?

- A. Use an Amazon Athena SQL query with the Amazon Athena DynamoDB connector to calculate performance metrics on a recurring schedule.
- B. Use an AWS Glue job with the AWS Glue DynamoDB export connector to calculate performance metrics on a recurring schedule. **Most Voted**
- C. Use an Amazon Redshift COPY command to calculate performance metrics on a recurring schedule.
- D. Use an Amazon EMR job with an Apache Hive external table to calculate performance metrics on a recurring schedule.

Correct Answer: B

Community vote distribution

B (90%)

10%

Comments

toyaji **Highly Voted** 1 year, 9 months ago

Selected Answer: B

DynamoDB export connector literally "exports" table snapshot to s3 as dynamoDB-json object, then process on it. So it does not affect on read / write capacity on dynamoDB itself.

But Athena query directly on dynamoDB so affects on read / write capacity

upvoted 9 times

GOTJ **Most Recent** 1 year, 4 months ago

Selected Answer: B

The trick here is to avoid, by any means, interact with the original DynamoDB table because doing so would affect the provisioned capacity. Option "B" is the only one that interacts with an EXPORTED COPY of the table. So, the question is: would provisioned capacity be affected by the DynamoDB export process? The answer is: NO! (<https://docs.aws.amazon.com/glue/latest/dg/aws-glue-programming-etl-connect-dynamodb-home.html>):

"DynamoDB export to S3 allows you to export both full and incremental data from your DynamoDB table. Exports do not consume any read capacity units (RCUs) and have no impact on table performance and availability"

upvoted 2 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: B

A - Athena allows you to use standard SQL, but it doesn't guarantee minimal impact on the table.

B - It minimizes direct read/write operations on the table because the exported data can be processed separately (for example, export to S3 then do the processing).

C - Cool but there's no native connector to directly pull data from DynamoDB. You must export the data to S3 or another source first, adding complexity.

D - EMR impacts throughput significantly.

upvoted 4 times

Anyio 1 year, 5 months ago

Selected Answer: B

The correct answer is B. Use an AWS Glue job with the AWS Glue DynamoDB export connector to calculate performance metrics on a recurring schedule.

Explanation:

Option B: Correct. AWS Glue offers an efficient method to extract, transform, and load (ETL) data from DynamoDB to Amazon S3, without affecting the table's provisioned throughput as significantly. Once data is in S3, it can be further processed or queried using AWS Glue or other analytics services, without impacting DynamoDB's immediate R/W provisioning.

upvoted 2 times

rockykrish 1 year, 6 months ago

Selected Answer: A

Explanation:

Amazon Athena with DynamoDB Connector:

Athena is a serverless interactive query service that allows you to run SQL queries directly on data in various storage systems, including DynamoDB, through the Athena DynamoDB connector.

This setup minimizes the impact on the provisioned read and write capacity of the DynamoDB table because the connector reads data efficiently without directly querying the table.

Performance Metrics Calculation:

Athena provides SQL capabilities to compute performance metrics on the data fetched via the DynamoDB connector.

The queries can be scheduled using Amazon EventBridge or other automation tools to run daily.

Minimal Operational Overhead:

Athena requires no infrastructure to manage, making it a low-maintenance solution.

It is cost-effective since you pay only for the queries you run.

upvoted 1 times

tonybuivannghia 1 year, 7 months ago

Selected Answer: A

I think A is correct. We can query the data from DynamoDB by Amazon Athena DynamoDB connector directly, not via S3 Bucket.

upvoted 2 times

Salilgen 1 year, 5 months ago

You can query the data from DynamoDB by Amazon Athena DynamoDB connector directly but this effect on the table's provisioned read and write capacity.

upvoted 1 times

tonybuivannghia 1 year, 7 months ago

<https://docs.aws.amazon.com/athena/latest/ug/connectors-dynamodb.html>

upvoted 1 times

spoved 1 year, 8 months ago

Selected Answer: B

<https://aws.amazon.com/blogs/big-data/accelerate-amazon-dynamodb-data-access-in-aws-glue-jobs-using-the-new-aws-glue-dynamodb-elt-connector/>

upvoted 3 times

JoeTromundo 1 year, 8 months ago

Selected Answer: B

The DynamoDB export connector allows you to export data from DynamoDB to other storage solutions like Amazon S3 without consuming the table's provisioned read capacity, ensuring minimal impact on the performance of the table.

upvoted 3 times

Bogey 1 year, 9 months ago

B.

Instead, the new AWS Glue DynamoDB export connector reads DynamoDB data from the snapshot, which is exported from DynamoDB tables. This approach has following benefits:

It doesn't consume read capacity units of the source DynamoDB tables

upvoted 2 times

AbhiBK 1 year, 9 months ago

To calculate performance metrics for customer device data on a daily basis with minimal effect on the table's provisioned read and write capacity, the best solution would be:

A. Use an Amazon Athena SQL query with the Amazon Athena DynamoDB connector to calculate performance metrics on a recurring schedule. This approach allows you to run SQL queries directly on the data stored in DynamoDB without impacting the provisioned throughput, as Athena queries are serverless and do not consume DynamoDB read or write capacity¹.

upvoted 2 times

RealPro111 1 year, 9 months ago

Selected Answer: B

The right answer is B

upvoted 3 times

siheom 1 year, 9 months ago

VOTE B

upvoted 2 times

ksdpmx 1 year, 9 months ago

why is B wrong.. Glue DynamoDB export connector will read data from PITR instead of DynamoDB directly..

upvoted 2 times

dhewa 1 year, 9 months ago

I go with A

upvoted 2 times

A solutions architect is designing the cloud architecture for a new stateless application that will be deployed on AWS. The solutions architect created an Amazon Machine Image (AMI) and launch template for the application.

Based on the number of jobs that need to be processed, the processing must run in parallel while adding and removing application Amazon EC2 instances as needed. The application must be loosely coupled. The job items must be durably stored.

Which solution will meet these requirements?

A. Create an Amazon Simple Notification Service (Amazon SNS) topic to send the jobs that need to be processed. Create an Auto Scaling group by using the launch template with the scaling policy set to add and remove EC2 instances based on CPU usage.

B. Create an Amazon Simple Queue Service (Amazon SQS) queue to hold the jobs that need to be processed. Create an Auto Scaling group by using the launch template with the scaling policy set to add and remove EC2 instances based on network usage.

C. Create an Amazon Simple Queue Service (Amazon SQS) queue to hold the jobs that need to be processed. Create an Auto Scaling group by using the launch template with the scaling policy set to add and remove EC2 instances based on the number of items in the SQS queue. **Most Voted**

D. Create an Amazon Simple Notification Service (Amazon SNS) topic to send the jobs that need to be processed. Create an Auto Scaling group by using the launch template with the scaling policy set to add and remove EC2 instances based on the number of messages published to the SNS topic.

Correct Answer: C

Community vote distribution

C (100%)

Comments

LeonSauveterre 1 year, 5 months ago

Selected Answer: C

SQS for sure, but I actually swayed between B and C. After a while, I can say now I'm sure that scaling based on network usage doesn't directly have anything to do with the number of jobs in the queue. Or worse, this could result in burst-caused downtime or other poor scaling decisions.

upvoted 3 times

dhewa 1 year, 9 months ago

Answer is C: SQS is your first cue then scaling based on the number of requests

upvoted 3 times

A global ecommerce company uses a monolithic architecture. The company needs a solution to manage the increasing volume of product data. The solution must be scalable and have a modular service architecture. The company needs to maintain its structured database schemas. The company also needs a storage solution to store product data and product images.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Use an Amazon EC2 instance in an Auto Scaling group to deploy a containerized application. Use an Application Load Balancer to distribute web traffic. Use an Amazon RDS DB instance to store product data and product images.
- B. Use AWS Lambda functions to manage the existing monolithic application. Use Amazon DynamoDB to store product data and product images. Use Amazon Simple Notification Service (Amazon SNS) for event-driven communication between the Lambda functions.
- C. Use Amazon Elastic Kubernetes Service (Amazon EKS) with an Amazon EC2 deployment to deploy a containerized application. Use an Amazon Aurora cluster to store the product data. Use AWS Step Functions to manage workflows. Store the product images in Amazon S3 Glacier Deep Archive.
- D. Use Amazon Elastic Container Service (Amazon ECS) with AWS Fargate to deploy a containerized application. Use Amazon RDS with a Multi-AZ deployment to store the product data. Store the product images in an Amazon S3 bucket. **Most Voted**

Correct Answer: D

Community vote distribution

D (93%)

7%

Comments

LeonSauveterre **Highly Voted** 1 year, 5 months ago

Selected Answer: D

Based on my personal experience, "monolith" or "monolithic" is almost always ECS + Fargate. For your reference only ;)
upvoted 7 times

trinh_le **Most Recent** 1 year, 5 months ago

Selected Answer: C

Keyword scalable. So must use serverless like lambda or fargate (B or D)
Keyword modular => ECS => D
upvoted 1 times

GOTJ 1 year, 4 months ago

I think you are taking "modular" in the wrong way. I think they mean "modular" as an alternative to "monolithic". In other words, "modular" --> "Micro services" --> ECS/EKS
upvoted 1 times

martinadurcakova1 1 year, 8 months ago

Selected Answer: D

D is right
upvoted 2 times

dhewa 1 year, 9 months ago

Selected Answer: D

D gives us a server-less solution which is what we want.
upvoted 4 times

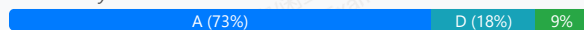
A company is migrating an application from an on-premises environment to AWS. The application will store sensitive data in Amazon S3. The company must encrypt the data before storing the data in Amazon S3.

Which solution will meet these requirements?

- A. Encrypt the data by using client-side encryption with customer managed keys. **Most Voted**
- B. Encrypt the data by using server-side encryption with AWS KMS keys (SSE-KMS).
- C. Encrypt the data by using server-side encryption with customer-provided keys (SSE-C).
- D. Encrypt the data by using client-side encryption with Amazon S3 managed keys.

Correct Answer: A

Community vote distribution



Comments

asosa_adorati 11 months, 3 weeks ago

Selected Answer: D

For D is the answer...where it says the customer has their owns keys??? Here it is the explanation...

Client-side encryption with Amazon S3 involves encrypting data on the client-side (your application) before uploading it to Amazon S3. This provides an extra layer of security as the data is encrypted before it ever leaves your control. When using S3 managed keys for client-side encryption, the key material and encryption key are managed by AWS, but the encryption process itself is handled by the S3 Encryption Client within your application

upvoted 2 times

Cpso 1 year, 6 months ago

Selected Answer: A

Question is unclear. "before" store= before send api or before save?

upvoted 2 times

Tygrins 1 year, 7 months ago

To meet the requirement of encrypting sensitive data before storing it in Amazon S3, the company should use Server-Side Encryption with Customer Provided Keys (SSE-C) or Client-Side Encryption. However, the most common and effective approach in AWS is Server-Side Encryption with AWS Key Management Service (SSE-KMS).

upvoted 1 times

JoeTromundo 1 year, 8 months ago

Selected Answer: A

If the answer is B, how will the data be encrypted before being stored in Amazon S3? It has to be client-side encryption.

upvoted 2 times

dhewa 1 year, 9 months ago

Selected Answer: B

AWS handles the encryption and decryption process, simplifying the management of encryption keys.

upvoted 1 times

George1990 1 year, 10 months ago

Correct is A

upvoted 3 times

JunsK1e 1 year, 10 months ago

Selected Answer: A

Before you store the data in S3 you can use client side encryption

upvoted 4 times

A company wants to create an Amazon EMR cluster that multiple teams will use. The company wants to ensure that each team's big data workloads can access only the AWS services that each team needs to interact with. The company does not want the workloads to have access to Instance Metadata Service Version 2 (IMDSv2) on the cluster's underlying EC2 instances.

Which solution will meet these requirements?

- A. Configure interface VPC endpoints for each AWS service that the teams need. Use the required interface VPC endpoints to submit the big data workloads.
- B. Create EMR runtime roles. Configure the cluster to use the runtime roles. Use the runtime roles to submit the big data workloads. **Most Voted**
- C. Create an EC2 IAM instance profile that has the required permissions for each team. Use the instance profile to submit the big data workloads.
- D. Create an EMR security configuration that has the EnableApplicationScopedIAMRole option set to false. Use the security configuration to submit the big data workloads.

Correct Answer: B

Community vote distribution

B (100%)

Comments

FlyingHawk 1 year, 4 months ago

Selected Answer: B

When you specify a runtime role for an Amazon EMR step, the jobs or queries that you submit can only access AWS resources that the policies attached to the runtime role allow. These jobs and queries can't access the Instance Metadata Service on the EC2 instances of the cluster or use the EC2 instance profile of the cluster to access any AWS resources.

<https://aws.amazon.com/blogs/big-data/introducing-runtime-roles-for-amazon-emr-steps-use-iam-roles-and-aws-lake-formation-for-access-control-with-amazon-emr/>

<https://docs.aws.amazon.com/emr/latest/ManagementGuide/emr-steps-runtime-roles.html>

upvoted 3 times

EllenLiu 1 year, 5 months ago

Selected Answer: B

runtime role is used to resolve the issue that there is only union permissions for all jobs and queries we can use. just like it mentions, for ec2, IMDSv2 is mandatory, however we can use runtime role to access instance metadata on ec2 avoid using IMDSv2

upvoted 1 times

martinadurcakova1 1 year, 8 months ago

Selected Answer: B

B. Creating EMR runtime roles and configuring the cluster to use them is the correct solution. EMR runtime roles allow you to grant specific permissions to the big data workloads, ensuring that each team's workloads can only access the required AWS services. Additionally, the runtime roles can be configured to disable access to IMDSv2, meeting the requirement.

upvoted 2 times

dhewa 1 year, 9 months ago

Selected Answer: B

This approach avoids the need for workloads to access the Instance Metadata Service (IMDSv2) on the underlying EC2 instances, as the permissions are managed through the runtime roles.

upvoted 2 times

A solutions architect is designing an application that helps users fill out and submit registration forms. The solutions architect plans to use a two-tier architecture that includes a web application server tier and a worker tier.

The application needs to process submitted forms quickly. The application needs to process each form exactly once. The solution must ensure that no data is lost.

Which solution will meet these requirements?

- A. Use an Amazon Simple Queue Service (Amazon SQS) FIFO queue between the web application server tier and the worker tier to store and forward form data. **Most Voted**
- B. Use an Amazon API Gateway HTTP API between the web application server tier and the worker tier to store and forward form data.
- C. Use an Amazon Simple Queue Service (Amazon SQS) standard queue between the web application server tier and the worker tier to store and forward form data.
- D. Use an AWS Step Functions workflow. Create a synchronous workflow between the web application server tier and the worker tier that stores and forwards form data.

Correct Answer: A

Community vote distribution

A (100%)

Comments

blehbleh 1 year, 8 months ago

Selected Answer: A

Unlike standard queues, FIFO queues don't introduce duplicate messages. FIFO queues help you avoid sending duplicates to a queue. If you retry the SendMessage action within the 5-minute deduplication interval, Amazon SQS doesn't introduce any duplicates into the queue.

<https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/FIFO-queues-exactly-once-processing.html>

upvoted 3 times

dhewa 1 year, 9 months ago

Selected Answer: A

Keywords in the qsn (quickly and once) = SQS FIFO

upvoted 3 times

komorebi 1 year, 10 months ago

Selected Answer: A

Answer is A

upvoted 3 times

A finance company uses an on-premises search application to collect streaming data from various producers. The application provides real-time updates to search and visualization features.

The company is planning to migrate to AWS and wants to use an AWS native solution.

Which solution will meet these requirements?

- A. Use Amazon EC2 instances to ingest and process the data streams to Amazon S3 buckets for storage. Use Amazon Athena to search the data. Use Amazon Managed Grafana to create visualizations.
- B. Use Amazon EMR to ingest and process the data streams to Amazon Redshift for storage. Use Amazon Redshift Spectrum to search the data. Use Amazon QuickSight to create visualizations.
- C. Use Amazon Elastic Kubernetes Service (Amazon EKS) to ingest and process the data streams to Amazon DynamoDB for storage. Use Amazon CloudWatch to create graphical dashboards to search and visualize the data.
- D. Use Amazon Kinesis Data Streams to ingest and process the data streams to Amazon OpenSearch Service. Use OpenSearch Service to search the data. Use Amazon QuickSight to create visualizations. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

Gbemi **Highly Voted** 1 year, 10 months ago

Answer is D

using Amazon Kinesis Data Streams, Amazon OpenSearch Service, and Amazon QuickSight provides a comprehensive and AWS-native solution that meets the requirements of real-time data ingestion, search, and visualization

upvoted 6 times

dhewa **Most Recent** 1 year, 9 months ago

Selected Answer: D

Keyword "streaming" hence Amazon Kinesis Data Streams as it is designed for real-time data ingestion and processing, making it ideal for collecting streaming data from various producers.

upvoted 2 times

A company currently runs an on-premises application that uses ASP.NET on Linux machines. The application is resource-intensive and serves customers directly.

The company wants to modernize the application to .NET. The company wants to run the application on containers and to scale based on Amazon CloudWatch metrics. The company also wants to reduce the time spent on operational maintenance activities.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Use AWS App2Container to containerize the application. Use an AWS CloudFormation template to deploy the application to Amazon Elastic Container Service (Amazon ECS) on AWS Fargate. **Most Voted**
- B. Use AWS App2Container to containerize the application. Use an AWS CloudFormation template to deploy the application to Amazon Elastic Container Service (Amazon ECS) on Amazon EC2 instances.
- C. Use AWS App Runner to containerize the application. Use App Runner to deploy the application to Amazon Elastic Container Service (Amazon ECS) on AWS Fargate.
- D. Use AWS App Runner to containerize the application. Use App Runner to deploy the application to Amazon Elastic Kubernetes Service (Amazon EKS) on Amazon EC2 instances.

Correct Answer: A

Community vote distribution

A (100%)

Comments

LeonSauveterre 1 year, 5 months ago

Selected Answer: A

App2Container: A tool to automatically containerize .NET applications.

Fargate: A serverless compute engine for containers, eliminating the need to manage underlying EC2 instances. It integrates with CloudWatch for scaling based on custom metrics.

App Runner: A fully managed service designed for simple application deployment and scaling, but it also deploys applications *directly*, which means it doesn't deploy to ECS or EKS.

upvoted 3 times

JoeTromundo 1 year, 8 months ago

Selected Answer: A

Meets all requirements.

upvoted 2 times

dhewa 1 year, 9 months ago

Selected Answer: A

A gives us a serverless solution

upvoted 2 times

A company is designing a new internal web application in the AWS Cloud. The new application must securely retrieve and store multiple employee usernames and passwords from an AWS managed service.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Store the employee credentials in AWS Systems Manager Parameter Store. Use AWS CloudFormation and the BatchGetSecretValue API to retrieve usernames and passwords from Parameter Store.
- B. Store the employee credentials in AWS Secrets Manager. Use AWS CloudFormation and AWS Batch with the BatchGetSecretValue API to retrieve the usernames and passwords from Secrets Manager.
- C. Store the employee credentials in AWS Systems Manager Parameter Store. Use AWS CloudFormation and AWS Batch with the BatchGetSecretValue API to retrieve the usernames and passwords from Parameter Store.
- D. Store the employee credentials in AWS Secrets Manager. Use AWS CloudFormation and the BatchGetSecretValue API to retrieve the usernames and passwords from Secrets Manager. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

JoeTromundo **Highly Voted** 1 year, 8 months ago

Selected Answer: D

Option D is similar to option B, but option B unnecessarily introduces AWS Batch into the solution. AWS Batch is designed for executing batch jobs and is not required for the use case of retrieving secrets in a web application. This adds complexity and overhead without benefit.

upvoted 10 times

FlyingHawk **Most Recent** 1 year, 4 months ago

Selected Answer: D

BatchGetSecretValue API is with Secrets Manager, not related to AWS Batch, it allows you to retrieve 1 to 20 secrets via a single API call.
<https://aws.amazon.com/blogs/security/how-to-use-the-batchgetsecretsvalue-api-to-improve-your-client-side-applications-with-aws-secrets-manager/>

upvoted 2 times

spoved 1 year, 8 months ago

Selected Answer: D

https://docs.aws.amazon.com/secretsmanager/latest/apireference/API_BatchGetSecretValue.html

upvoted 4 times

viejito 1 year, 9 months ago

respuesta correcta D : La opción D proporciona una solución adecuada y segura para gestionar y recuperar secretos, con un enfoque en la menor sobrecarga operativa posible al utilizar AWS Secrets Manager junto con la API correcta y la integración con AWS CloudFormation.

upvoted 1 times

A company that is in the ap-northeast-1 Region has a fleet of thousands of AWS Outposts servers. The company has deployed the servers at remote locations around the world. All the servers regularly download new software versions that consist of 100 files. There is significant latency before all servers run the new software versions.

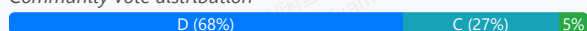
The company must reduce the deployment latency for new software versions.

Which solution will meet this requirement with the LEAST operational overhead?

- A. Create an Amazon S3 bucket in ap-northeast-1. Set up an Amazon CloudFront distribution in ap-northeast-1 that includes a CachingDisabled cache policy. Configure the S3 bucket as the origin. Download the software by using signed URLs.
- B. Create an Amazon S3 bucket in ap-northeast-1. Create a second S3 bucket in the us-east-1 Region. Configure replication between the buckets. Set up an Amazon CloudFront distribution that uses ap-northeast-1 as the primary origin and us-east-1 as the secondary origin. Download the software by using signed URLs.
- C. Create an Amazon S3 bucket in ap-northeast-1. Configure Amazon S3 Transfer Acceleration. Download the software by using the S3 Transfer Acceleration endpoint.
- D. Create an Amazon S3 bucket in ap-northeast-1. Set up an Amazon CloudFront distribution. Configure the S3 bucket as the origin. Download the software by using signed URLs. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

JoeTromundo **Highly Voted** 1 year, 8 months ago

Selected Answer: D

By setting up a CloudFront distribution with the S3 bucket as the origin, the software versions can be cached at edge locations close to the remote AWS Outposts servers. This reduces the latency of downloading new software versions because the servers can access the content from the nearest CloudFront edge location instead of downloading it directly from the S3 bucket in the ap-northeast-1 Region.

About option C: S3 Transfer Acceleration optimizes the upload and download of files to S3 by routing through optimized network paths. However, it is primarily designed to improve performance when transferring data over long distances to S3 (uploads). It does not provide the same level of global caching and latency reduction as CloudFront for large-scale distribution.

upvoted 7 times

ashishs174 **Highly Voted** 1 year, 8 months ago

Option C is correct.

Option D brings additional overhead in terms of setting up, so opted out.

Also from google : If you want to download software quickly from a large distance, you should generally use S3 Transfer Acceleration to download from your S3 bucket, as it leverages Amazon's global edge network to significantly speed up long-distance data transfers, making it ideal for large file downloads across continents; while CloudFront is better suited for delivering smaller content like website assets to users globally, where faster response times and caching are critical

upvoted 6 times

In2718 **Most Recent** 9 months, 2 weeks ago

Selected Answer: D

But you will need to configure for caching issues...

Best practice: Immutable objects + versioned paths (no invalidations)

Name files by version (or content hash) and never overwrite:

s3://updates/myapp/1.2.3/bin/fileA

s3://updates/myapp/1.2.3/bin/fileB

Publish a tiny manifest (e.g., latest.json) that tells clients which version to fetch:

```
{ "version": "1.2.3", "files": ["bin/fileA", "bin/fileB", ...] }
```

Set short TTL or Cache-Control: max-age=60 on the manifest so Outposts servers quickly see the new version.
Set long TTL on the versioned files (they never change), maximizing cache hit ratio.

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Selected Answer: D

Using Amazon CloudFront as a global content delivery network (CDN) caches software updates closer to the remote Outposts servers at AWS edge locations. This significantly reduces latency compared to downloading files directly from S3.

- ✓ Global Acceleration – CloudFront caches the software in edge locations worldwide, reducing latency for remote locations.
- ✓ Efficient Distribution – Servers in different regions fetch files from the closest CloudFront edge cache instead of always pulling from ap-northeast-1.
- ✓ Signed URLs for Security – Ensures only authorized servers can download the software.
- ✓ Low Operational Overhead – No need for replication, manual distribution, or complex networking setups.

upvoted 1 times

FlyingHawk 1 year, 4 months ago

C. S3 Transfer Acceleration ✗ Improves upload speed but does not help with global caching. Each Outpost server still downloads directly from ap-northeast-1, causing high latency.

upvoted 1 times

dfgdsfgfdgreg 1 year, 4 months ago

Selected Answer: C

Including "global" makes CloudFront stand out as the obvious choice. But S3 global acceleration requires the least operational overhead while also assisting with latency issues. The mention of hundreds of files also points towards s3 acceleration.

upvoted 1 times

Saliigen 1 year, 5 months ago

Selected Answer: C

IMO answer is C.

A is out because disabling caching, it involves higher latency than C

B is out because it doesn't meet the requirements: adds complexity without significant latency reduction

D meets the requirements but it requires more operational overhead than C

upvoted 1 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: D

A - CachingDisabled is the polar opposite of our goal.

B - CloudFront with dual-region origins adds unnecessary complexity without significant latency reduction.

C - Faster upload/download speeds compared to direct S3 access, but the distribution latency is still there!

D - Looks good. Just one small thing: files may take a little longer to propagate to edge locations initially, but this can be mitigated by configuring an appropriate cache invalidation policy. Once cached at edge locations, subsequent downloads would be much faster.

upvoted 1 times

deacon967 1 year, 5 months ago

Selected Answer: C

C : all other options use CloudFront. CloudFront is a CDN, not needed for software download.

upvoted 1 times

RamanAgarwal 1 year, 5 months ago

Selected Answer: C

Should use S3 Transfer acceleration to avoid CF setup overhead. It will help with improving download speed

upvoted 1 times

JA2018 1 year, 5 months ago

Selected Answer: A

What about A?

1. Minimized operational overhead:

This option provides the least operational overhead because it only requires setting up a single S3 bucket and a CloudFront distribution in the same region (ap-northeast-1), with a simple configuration to disable caching on the CloudFront distribution, ensuring fresh software updates are always delivered to Outposts servers.

2. Reduced latency:

By using a CloudFront distribution with a CachingDisabled policy, the software updates will be delivered from the closest edge location to each Outpost server, significantly reducing latency due to geographical proximity.

3. Signed URLs for security:

Using signed URLs ensures that only authorized users can access the software updates, enhancing security.

upvoted 1 times

dhewa 1 year, 9 months ago

Selected Answer: D

This approach leverages the global presence of CloudFront to minimize latency and simplifies the deployment process, reducing operational overhead.

upvoted 2 times

nebajp 1 year, 10 months ago

Selected Answer: C

Correct answer is C

Because for new software version we do not use Cloudfront,

upvoted 2 times

pelrock 1 year, 10 months ago

Selected Answer: D

The question asks for the solution with the least operational overhead. Option D has minimal setup and maintenance requirements, as CloudFront is a managed service that handles caching and distribution across the globe, reducing latency without requiring additional configurations or services.

upvoted 3 times

A company currently runs an on-premises stock trading application by using Microsoft Windows Server. The company wants to migrate the application to the AWS Cloud.

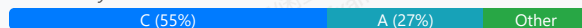
The company needs to design a highly available solution that provides low-latency access to block storage across multiple Availability Zones.

Which solution will meet these requirements with the LEAST implementation effort?

- A. Configure a Windows Server cluster that spans two Availability Zones on Amazon EC2 instances. Install the application on both cluster nodes. Use Amazon FSx for Windows File Server as shared storage between the two cluster nodes.
- B. Configure a Windows Server cluster that spans two Availability Zones on Amazon EC2 instances. Install the application on both cluster nodes. Use Amazon Elastic Block Store (Amazon EBS) General Purpose SSD (gp3) volumes as storage attached to the EC2 instances. Set up application-level replication to sync data from one EBS volume in one Availability Zone to another EBS volume in the second Availability Zone.
- C. Deploy the application on Amazon EC2 instances in two Availability Zones. Configure one EC2 instance as active and the second EC2 instance in standby mode. Use an Amazon FSx for NetApp ONTAP Multi-AZ file system to access the data by using Internet Small Computer Systems Interface (iSCSI) protocol. **Most Voted**
- D. Deploy the application on Amazon EC2 instances in two Availability Zones. Configure one EC2 instance as active and the second EC2 instance in standby mode. Use Amazon Elastic Block Store (Amazon EBS) Provisioned IOPS SSD (io2) volumes as storage attached to the EC2 instances. Set up Amazon EBS level replication to sync data from one io2 volume in one Availability Zone to another io2 volume in the second Availability Zone.

Correct Answer: C

Community vote distribution



Comments

LeonSauveterre **Highly Voted** 1 year, 5 months ago

Selected Answer: C

Keep in mind: WE NEED LOW LATENCY BLOCK STORAGE.

A - FSx for Windows File Server is not designed for block storage. It is optimized for shared file storage.

B - God this requires high implementation effort.

C - FSx for NetApp ONTAP supports block storage (via iSCSI) and provides Multi-AZ fault tolerance out of the box.

D - EBS io2 volumes with replication across AZs are hard and complex to configure and manage. Even if you think this is acceptable, still it's definitely much harder than option C.

upvoted 7 times

LeonSauveterre 1 year, 5 months ago

I know it is almost correct to look for "FSx for Windows" when you spot "Microsoft Windows Server" in the question, but that's far from enough to get the correct answer.

upvoted 1 times

Anyio **Most Recent** 1 year, 5 months ago

Selected Answer: C

The correct answer is C.

Explanation:

Option C: Correct. Amazon FSx for NetApp ONTAP provides a managed file storage solution that supports iSCSI protocol for block storage access, enabling low-latency, highly available access to shared storage across multiple Availability Zones. This solution requires less implementation effort

because Amazon FSx manages the complex underlying multi-AZ configuration and data replication automatically.

Option A: Incorrect. While Amazon FSx for Windows File Server provides a managed shared storage solution, it is primarily used for SMB-based file shares rather than block storage needs like those typically demanded by stock trading applications. Also, it might introduce additional latency for block-level access.

upvoted 3 times

EllenLiu 1 year, 5 months ago

Selected Answer: C

block storage => FSx for NetApp ONTAP not only provides iSCSI(for block) support, but also offers comprehensive protocol support including SMB and NFS.

upvoted 2 times

ARV14 1 year, 6 months ago

Selected Answer: C

FSx windows file server does not provide block storage, also Ontap provides very low latency and provides various compatible storage including block

upvoted 1 times

rosanna 1 year, 6 months ago

Selected Answer: D

block means EBS, and I've searched but I don't see FSx supporting shared block storage, not that I know of, and thinking about if it did exist, I would cost more than a replica- both operationally and money wise. And mostly we don't need complex methods so I'd go for D

upvoted 1 times

ckhemani 1 year, 6 months ago

Selected Answer: A

Answer A. Although they are looking for Block storage which is EBS but EBS can't be shared across two availability zones

Multi-Attach feature: While you can use the "Multi-Attach" feature to attach a single EBS volume to multiple instances within the same Availability Zone, this does not allow sharing across different Availability Zones.

Configuring NETAPP is not least operational.

upvoted 1 times

Bwhizzy 1 year, 7 months ago

Selected Answer: A

Answer is A.

Amazon FSx for Windows File Server provides fully managed shared storage designed for Windows workloads. It offers seamless integration with Windows applications and supports Windows Server Failover Clustering (WSFC), making it the ideal choice for clustering applications.

High availability is provided as the cluster spans multiple Availability Zones, ensuring that the application continues to function in case of an AZ failure.

Low-latency access to shared block storage is achievable with FSx, and it reduces the complexity compared to setting up replication between multiple Amazon EBS volumes.

Least implementation effort: You don't need to manage block-level replication or create complex replication setups between AZs, as FSx for Windows File Server handles this automatically. It simplifies storage management and offers native Windows support.

upvoted 2 times

martinadurcakova1 1 year, 8 months ago

Using Amazon FSx for Windows File Server would not be the best solution, as the company requires block storage, not file storage.

D. Deploying the application on EC2 instances and using Amazon EBS Provisioned IOPS SSD (io2) volumes with EBS-level replication is the solution with the least implementation effort for the following reasons:

EBS volumes provide the required block storage for the application.

Using EBS-level replication to sync data between Availability Zones is a built-in feature, requiring less implementation effort compared to setting up custom replication mechanisms.

Provisioned IOPS SSD (io2) volumes ensure the low-latency access to the block storage required by the application.

By using Amazon EBS Provisioned IOPS SSD (io2) volumes with EBS-level replication, the company can meet the requirements for high availability and low-latency access to block storage with the least implementation effort.

upvoted 4 times

spoved 1 year, 8 months ago

Selected Answer: D

block storage -> B or D

io1, io2 support EBS Multi-Attach -> D

upvoted 3 times

HappyG 1 year, 8 months ago

Selected Answer: C

Key word in the question is block storage which eliminates A as a possibility. FSx for NetApp ONTAP provides low-latency block storage, multi-AZ high availability, and requires the least implementation effort because it is a fully managed service with built-in replication and failover capabilities.

This makes it the most suitable and cost-effective option for the company's needs.

upvoted 2 times

trongod05 1 year, 8 months ago

Amazon FSx for NetApp ONTAP also provides shared block storage over the iSCSI and NVMe-over-TCP protocols. Seems to be a complete match there.

upvoted 2 times

JoeTromundo 1 year, 8 months ago

Selected Answer: A

For those who are choosing option other than A: what part of "with the LEAST implementation effort" do you not understand?

upvoted 2 times

HappyG 1 year, 8 months ago

The part where it calls for block storage.

upvoted 2 times

ashishs174 1 year, 8 months ago

Answer is B

EFS is not Block Storage, EBS is.(Thus, A and C are out)

upvoted 1 times

rpmaws 1 year, 9 months ago

Selected Answer: C

due to this line in question "low-latency access to block storage" C looks appropriate.

upvoted 3 times

toyaji 1 year, 9 months ago

Selected Answer: A

A - Windows cluster across two availability zones satisfies highly available condition and FSx satisfies storage access from multiple ec2 especially for Windows server.

B - "Set up application-level replication to sync data" not for LEAST effort condition

C, D - "in standby mode" need to be activated manually so its not highly available

upvoted 2 times

elmyth 1 year, 9 months ago

Selected Answer: B

Block storage

upvoted 2 times

elmyth 1 year, 9 months ago

Changed for C, bc application-level replication is more difficult to implement.

upvoted 2 times

dhewa 1 year, 9 months ago

Selected Answer: A

Windows = FSX

upvoted 2 times

A company is designing a web application with an internet-facing Application Load Balancer (ALB).

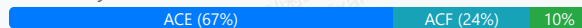
The company needs the ALB to receive HTTPS web traffic from the public internet. The ALB must send only HTTPS traffic to the web application servers hosted on the Amazon EC2 instances on port 443. The ALB must perform a health check of the web application servers over HTTPS on port 8443.

Which combination of configurations of the security group that is associated with the ALB will meet these requirements? (Choose three.)

- A. Allow HTTPS inbound traffic from 0.0.0.0/0 for port 443. **Most Voted**
- B. Allow all outbound traffic to 0.0.0.0/0 for port 443.
- C. Allow HTTPS outbound traffic to the web application instances for port 443. **Most Voted**
- D. Allow HTTPS inbound traffic from the web application instances for port 443.
- E. Allow HTTPS outbound traffic to the web application instances for the health check on port 8443. **Most Voted**
- F. Allow HTTPS inbound traffic from the web application instances for the health check on port 8443.

Correct Answer: ACE

Community vote distribution



Comments

In2718 **Highly Voted** 10 months ago

Selected Answer: ACF

I agree, one of the worst written questions, which I hope not to see on the real test. Terrible question. Terribly written answers.
upvoted 5 times

spoved **Most Recent** 1 year, 8 months ago

Selected Answer: ACE

<https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-update-security-groups.html>
The following rules are recommended for an internet-facing load balancer.
upvoted 4 times

JoeTromundo 1 year, 8 months ago

Selected Answer: ACE

This question is poorly worded: assuming that, by default in security groups, all OUTBOUND connections are ALLOWED and all INBOUND connections are DENIED, options C and E would not even need to be configured. What would be needed is to create a security group for the EC2 instances allowing INBOUND connections from the ALB security group to the EC2 instances security group on ports 443 and 8443.
upvoted 4 times

dhewa 1 year, 9 months ago

Selected Answer: ACE

A. This allows the ALB to receive HTTPS traffic from the public internet.
C. This ensures that the ALB can send HTTPS traffic to the web application servers.
E. This allows the ALB to perform health checks on the web application servers over HTTPS on port 8443.
upvoted 4 times

dhewa 1 year, 9 months ago

Selected Answer: ADE

A. This allows the ALB to receive HTTPS traffic from the public internet.
C. This ensures that the ALB can send HTTPS traffic to the web application servers.
E. This allows the ALB to perform health checks on the web application servers over HTTPS on port 8443.

upvoted 2 times

aragon_saa 1 year, 9 months ago

Selected Answer: ACE

Answer is ACE

upvoted 2 times

A company hosts an application on AWS. The application gives users the ability to upload photos and store the photos in an Amazon S3 bucket. The company wants to use Amazon CloudFront and a custom domain name to upload the photo files to the S3 bucket in the eu-west-1 Region.

Which solution will meet these requirements? (Choose two.)

A. Use AWS Certificate Manager (ACM) to create a public certificate in the us-east-1 Region. Use the certificate in CloudFront.

Most Voted

B. Use AWS Certificate Manager (ACM) to create a public certificate in eu-west-1. Use the certificate in CloudFront.

C. Configure Amazon S3 to allow uploads from CloudFront. Configure S3 Transfer Acceleration.

D. Configure Amazon S3 to allow uploads from CloudFront origin access control (OAC). Most Voted

E. Configure Amazon S3 to allow uploads from CloudFront. Configure an Amazon S3 website endpoint.

Correct Answer: AD

Community vote distribution

AD (79%)

BD (17%) 4%

Comments

JoeTromundo Highly Voted 1 year, 8 months ago

Selected Answer: AD

Amazon CloudFront requires an SSL/TLS certificate to use HTTPS with a custom domain name. This certificate MUST be provisioned in the us-east-1 Region, regardless of where your content is hosted. This is because CloudFront only supports certificates in the us-east-1 Region for use with custom domain names.

Origin Access Control (OAC) is a feature that allows you to securely upload content to an S3 bucket using CloudFront. It provides fine-grained access control and ensures that only CloudFront can upload files to the S3 bucket, preventing direct access. Configuring S3 to allow uploads from CloudFront using OAC ensures that only CloudFront can interact with the S3 bucket, adding an extra layer of security.

upvoted 10 times

George1990 Highly Voted 1 year, 10 months ago

Correct is BD

upvoted 6 times

yangbo Most Recent 1 year, 3 months ago

Selected Answer: AC

OAC do not support uploads from CloudFront to the S3 bucket.

upvoted 1 times

TicDcNess 1 year, 8 months ago

AWS Region for AWS Certificate Manager

To use a certificate in AWS Certificate Manager (ACM) to require HTTPS between viewers and CloudFront, make sure you request (or import) the certificate in the US East (N. Virginia) Region (us-east-1)

upvoted 2 times

rpmaws 1 year, 8 months ago

Selected Answer: AD

cloud front require all SSL certificate to be in us-east region regardless the origin location of the site server.

upvoted 2 times

dhewa 1 year, 9 months ago

Selected Answer: AD

A. Use AWS Certificate Manager (ACM) to create a public certificate in the us-east-1 Region. Use the certificate in CloudFront: CloudFront requires the certificate to be in the us-east-1 Region for custom domain names.

D. Configure Amazon S3 to allow uploads from CloudFront origin access control (OAC): This ensures secure uploads from CloudFront to the S3 bucket.

upvoted 3 times

komorebi 1 year, 10 months ago

Selected Answer: AD

Answer is AD

upvoted 4 times

JunsK1e 1 year, 10 months ago

Selected Answer: BD

BD correct answer

upvoted 4 times

A weather forecasting company collects temperature readings from various sensors on a continuous basis. An existing data ingestion process collects the readings and aggregates the readings into larger Apache Parquet files. Then the process encrypts the files by using client-side encryption with KMS managed keys (CSE-KMS). Finally, the process writes the files to an Amazon S3 bucket with separate prefixes for each calendar day.

The company wants to run occasional SQL queries on the data to take sample moving averages for a specific calendar day.

Which solution will meet these requirements MOST cost-effectively?

- A. Configure Amazon Athena to read the encrypted files. Run SQL queries on the data directly in Amazon S3. **Most Voted**
- B. Use Amazon S3 Select to run SQL queries on the data directly in Amazon S3.
- C. Configure Amazon Redshift to read the encrypted files. Use Redshift Spectrum and Redshift query editor v2 to run SQL queries on the data directly in Amazon S3.
- D. Configure Amazon EMR Serverless to read the encrypted files. Use Apache SparkSQL to run SQL queries on the data directly in Amazon S3.

Correct Answer: A

Community vote distribution

A (71%)

D (29%)

Comments

andreea10 10 months ago

Selected Answer: D

A (Athena) / C (Redshift Spectrum) / B (S3 Select): they can't read client-side-encrypted objects, so they'd fail unless you first re-upload plaintext or switch to SSE—extra work/cost the question doesn't allow.

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Selected Answer: A

<https://aws.amazon.com/blogs/storage/how-to-optimize-querying-your-data-in-amazon-s3/>

upvoted 1 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: A

Seems like "sample moving averages" is a typo of "simple moving averages". A simple moving average is formed by computing the average price of a security over a specific number of periods, but let's just ignore it if you don't care one bit haha

A - Athena is serverless, works perfectly with Parquet files, can handle encrypted data, and super cost-effective because you only pay for the amount of data you query.

B - Amazon S3 Select is fast for simple filtering or lightweight queries, not for complex queries like calculating moving averages.

C - Redshift with Redshift Spectrum is extremely powerful and competent to crunch heavy-duty data, so overkill for occasional SQL queries.

D - EMR is scalable and fast but is usually for heavy data pipelines, not casual SQL analysis. Like C, this is overkill.

upvoted 3 times

EllenLiu 1 year, 5 months ago

Selected Answer: A

it is a simple SQL just in S3, believe it is a use case of Athena which supports client-side decryption

upvoted 1 times

Cpso 1 year, 6 months ago

Selected Answer: D

Key: CSE-KMS

A,B,C not support decrypted by CSE-KMS

upvoted 3 times

Ronanrh 1 year, 8 months ago

A. Athena would be overkill and more expensive for simple queries on specific files.

C. Amazon Redshift with Redshift Spectrum would be significantly more complex and costly to set up and maintain for occasional queries.

D. EMR Serverless with Apache SparkSQL would also be more complex and likely more expensive for this use case.

S3 Select provides the right balance of functionality and cost-effectiveness for the described scenario, making it the most suitable choice

upvoted 2 times

elmyth 1 year, 7 months ago

S3 Select is deprecated, right answer is A, clue word is "occasional", otherwise it would be Redshift.

upvoted 3 times

FlyingHawk 1 year, 4 months ago

Thank for sharing this, I just checked,

After careful consideration, we have made the decision to close new customer access to Amazon S3 Select and Amazon S3 Glacier Select, effective July 25, 2024. Amazon S3 Select and Amazon S3 Glacier Select existing customers can continue to use the service as usual.

upvoted 1 times

dhewa 1 year, 9 months ago

Selected Answer: A

A is my choice.

upvoted 2 times

swati1508 1 year, 10 months ago

Selected Answer: A

Athena for sql

upvoted 3 times